Summary Report, v2: Auditing a Deployed LLM Judge for Data Matching: Is Any Weighting of Its Ingredients Better Than the One It Ships With?

2026

1 Setup

1.1 Suite and pools

- 31 datasets: 11 entity matching (EM), 6 schema matching (SM), 11 table question answering and understanding (TQA), 3 financial documents (FIN). Each adapter reads the official test or dev split and emits items with a question, a context (table, document, or record pair), a gold answer with aliases, a list gold where the answer is a set, an answer type (numeric / boolean / text / free-form) and the dataset's native metric.
- Pools: 150 items per dataset by seeded stratified sampling (strata: question type, sub-task, database, product category, match label); 100 flagged for answering (OfficeQA Pro V2: all 90; Alaska schema: 93). The remaining 50 are a seeded reserve and are not used by any step. Total answered items: 3,083 per model.
- Entity-matching pools are 50/50 match / non-match (Papadakis–Christen keeps its real 1.8 % match ratio). Schema matching is framed as one question per source column with a candidate list that always contains the gold. BIRD is graded on the executed result of the gold SQL, not on SQL text.
- Official census: every dataset's official population is counted by the same strata, so that pool numbers can be re-weighted to the official split (Section 1.5).

1.2 Collecting answers

- One prompt per item: context, question, and an answer-format line requesting `FINAL ANSWER: <answer>`.
- Models: Claude Opus 5, Claude Sonnet 5, GPT-5.5, GPT-5.4-mini, Gemini 3.8 Flash, Grok 4, DeepSeek-chat, DeepSeek-R1, gpt-oss-120b (Together), Llama 3.3 70B (Together). Gemini 3.1 Pro is collected separately at 240 items a day by a scheduled GitHub Actions workflow and enters the study automatically at 90 % suite coverage.
- Decoding: temperature 0 where the API accepts it (Gemini, Grok, DeepSeek, Llama, gpt-oss). Claude 5 reports the parameter deprecated and GPT-5.5 rejects any non-default value; those ran at provider defaults. Every cached row records which applied.
- Output budget 700 tokens. A reply cut off for length with no final-answer line is re-asked at 2,500 tokens, then at 8,000. The trigger is mechanical (finish reason plus absence of the final

line), never correctness. Originals are archived with a reason; replacement rows carry `attempt` = 2 or 3. Replies still unterminated at 8,000 tokens are scored as given (effectively failures).
- Every answer is cached with tokens, latency, cost, finish reason and decoding; re-runs skip cached items. Cost: USD 161 for ten models (Claude Opus 5: USD 99).

1.3 Scoring an answer

- Extraction: the `FINAL ANSWER:` line; else a boxed expression; else the last short line of the reply.
- Normalisation: Unicode NFKC, lower-case, canonical numbers ("1,577.00" → "1577", "(12.6)" → "−12.6"), currency symbols and punctuation removed, articles dropped, whitespace collapsed.
- Units and scale: every number is parsed with sign, percent marker (%, "percent", "pp") and scale word (thousand / million / billion / trillion and abbreviations). A parsed number exposes its equivalent readings (raw, scaled, ÷100 if marked percent, ×100 if an unmarked fraction); the relative error between prediction and gold is the minimum over all pairs of readings. Real-unit conversions (km ↔ miles) are not attempted.
- Aliases: each item carries the primary gold plus acceptable alternatives supplied by the adapter (e.g. *yes* ≡ *match* ≡ 1; several legitimate target columns in schema matching; comma- and pipe-joined forms of list golds). Every component is computed against each alias and the best-scoring alias is kept. Booleans are canonicalised before comparison.
- Components per answer: token F1, numeric decay $e^{-2.5 \cdot \text{rel. err}}$ (error clipped at 2; falls back to exact match for non-numeric golds), token precision, token recall; also exact match, numeric tolerance (1 %), ROUGE-L, and a hedge flag (several distinct numeric candidates, "A or B", or an enumeration of short candidates).
- Native metric per dataset (baseline column): numeric tolerance for numeric golds, exact match for text and boolean golds, ROUGE-L for free-form (FeTaQA).
- Composite under weighting w on the simplex: $R_w = \sum_k w_k m_k$, $k \in \{\text{F1, decay, precision, recall}\}$. v1 is $w = (0.35, 0.35, 0.15, 0.15)$.
- Dataset score of a model: mean R_w over the items that every model answered. A model is in the study only if it covered ≥ 90 % of the suite; a dataset is used only with ≥ 20 answered items.

1.4 Evaluating a weighting

1. **Agreement $J(w)$.** Rank the ten models inside each dataset under w; pool the 31 rankings by Borda; J is the mean Kendall τ_b between each dataset's ranking and the pooled one. J measures consistency between datasets, not correctness. Computed for all 1,771 weightings of the step-0.05 simplex lattice (v1, the four corners and the uniform mix are lattice points), and also under Kemeny, Copeland and Bradley–Terry pooling.
2. **Correctness $\rho(w)$.** 31,380 synthetic answers of known utility generated from the golds of the answered items by typed operators (identity, alias, unit variant, typo, rounding, magnitude error, sign flip, wrong value, partial list, superset, hedge, verbose, no answer, ...). Reported: Spearman ρ between R_w and utility, AUC correct vs wrong, mean score of wrong answers, hedge gap; the (J, ρ) Pareto front and its knee.
3. **Honest selection.** Nested item bootstrap (B = 1,000): choose the best w on in-bag items, score it out of bag. Reported: the optimism gap, the paired out-of-bag difference to v1 with a 95 % interval, and the plateau (weightings not significantly below the best, paired bootstrap $p > .05$) plus a practical plateau (within 0.02 of the best).
4. **Transfer.** Leave-one-dataset-out and leave-one-family-out: re-fit w on the rest, measure τ of the held-out dataset's ranking to the pooled rest; per-family optima; sensitivity to one lattice

step (0.05).

5. **Designs.** Uniform-lattice, Dirichlet-snapped and Latin-hypercube samples of 100 weightings, 20 seeds each, compared with the exhaustive optimum (regret, rank of the found best, centred L_2 discrepancy, minimum pairwise L_1).

1.5 From scores to a leaderboard

- Rules reported: exact match only, v1, the five best by J, the best by ρ, the (J, ρ) Pareto knee.
- Pooling: Bradley–Terry strengths from item-level pairwise wins (MLE, ridge 0.01, standard errors from the observed information) as the headline; Borda, mean-z, Kemeny (with cost), Copeland, reciprocal-rank fusion and family-balanced z as checks.
- Uncertainty: 1,000 item bootstraps of the Borda and Bradley–Terry rankings (rank intervals, probability of first place), saved in full.
- Significance: Friedman with the Iman–Davenport correction across datasets, Nemenyi critical difference, Holm-corrected pairwise Wilcoxon; paired permutation and Wilcoxon per dataset and model pair (Holm within dataset); Kendall's W among datasets; majority-class baselines for boolean sets.
- Official-split estimates: the same answers re-weighted by official stratum share / sample share (post-stratification) with a weighted item bootstrap; positive-class precision, recall and F1 at the official match ratio for the matching sets.
- Diagnostics: pairwise τ between dataset rankings; floor, ceiling and self-stability flags per dataset.
- Reporting: one script regenerates the report from the cached answers in minutes, so text and numbers cannot disagree.

2 Results

All numbers are from the final run of 2026-09-27 (31 datasets × 10 models, 3,083 answered items per model). Table 1 gives the audit in one place; the subsections that follow give one figure or table per row.

2.1 The objective is flat

Figure 1 shows J for all 1,771 weightings. The whole lattice spans 0.39–0.42. The best weighting (decay 0.70, recall 0.30) reaches 0.421 and v1 reaches 0.404; 1,482 weightings are nominally better than v1, but the entire spread is smaller than the bootstrap noise of a single weighting. No single weight has a slope worth acting on (Figure 2).

2.2 The best weighting does not survive honest selection

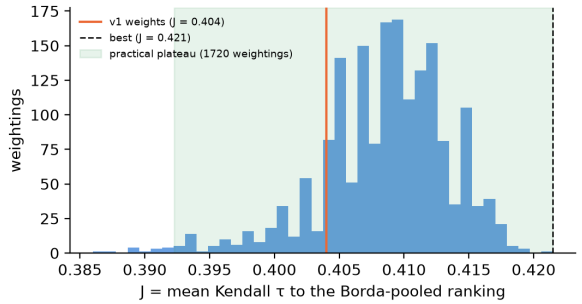
Figure 3 reports the nested item bootstrap. The weighting chosen on in-bag items scores 0.380 in bag but 0.331 out of bag (optimism gap 0.049), below v1's out-of-bag 0.339. The paired out-of-bag difference to v1 is -0.008 with a 95 % interval of $[-0.039, +0.013]$ ($p = 0.72$); the selected weighting is ahead of v1 in 28 % of resamples, and 608 distinct weightings were selected across the 1,000 resamples. The plateau of weightings not significantly below the best contains 1,765 of 1,771 points and v1 is inside it ($p = 0.66$).

Table 1: Audit summary. J = agreement objective (mean Kendall τ_b of each dataset's model ranking to the Borda pool); ρ = Spearman correlation of the composite with known utility on 31,380 synthetic anchors.

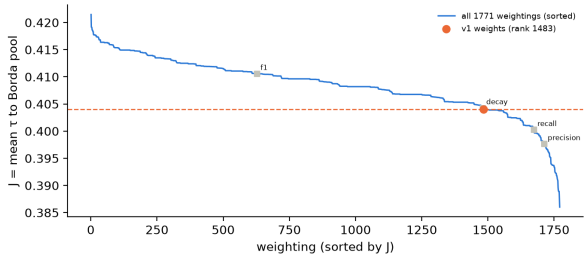
Exhaustive grid	1,771 weightings (step 0.05); best decay .70 / recall .30, $J = 0.421$; v1 $J = 0.404$ (rank 1,483 of 1,771)
J over the lattice	median 0.409, 90th percentile 0.415, 99th percentile 0.418: a flat landscape
Plateau (paired bootstrap $p > .05$ vs best)	1,765 of 1,771 (v1 inside, $p = 0.66$); practical plateau within 0.02: 1,720
Nested selection ($B = 1{,}000$)	in-bag J of the selected weighting 0.380, out-of-bag 0.331 (optimism gap 0.049); v1 out-of-bag 0.339
Selected vs v1, out of bag	-0.008, 95 % CI $[-0.039, +0.013]$, $p = 0.72$; 608 distinct weightings selected across resamples
Sampled designs (100 points, 20 seeds)	Dirichlet / LHS / uniform all reach 0.418 ± 0.001, regret ≤ 0.005: sampling would have sufficed
Objective under other pooling rules	Kemeny, Copeland and Bradley–Terry each pick a different "best"; v1's rank swings from 393 to 1,506: rank is noise
Transfer (leave-one-dataset-out)	held-out τ: LODO weights +0.34, full best +0.39, v1 +0.37
Leave-one-family-out	v1 within 0.02 of the held-out-fitted weights on EM, SM, TQA; FIN favours precision-heavy weights (+0.51 vs +0.43)
Anchors (31,380 synthetic answers)	v1 $\rho = 0.878$, best-ρ weighting 0.883, best-J weighting 0.845, exact match 0.828; AUC correct-vs-wrong 0.994 for every rule
J vs ρ across the lattice	-0.34 on this cache; +0.38 on the previous cache: unstable because J is flat
Sensitivity	one 0.05 step moves J by at most 0.007 around v1 and 0.005 around the best
Leaderboard	Friedman $p < 0.001$ under every rule; Kendall's W among datasets 0.26–0.28
Significant pairs (Holm)	Opus, Sonnet, GPT-5.5, Gemini Flash, DeepSeek-chat, DeepSeek-R1, GPT-5.4-mini, Grok 4 all > Llama 3.3 70B; Opus, Gemini Flash, GPT-5.5, Grok 4 > gpt-oss-120b
First place	Bradley–Terry: Gemini 3.8 Flash under v1 and the calibrated rules ($P(\text{first}) = 0.99$), Opus under exact match; Borda: GPT-5.5 under v1, Opus under exact match and the top-5 rules
Official-split comparability	accuracy moves by ≤ 0.09 after post-stratification; positive-class F1 on the balanced pools overstates official-ratio F1 by up to 0.19 (Amazon–Google)
Datasets at ceiling	Valentine, WDC LSPC, WDC Products (means 0.98): flagged; still inside the objective
Cost	USD 161 total (Opus 99)

Table 2: Sampled designs at a budget of 100 weightings (20 seeds each) against the exhaustive optimum.

Design	best J (mean $\pm$ sd)	regret mean / max	rank of found best (median / worst)	min pairwise L_1	centred L_2 discrepancy	share with a zero weight
Dirichlet-snapped	0.418 ± 0.001	0.004 / 0.005	14 / 49	0.10	0.538	0.27
Latin hypercube	0.418 ± 0.001	0.003 / 0.005	10 / 49	0.10	0.531	0.28
Uniform lattice	0.418 ± 0.001	0.003 / 0.005	6 / 42	0.10	0.526	0.45



(a) Distribution of J over the lattice with v1, the best and the plateau band.



(b) Sorted J with v1 and the four single-ingredient corners.

Figure 1: The agreement landscape over all 1,771 step-0.05 weightings.

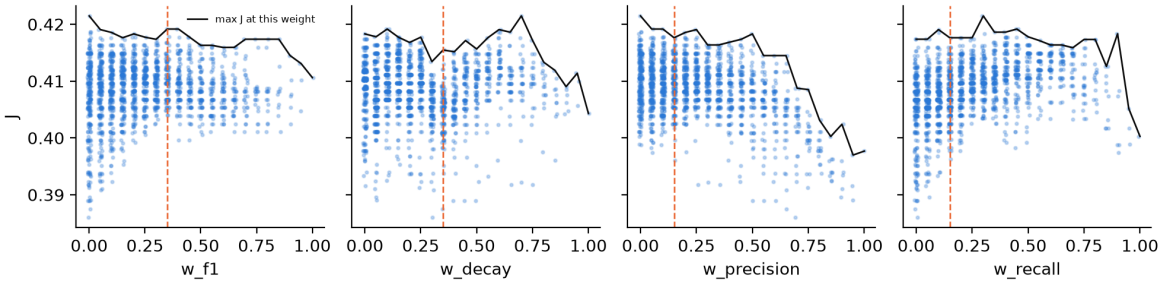
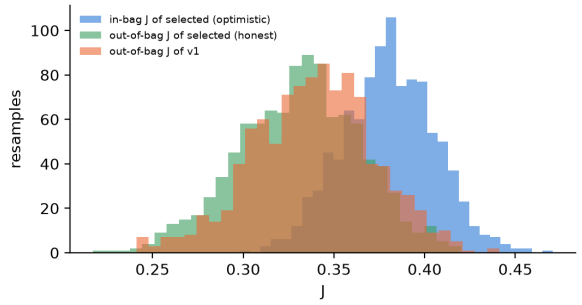
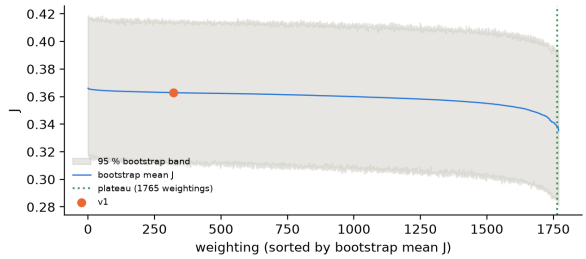


Figure 2: J against each of the four weights. Corners: F1 only 0.411, decay only 0.404, precision only 0.398, recall only 0.400; uniform 0.410.



(a) In-bag vs out-of-bag J of the selected weighting against v1.



(b) Bootstrap band over sorted weightings; the plateau boundary.

Figure 3: Honest selection: the optimism gap and the plateau.

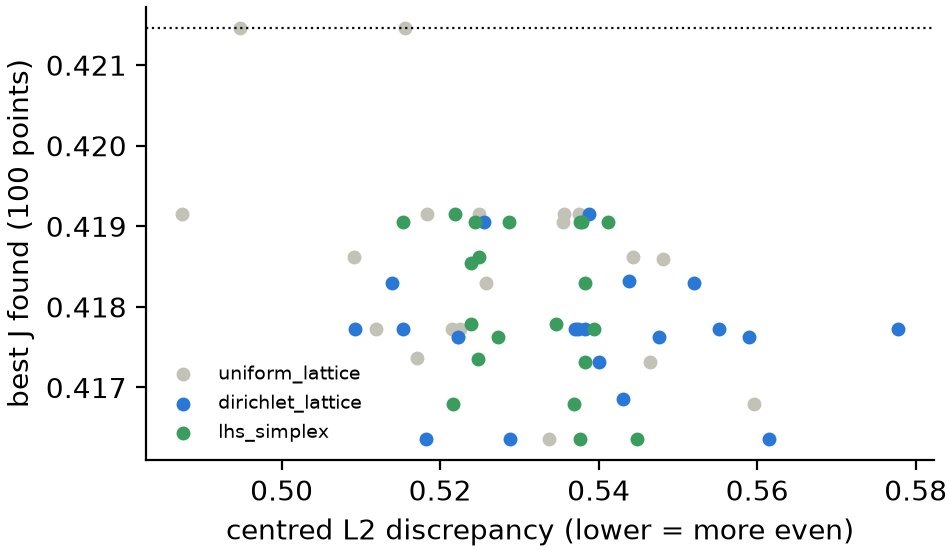


Figure 4: Any 100-point design finds the plateau; regret against the exhaustive optimum is at most 0.005.

2.3 The optimum does not transfer, and neither does the rank

Leave-one-dataset-out: weights re-fitted without the held-out dataset predict its ranking with mean $\tau = +0.34$, the full-data best with +0.39, v1 with +0.37 (Table 3 gives the family-level version). Per-family optima pull in different directions (Figure 5): decay-dominated for EM and SM, recall-dominated for TQA, precision-heavy for FIN, yet none is far from the plateau. Changing only the pooling rule inside the objective moves v1's rank from 393 (Bradley–Terry) to 1,506 (Kemeny) and changes which weighting is "best", so rank inside a flat objective carries no information.

Table 3: Leave-one-family-out. τ = agreement of the held-out family's pooled ranking with the pooled ranking of the remaining families.

Held-out family	LOFO weights (F1 / decay / prec / rec)	τ: LOFO weights	full best	v1
EM (11 sets)	0.20 / 0.00 / 0.50 / 0.30	+0.32	+0.29	+0.30
SM (6 sets)	0.00 / 0.10 / 0.00 / 0.90	+0.35	+0.37	+0.25
TQA (11 sets)	0.45 / 0.00 / 0.40 / 0.15	+0.38	+0.40	+0.38
FIN (3 sets)	0.00 / 0.95 / 0.05 / 0.00	+0.43	+0.51	+0.51

2.4 Agreement and correctness do not pull the same way

On 31,380 synthetic answers of known utility, v1 scores $\rho = 0.878$; the best weighting by ρ reaches 0.883 and the best by J only 0.845; exact match 0.828. Every rule separates correct from wrong answers with AUC 0.994. Across the lattice J and ρ correlate -0.34 on this cache and +0.38 on the previous one, which is what a flat J predicts. The Pareto knee (F1 0.05, decay 0.60, precision 0.25, recall 0.10) is within 0.002 of the best J and within 0.006 of the best ρ (Table 4, Figure 6).

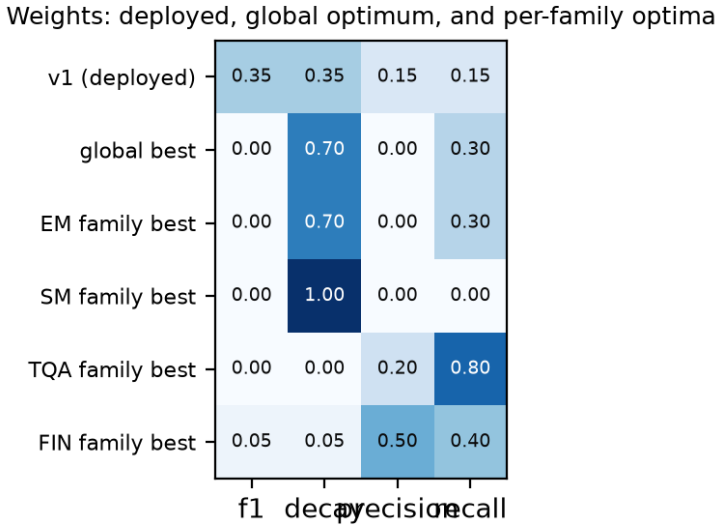


Figure 5: Deployed weights (v1), the global optimum and the per-family optima.

Table 4: Correctness on known-quality anchors (9,881 correct, 7,430 wrong, remainder partial). ρ_d = mean per-dataset ρ; wrong = mean score given to wrong answers.

Rule	F1 / decay / prec / rec	J	ρ	ρ_d	AUC	wrong	hedge gap
v1	0.35 / 0.35 / 0.15 / 0.15	0.404	0.878	0.868	0.994	0.034	−0.51
best J	0.00 / 0.70 / 0.00 / 0.30	0.421	0.845	0.840	0.994	0.037	−0.49
best ρ	0.00 / 0.35 / 0.60 / 0.05	0.398	0.883	0.871	0.994	0.034	−0.57
Pareto knee	0.05 / 0.60 / 0.25 / 0.10	0.419	0.877	0.866	0.994	0.036	−0.55
exact match	–	–	0.828	0.830	0.989	0.011	−0.81

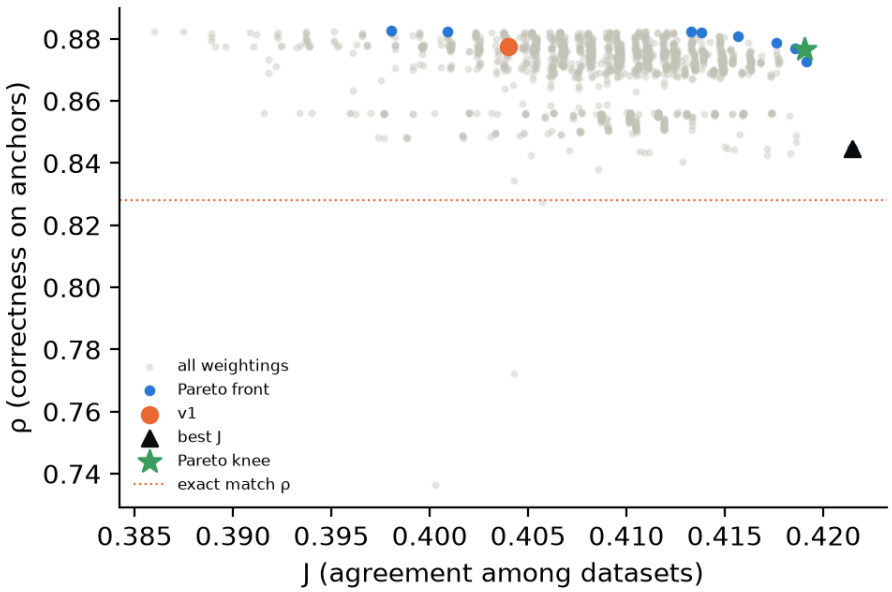


Figure 6: Agreement J against anchor correctness ρ for every weighting; v1, the best-J, best-ρ and knee points marked.

2.5 Leaderboard

Under every rule the Friedman test rejects equal models ($p < 0.001$) while datasets agree only moderately (Kendall's W 0.26–0.28). The ends of the ranking are stable: Llama 3.3 70B is last and gpt-oss-120b ninth under every rule and every pooling method, with Holm-corrected significance against the top group. The order inside the top three depends on the rule and the pooling method (Table 5, Table 6, Figure 7): Bradley–Terry puts Gemini 3.8 Flash first under v1 and all calibrated rules, Borda puts GPT-5.5 first under v1 and Opus first under the top-5 rules, and exact match puts Opus first under both.

Table 5: Leaderboard under the deployed rule (v1 weights). Bradley–Terry strength ± SE relative to Opus; Borda rank with 95 % bootstrap interval; mean composite score over the suite.

Rank	Model	BT strength	Borda rank [95 %]	mean score
1	Gemini 3.8 Flash	$+0.12 \pm 0.02$	3 [1, 3]	0.812
2	Claude Opus 5	$+0.10 \pm 0.00$	2 [1, 3]	0.811
3	GPT-5.5	$+0.05 \pm 0.02$	1 [1, 3]	0.808
4	Claude Sonnet 5	$+0.03 \pm 0.02$	4 [4, 6]	0.795
5	Grok 4	$+0.03 \pm 0.02$	5 [4, 6]	0.797
6	GPT-5.4-mini	-0.01 ± 0.02	8 [5, 8]	0.789
7	DeepSeek-chat	-0.02 ± 0.02	7 [5, 8]	0.783
8	DeepSeek-R1	-0.03 ± 0.02	6 [5, 8]	0.788
9	gpt-oss-120b	-0.10 ± 0.02	9 [9, 10]	0.767
10	Llama 3.3 70B	-0.17 ± 0.02	10 [9, 10]	0.739

Table 6: Bradley–Terry rank (Borda rank in brackets) under each reported rule. top1–top5 = the five best weightings by J; best-ρ = best by anchor correctness; knee = (J, ρ) Pareto knee.

Rule	Opus 5	Sonnet 5	DS-chat	DS-R1	Flash 3.8	5.4-mini	GPT-5.5	gpt-oss	Grok 4	Llama
exact match	1 (1)	5 (4)	8 (7)	7 (6)	3 (3)	6 (8)	2 (2)	9 (9)	4 (5)	10 (10)
v1	2 (2)	4 (4)	7 (7)	8 (6)	1 (3)	6 (8)	3 (1)	9 (9)	5 (5)	10 (10)
top1	2 (1)	4 (4)	7 (8)	6 (6)	1 (3)	8 (7)	3 (2)	9 (9)	5 (5)	10 (10)
top2	2 (1)	4 (4)	6 (7)	8 (6)	1 (3)	7 (8)	3 (2)	9 (9)	5 (5)	10 (10)
top3	2 (1)	4 (4)	6 (7)	8 (6)	1 (3)	7 (8)	3 (2)	9 (9)	5 (5)	10 (10)
top4	2 (1)	4 (4)	7 (8)	8 (6)	1 (3)	6 (7)	3 (2)	9 (9)	5 (5)	10 (10)
top5	2 (1)	4 (4)	7 (8)	6 (6)	1 (3)	8 (7)	3 (2)	9 (9)	5 (5)	10 (10)
best-ρ	2 (2)	4 (4)	8 (8)	7 (6)	1 (3)	5 (7)	3 (1)	9 (9)	6 (5)	10 (10)
knee	2 (1)	4 (4)	7 (8)	8 (6)	1 (3)	6 (7)	3 (2)	9 (9)	5 (5)	10 (10)

Kendall τ between the BT rankings of the rules: 0.78–1.00 (exact match vs the rest 0.78–0.82).

2.6 Comparability with published numbers

Post-stratification to the official split changes dataset accuracy by at most 0.09 (OpenSanctions pairs 0.82 → 0.73; TAT-QA 0.80 → 0.76; MMTU 0.87 → 0.83). For the matching sets, positive-class F1 computed on the balanced 50/50 pool overstates F1 at the official match ratio wherever the official split is sparse (Table 7, Figure 9).

2.7 Dataset diagnostics

Three datasets sit at ceiling (mean composite 0.98 with a spread of 0.02–0.04 across models): Valentine, WDC LSPC and WDC Products. They are flagged and remain inside the objective in

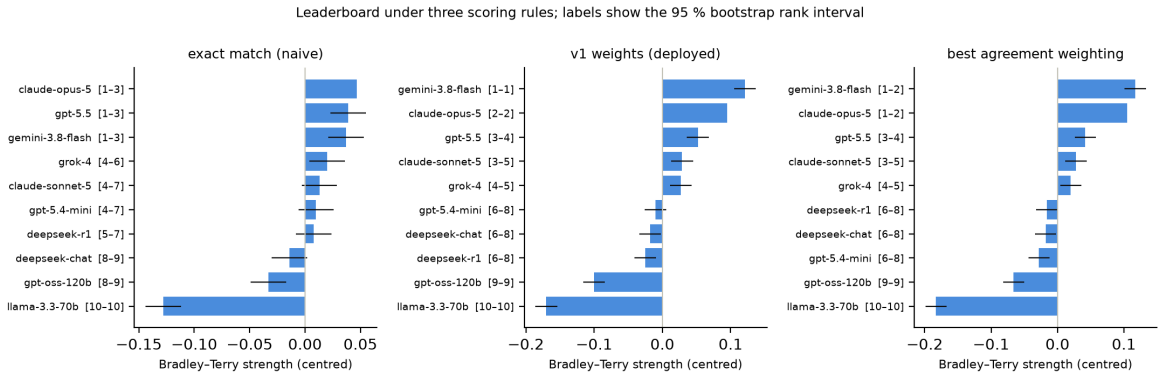


Figure 7: Bradley–Terry strengths ± SE with bootstrap rank intervals under exact match, v1 and the best-J weighting. The ends are stable; the top order is rule-dependent.

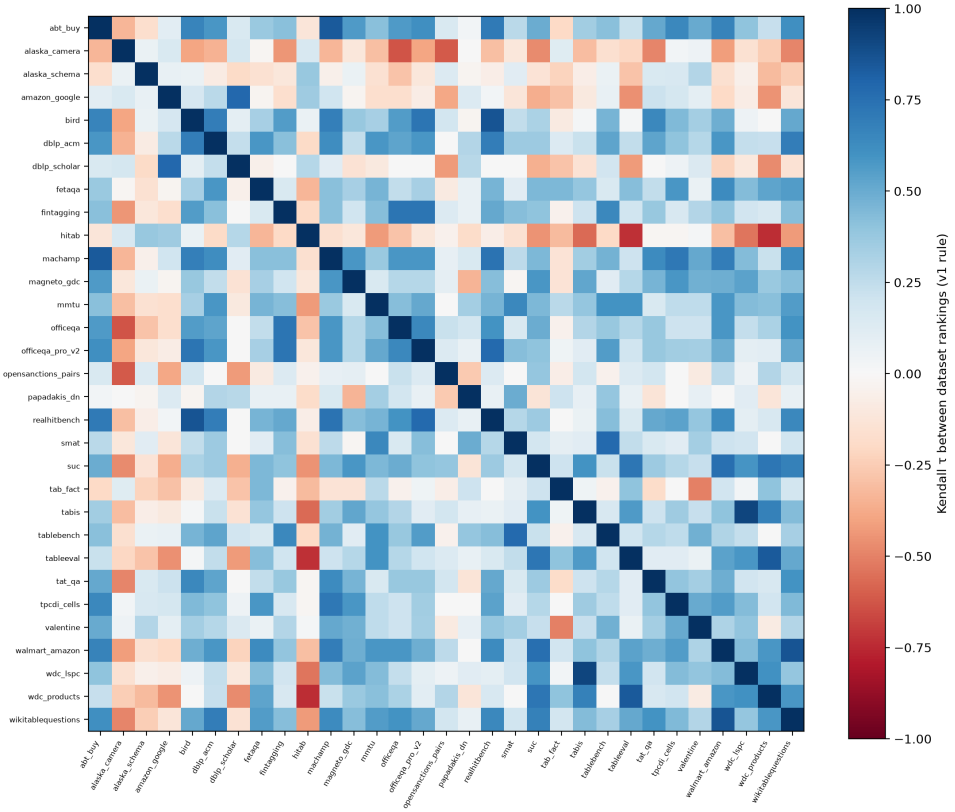


Figure 8: Kendall τ between the model rankings of each pair of datasets under v1. Datasets agree only moderately ($W \approx 0.27$).

Table 7: Matching sets under v1, averaged over the ten models: suite accuracy, post-stratified official accuracy, and positive-class F1 on the balanced pool vs at the official match ratio.

Dataset	official match share	suite acc.	official acc.	F1 (pool)	F1 (official ratio)
Abt–Buy	0.108	0.970	0.986	0.968	0.936
Amazon–Google	0.102	0.936	0.932	0.929	0.739
Walmart–Amazon	0.094	0.958	0.975	0.947	0.875
DBLP–ACM	0.180	0.986	0.991	0.985	0.976
DBLP–Scholar	0.186	0.949	0.966	0.948	0.910
WDC Products 2024	0.280	0.978	0.985	0.979	0.974
WDC LSPC 2017	0.264	0.982	0.983	0.983	0.971
Machamp	0.175	0.887	0.890	0.892	0.733
Papadakis–Christen Dn	0.018	0.941	0.988	0.833	0.683
OpenSanctions pairs	0.769	0.821	0.727	0.773	0.777
Alaska Camera	n/c	0.958	–	0.955	–

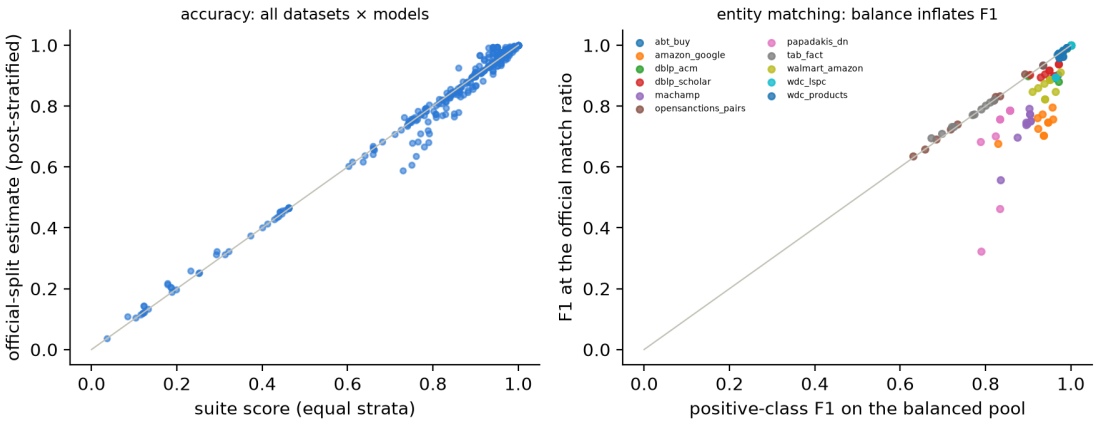


Figure 9: Suite score vs post-stratified official estimate per dataset, and matching F1 on the balanced pool vs at the official match ratio.

this run. Two sit near floor for every model (OfficeQA closed-book 0.13, OfficeQA Pro V2 0.14) and one is a known hard set (BIRD 0.19); none is excluded. Self-stability τ (rank agreement between split halves of a dataset) ranges from 0.58 (HiTab) to 0.93 (FeTaQA).

A All figures

Every figure that can be drawn from the saved outputs. Source data for each is in `data/`.

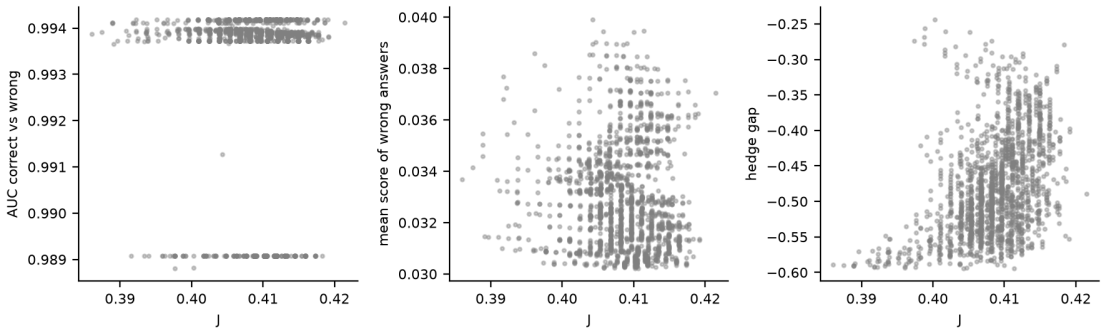


Figure 10: Three further anchor diagnostics against J for every weighting.

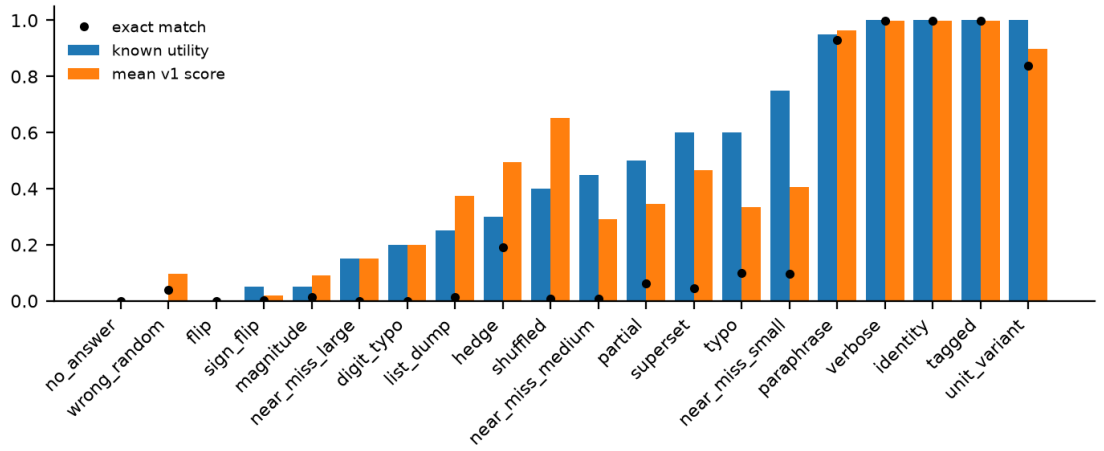


Figure 11: Synthetic anchors: known utility of each perturbation operator against the mean v1 score and exact match it receives.

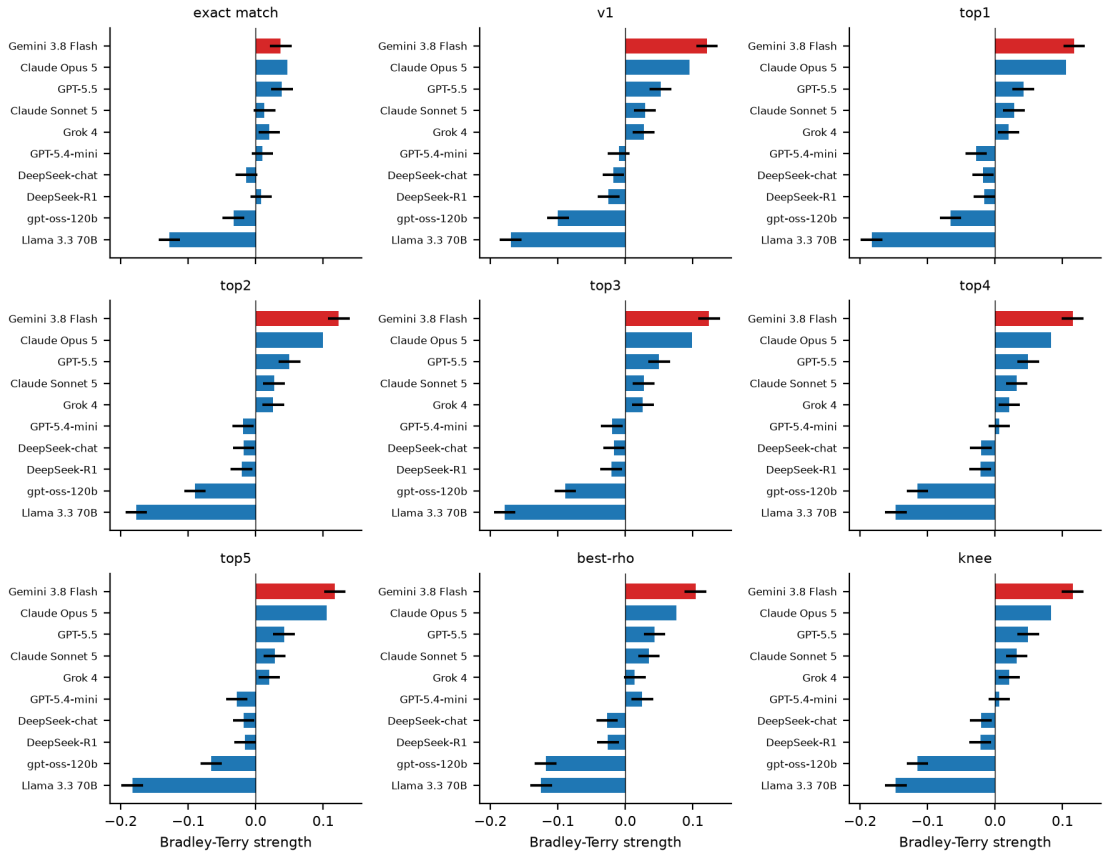


Figure 12: Bradley–Terry strengths (± SE vs reference) under every reported rule; models fixed in v1 order.

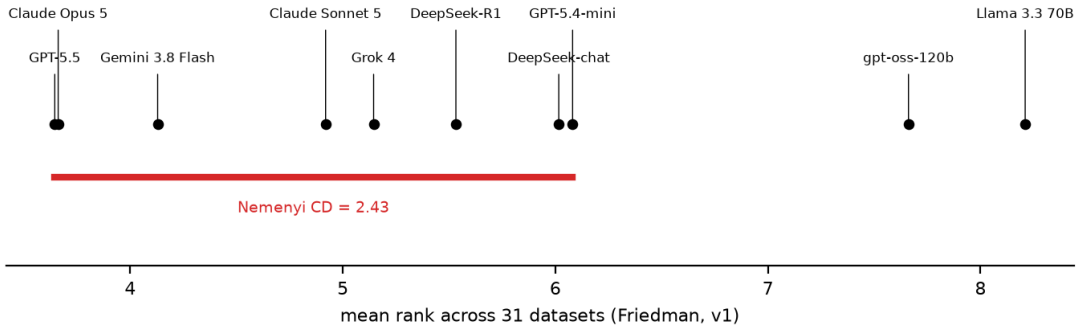


Figure 13: Critical-difference view of the Friedman test under v1: mean rank per model and the Nemenyi critical difference at $\alpha = .05$.

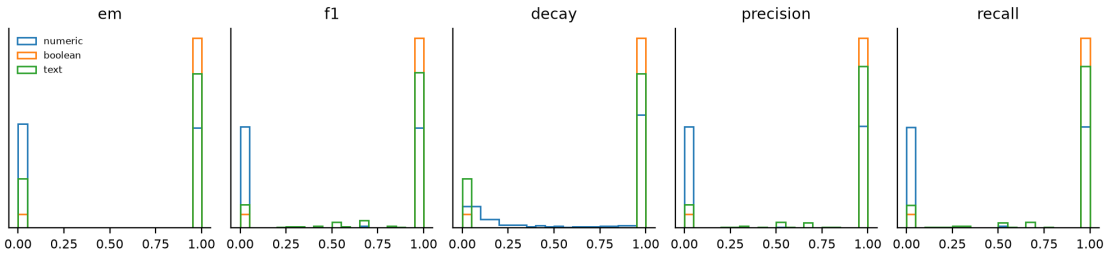


Figure 14: Distribution of each v1 ingredient over all answers, by answer type.

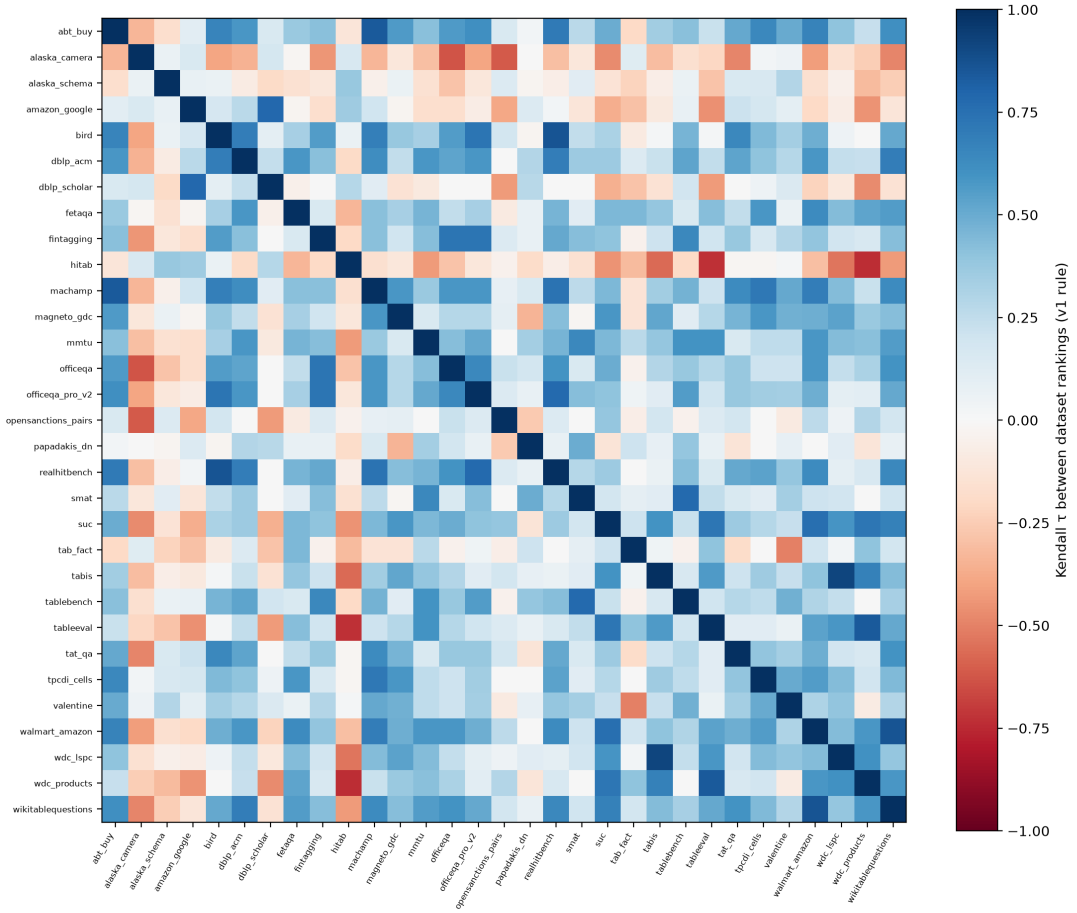


Figure 15: Kendall τ between dataset rankings under v1.

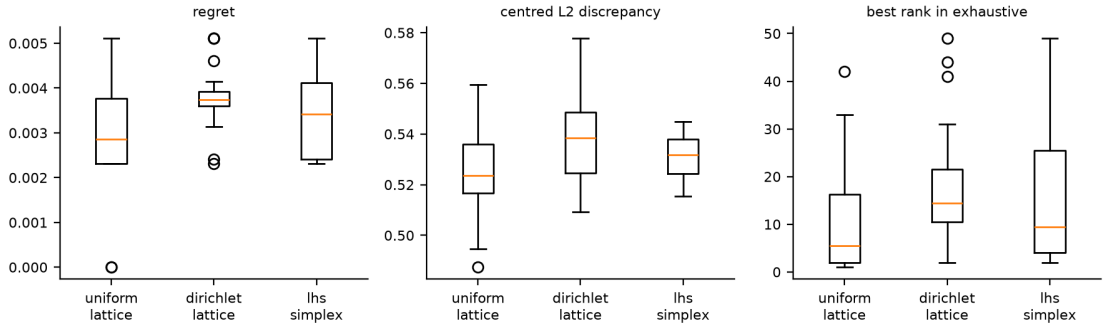


Figure 16: Sampled designs over 20 seeds: regret against the exhaustive optimum, centred L_2 discrepancy and rank of the found best.

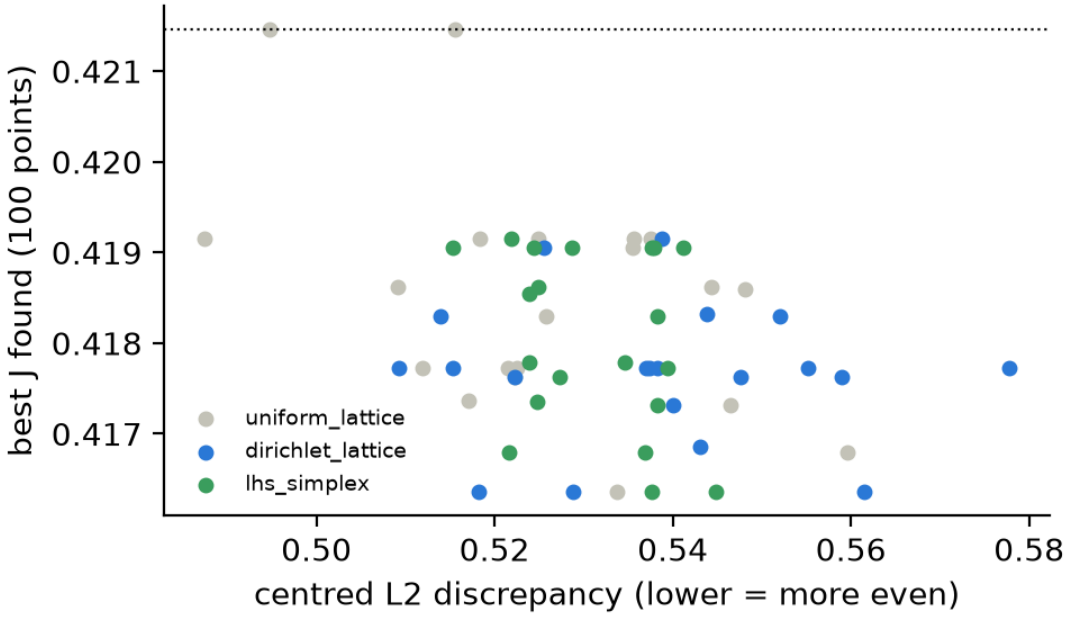


Figure 17: Sampled designs against the exhaustive optimum.

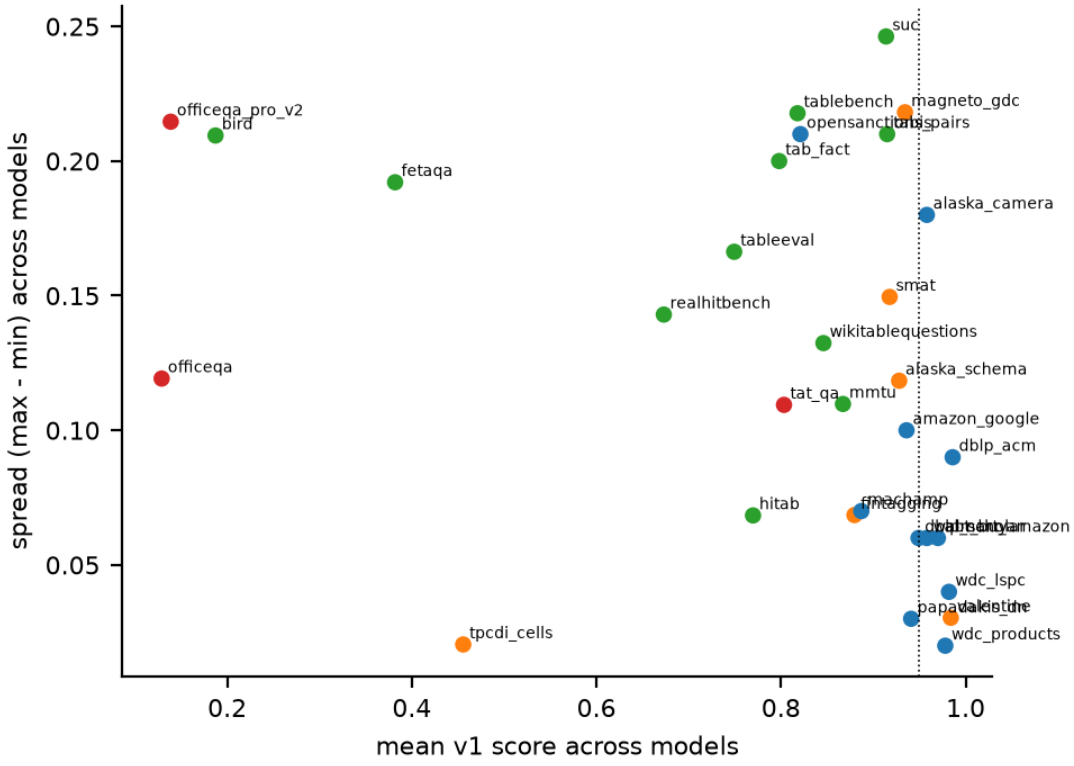


Figure 18: Dataset diagnostics: mean score against spread across models; dotted line marks the ceiling flag.

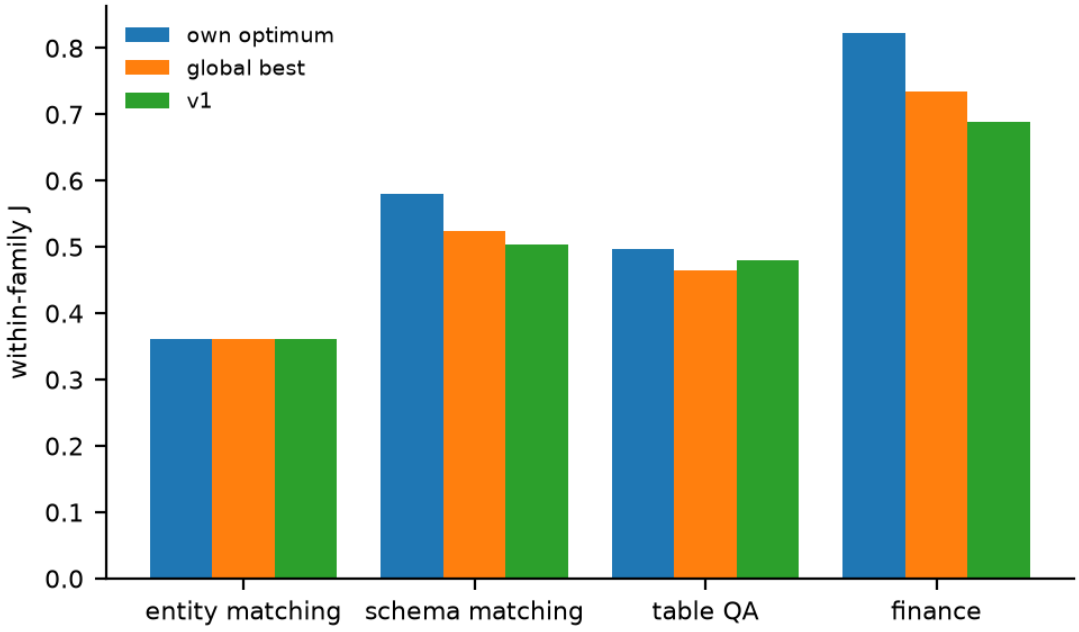


Figure 19: Within-family agreement J under each family's own optimum, the global best and v1.

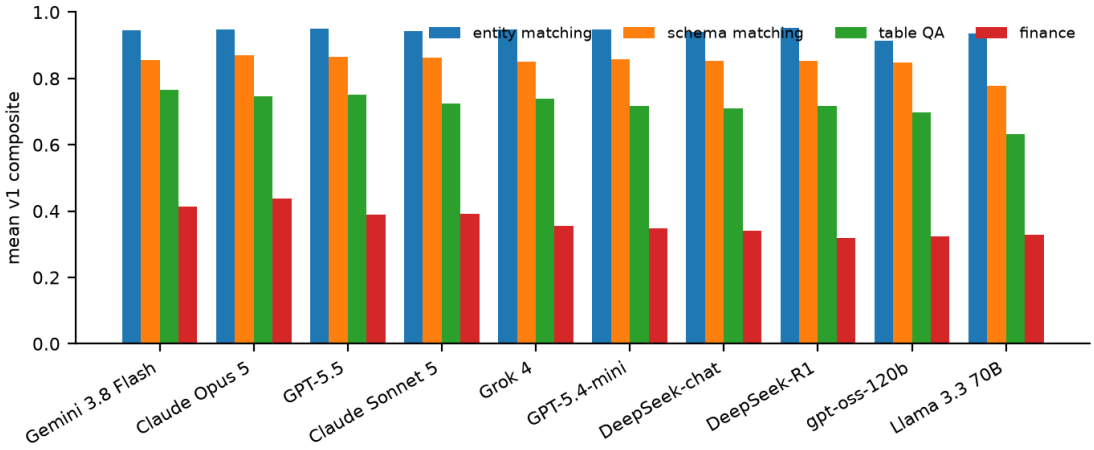


Figure 20: Mean v1 composite per model within each task family.

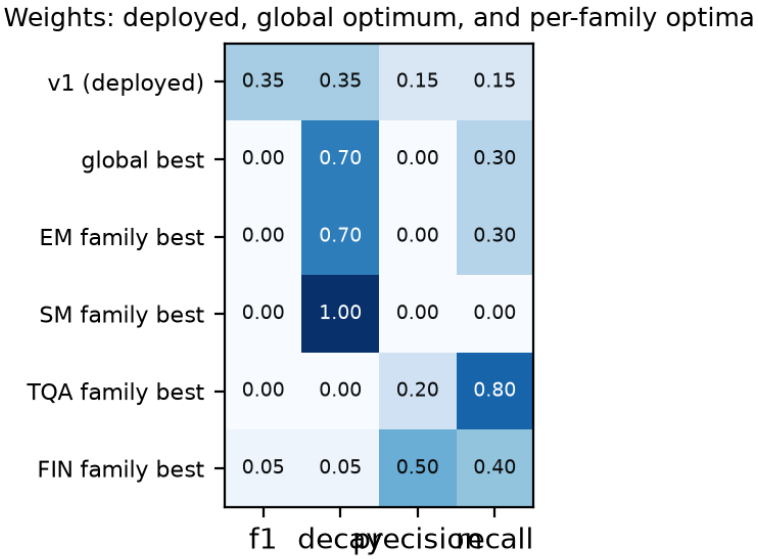


Figure 21: Deployed, global-optimum and per-family weights.

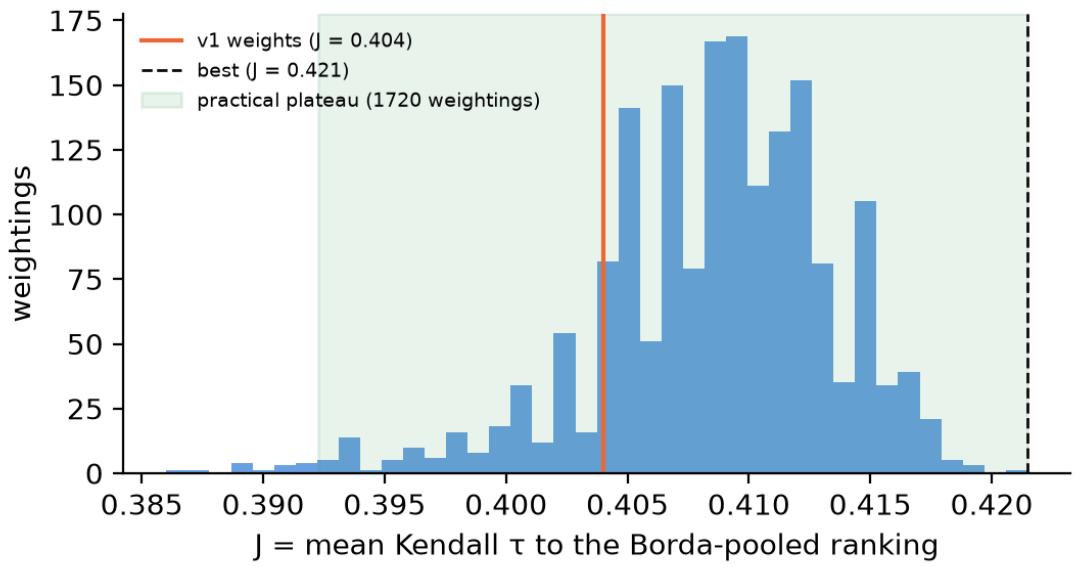


Figure 22: Distribution of J over the 1,771 weightings with v1, the best and the plateau band.

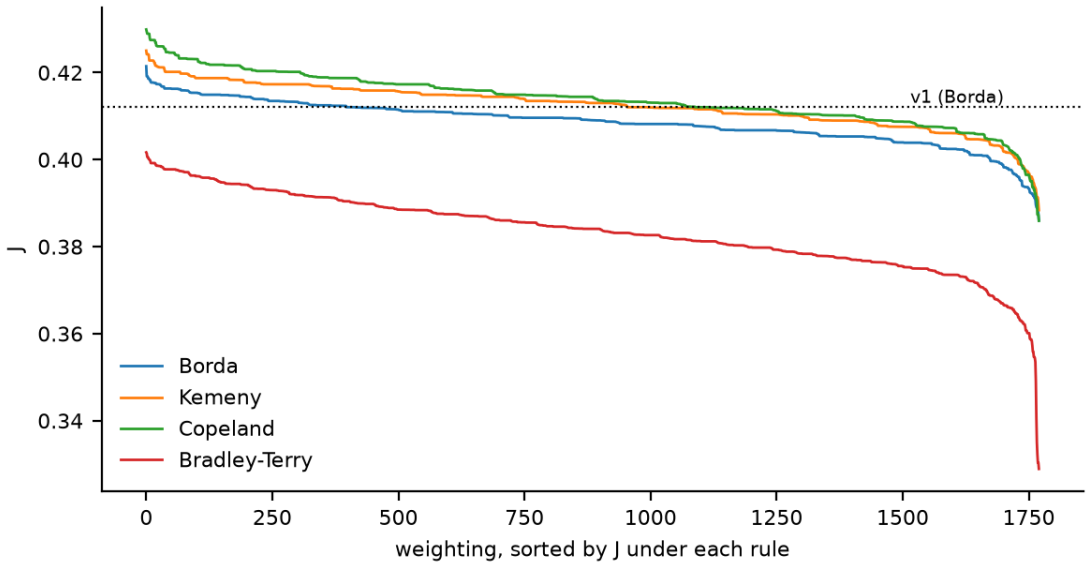


Figure 23: Agreement objective J over all 1,771 weightings under four pooling rules (each curve sorted separately). The landscape is flat under every rule.

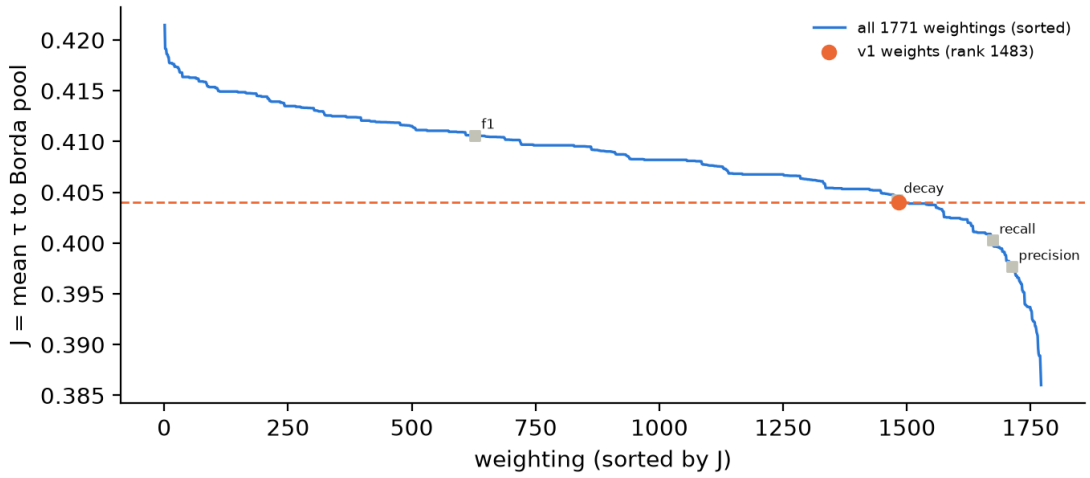


Figure 24: Sorted J with v1 and the four single-ingredient corners.

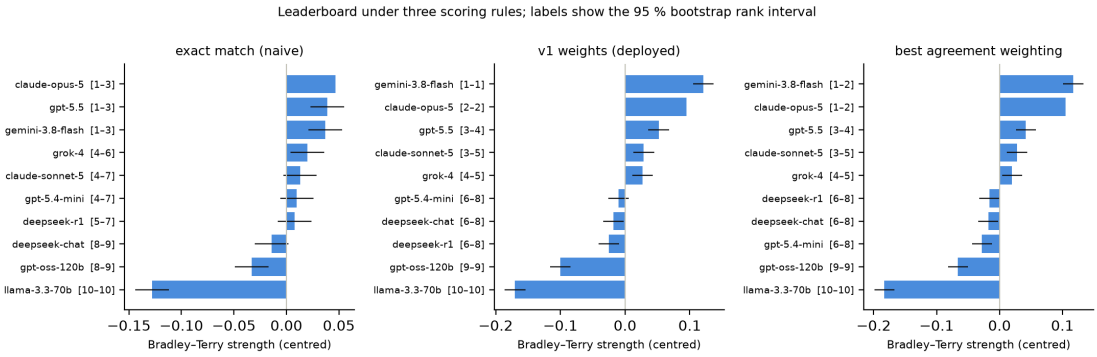


Figure 25: Bradley–Terry strengths with rank intervals under exact match, v1 and the best-J weighting.

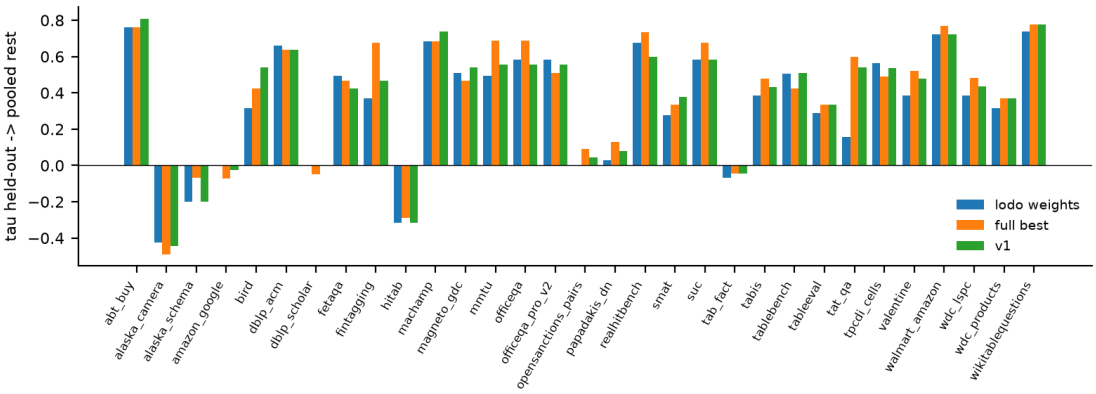


Figure 26: Leave-one-dataset-out: agreement of the held-out dataset with the pooled ranking of the rest, under weights fitted without it, the full-data best and v1.

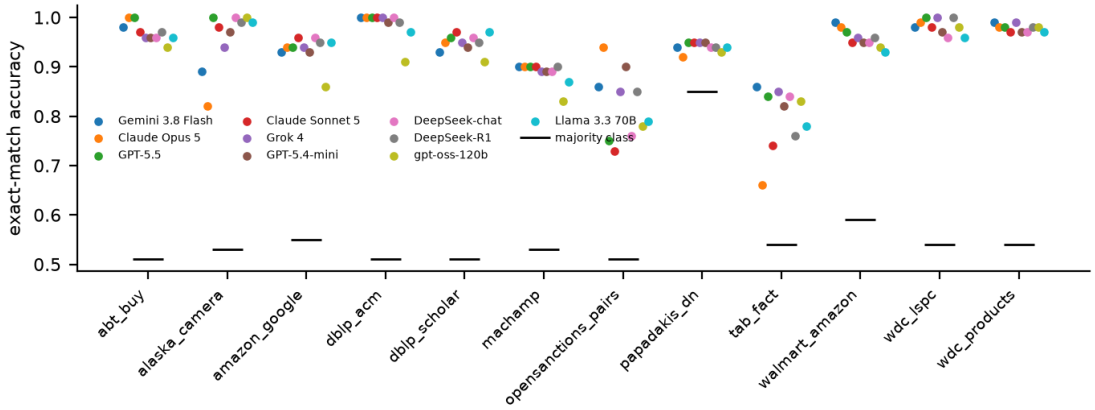


Figure 27: Exact-match accuracy of every model on the boolean datasets against the majority-class baseline of each pool.

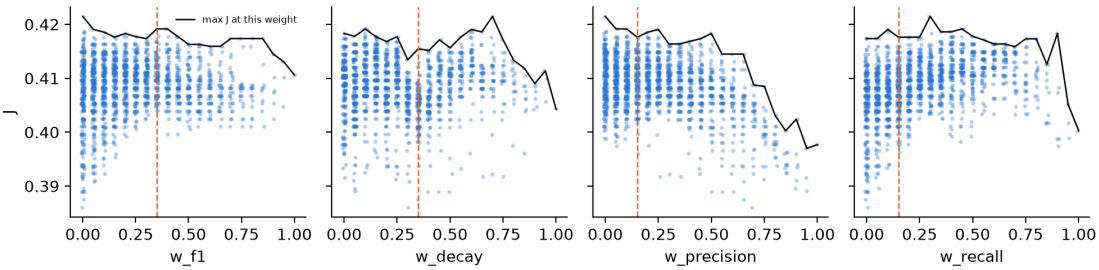


Figure 28: J against each of the four weights.

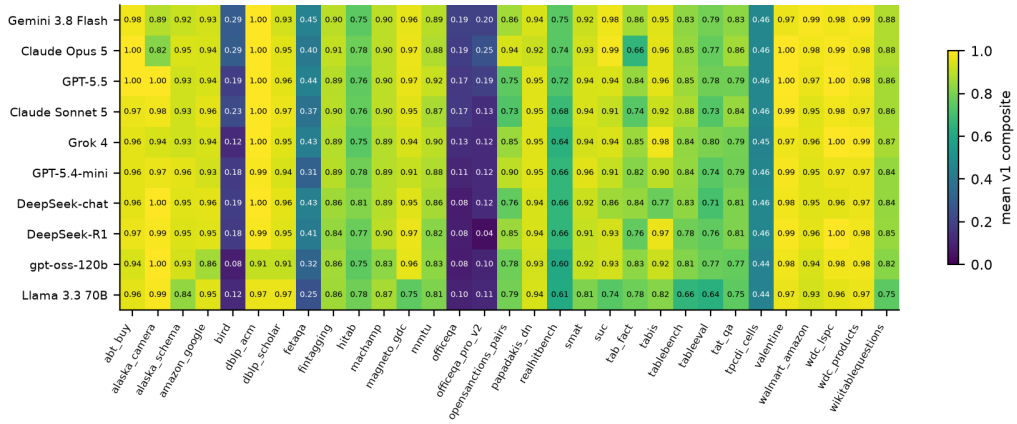


Figure 29: Mean v1 composite score per model and dataset (models in Bradley–Terry order under v1).

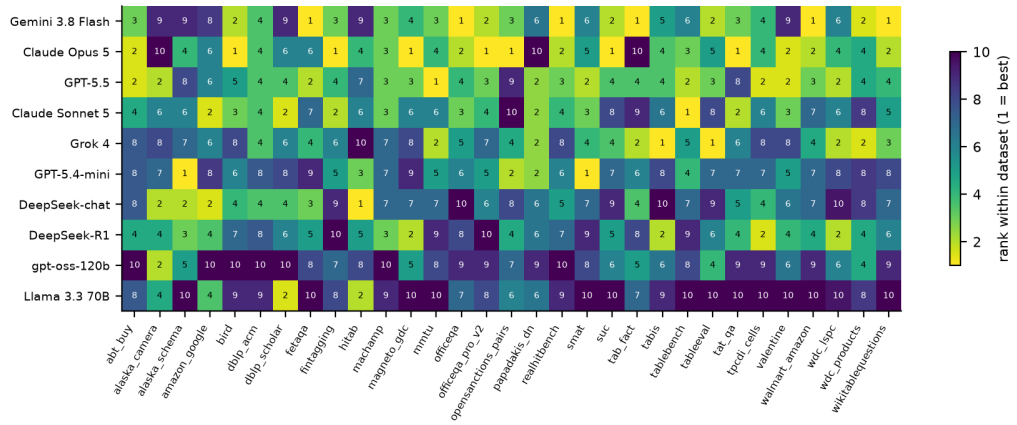


Figure 30: Rank of each model inside each dataset under v1 (ties by average rank).

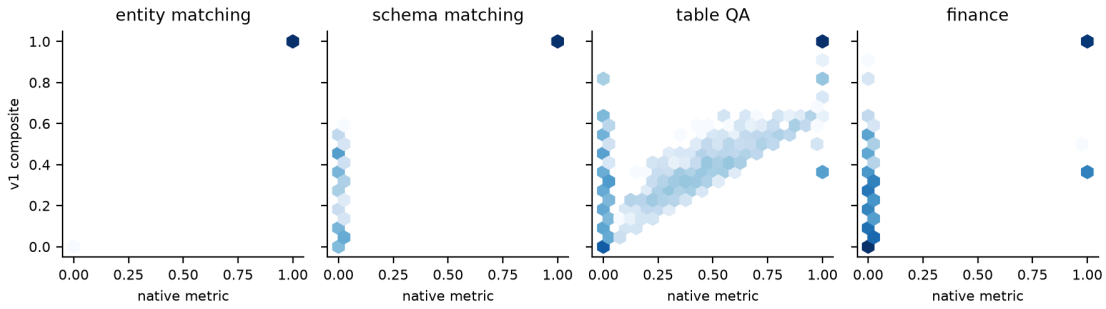


Figure 31: Per-answer native metric against the v1 composite, by family (log-count hexbins).

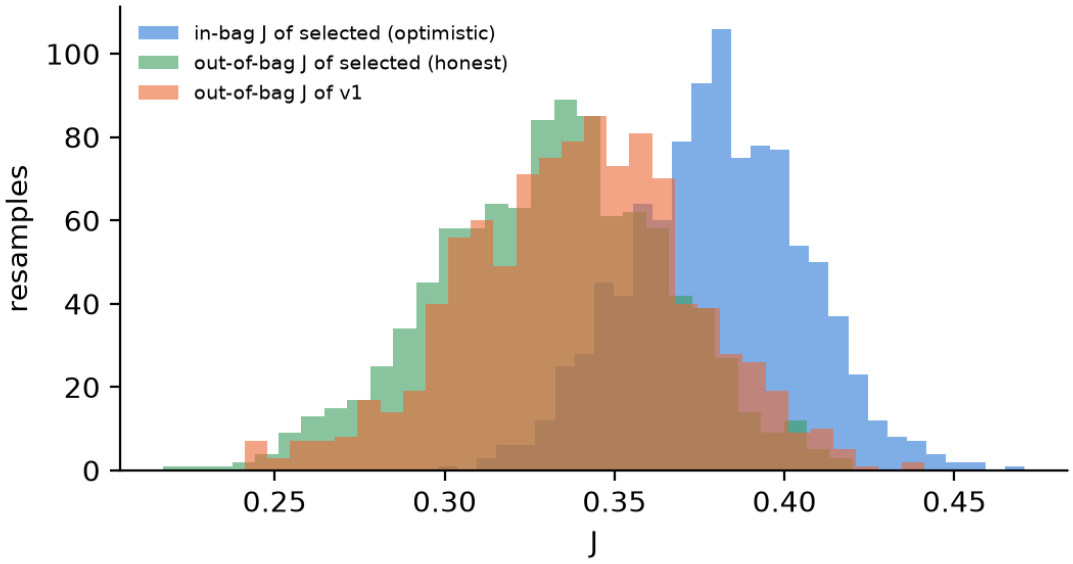


Figure 32: In-bag vs out-of-bag J of the selected weighting against v1.

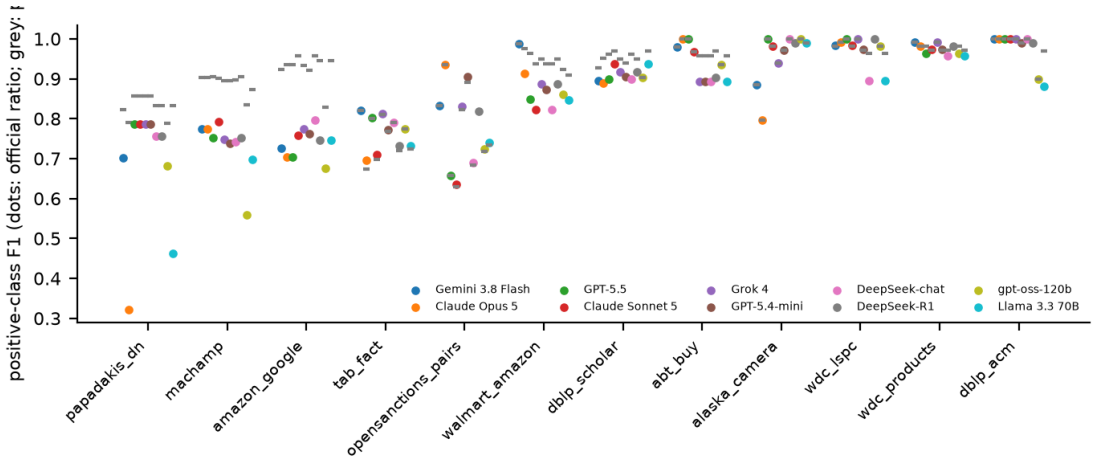


Figure 33: Positive-class F1 on the balanced pool (grey) and at the official match ratio (coloured) per model and matching dataset.

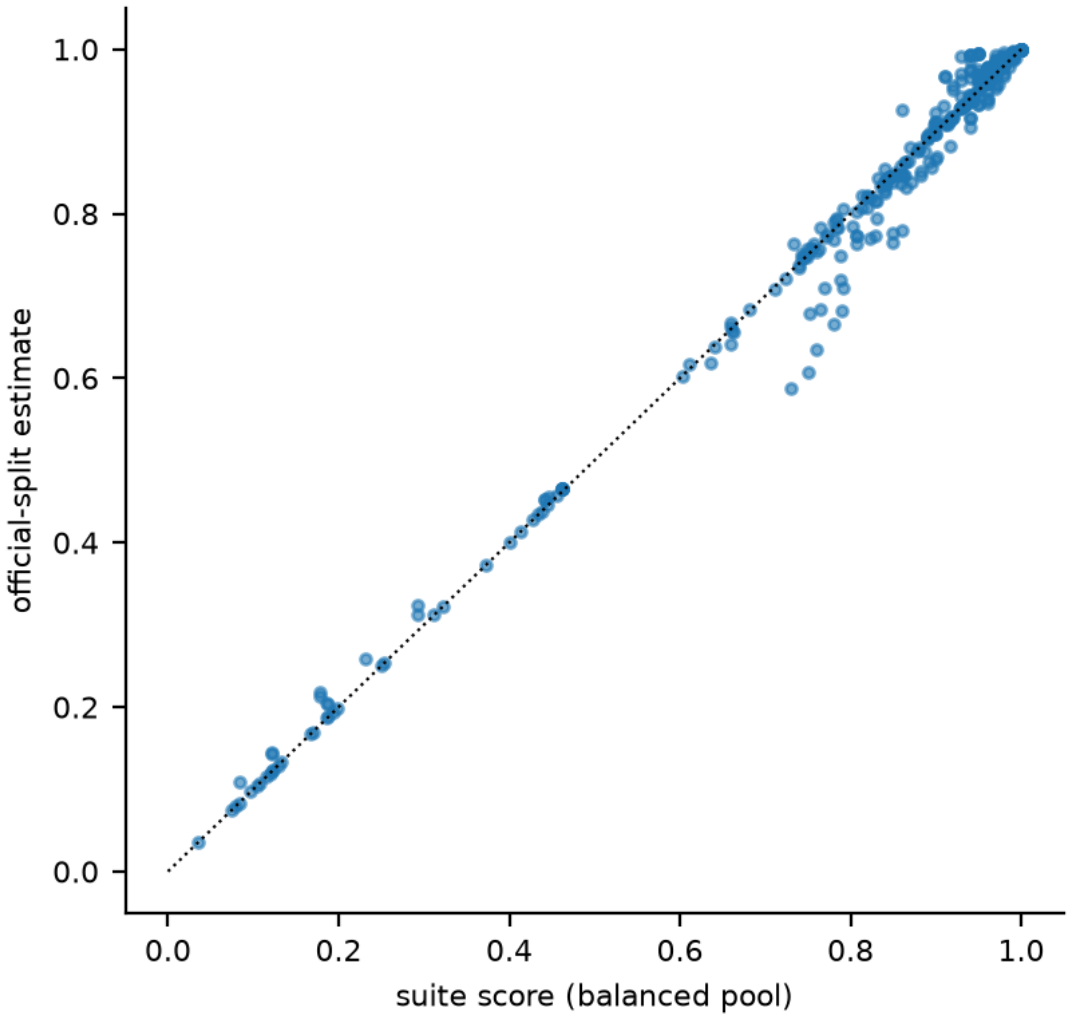


Figure 34: Suite score against post-stratified official-split estimate for every (dataset, model) pair under v1.

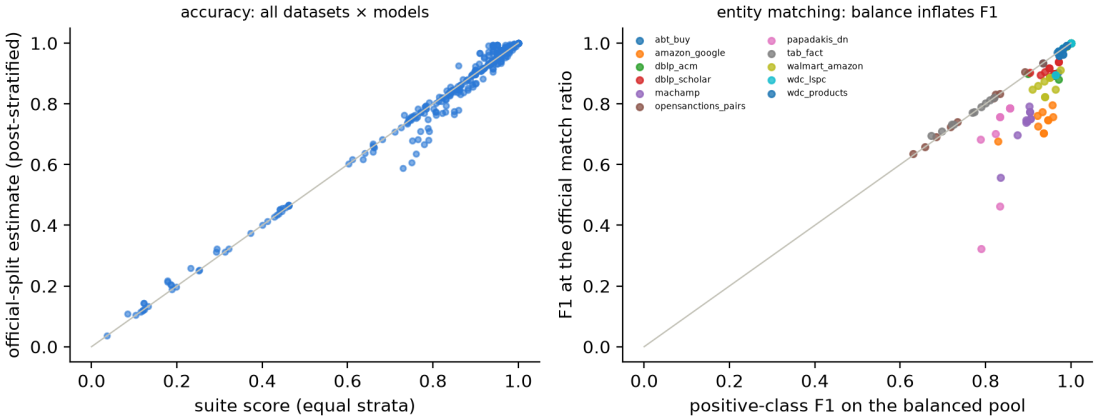


Figure 35: Suite score vs post-stratified official estimate; matching F1 at the official ratio.

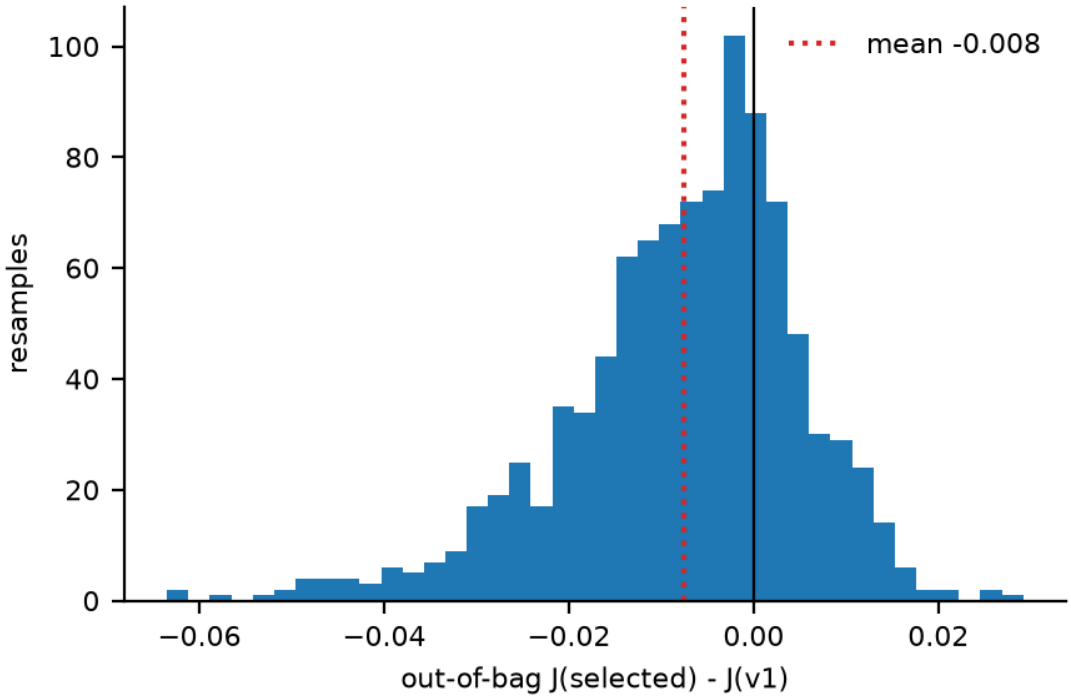


Figure 36: Paired out-of-bag difference between the selected weighting and v1 over 1,000 resamples.

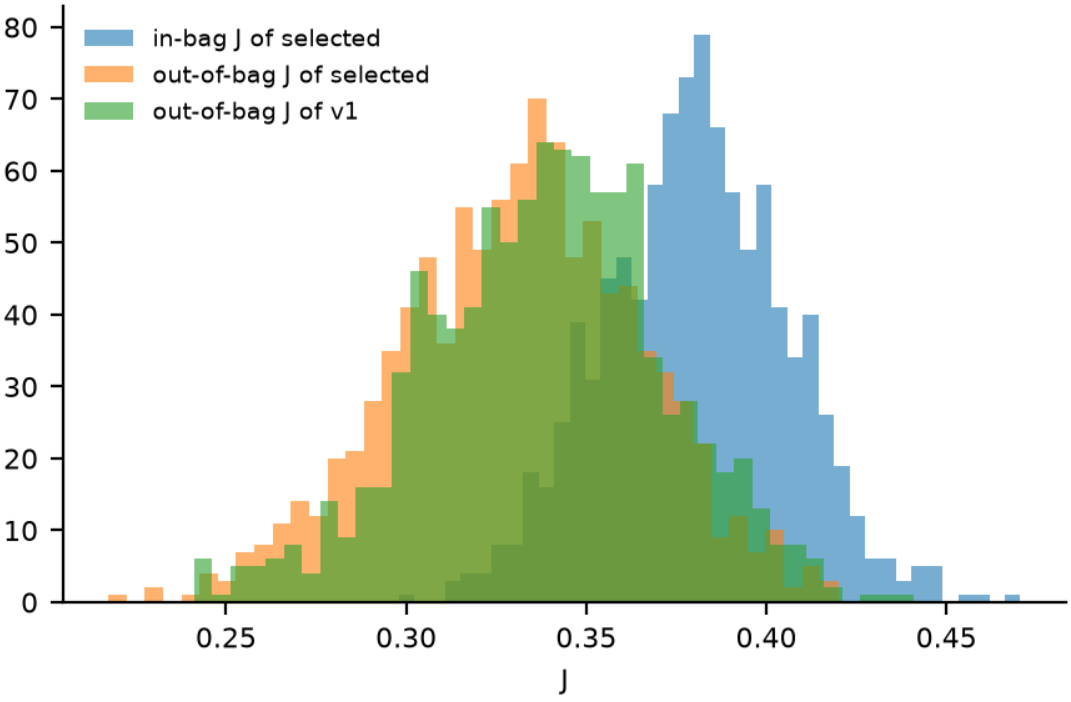


Figure 37: Distributions behind the optimism gap: in-bag J of the selected weighting, its out-of-bag J, and v1's out-of-bag J.

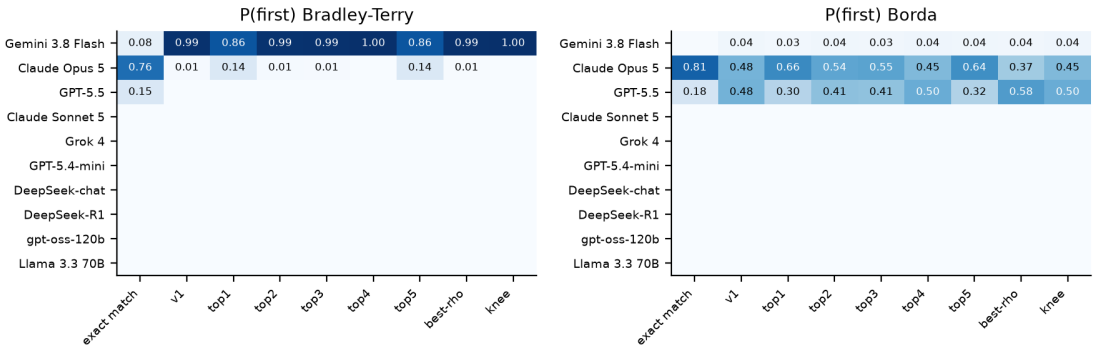


Figure 38: Bootstrap probability of first place for every model under every rule, by pooling method.

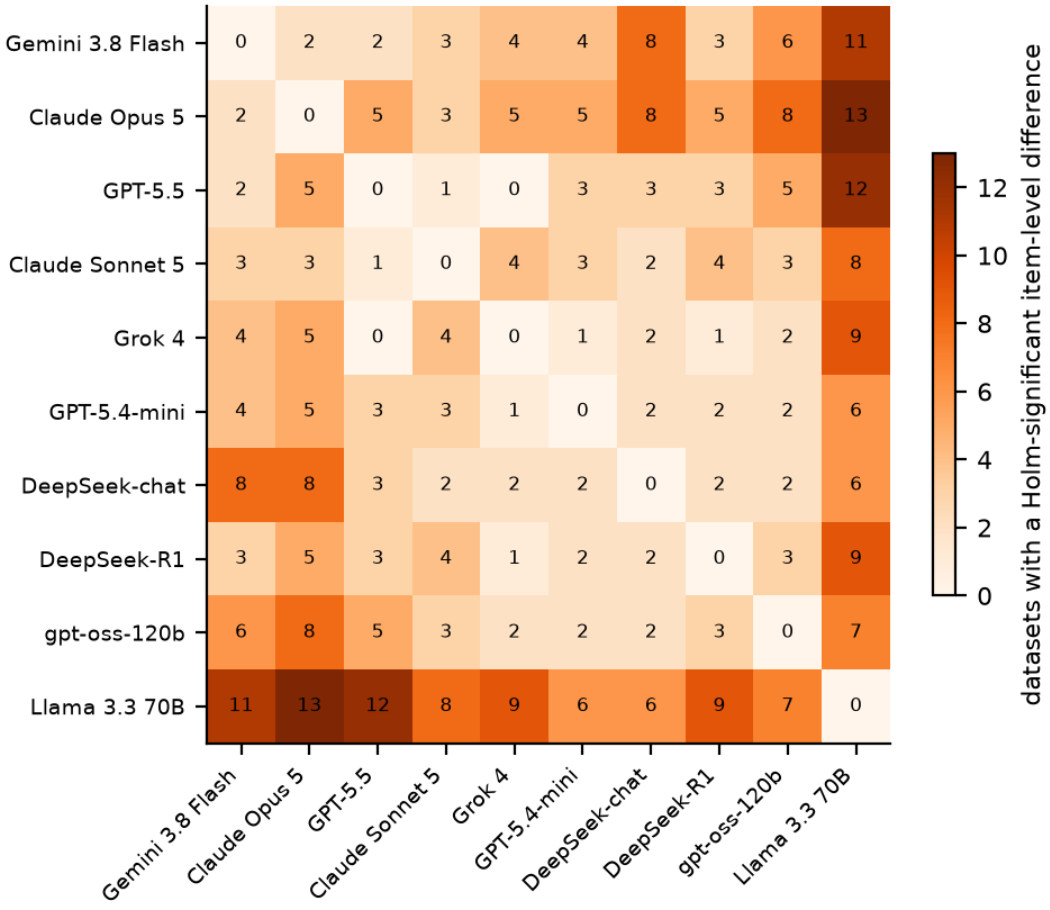


Figure 39: Number of datasets (of 31) in which the per-item paired test separates each pair of models under v1 (Holm within dataset).

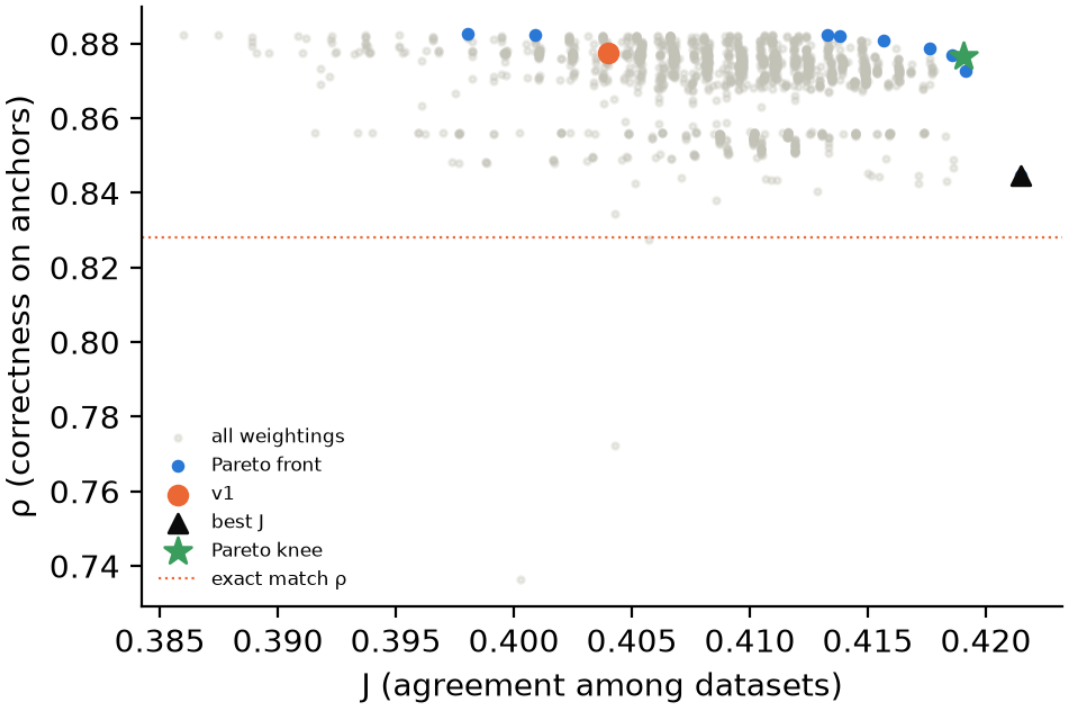


Figure 40: Agreement J against anchor correctness ρ for every weighting.

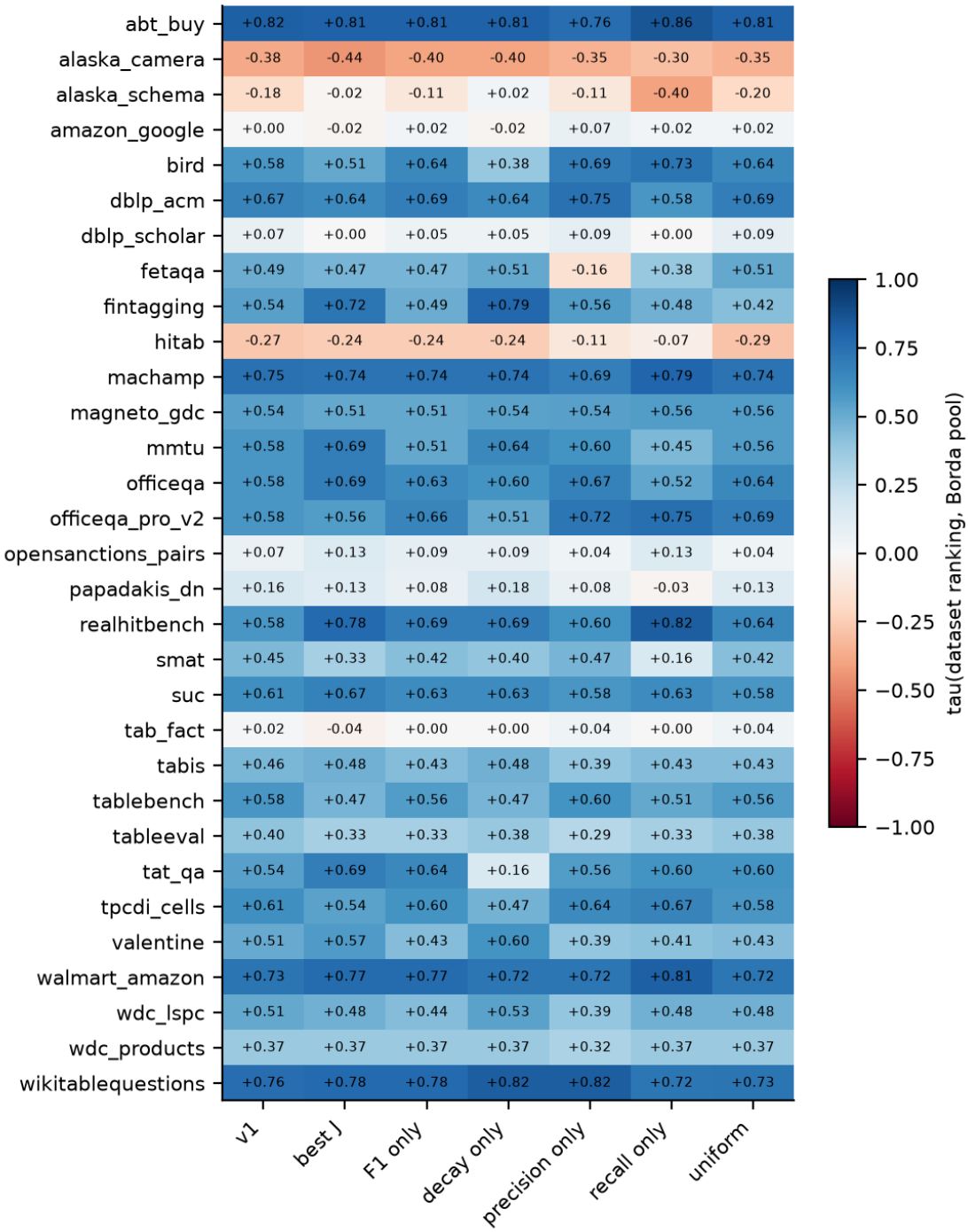


Figure 41: Per-dataset Kendall τ to the Borda pool under v1, the best weighting, the four single-ingredient corners and the uniform mix.

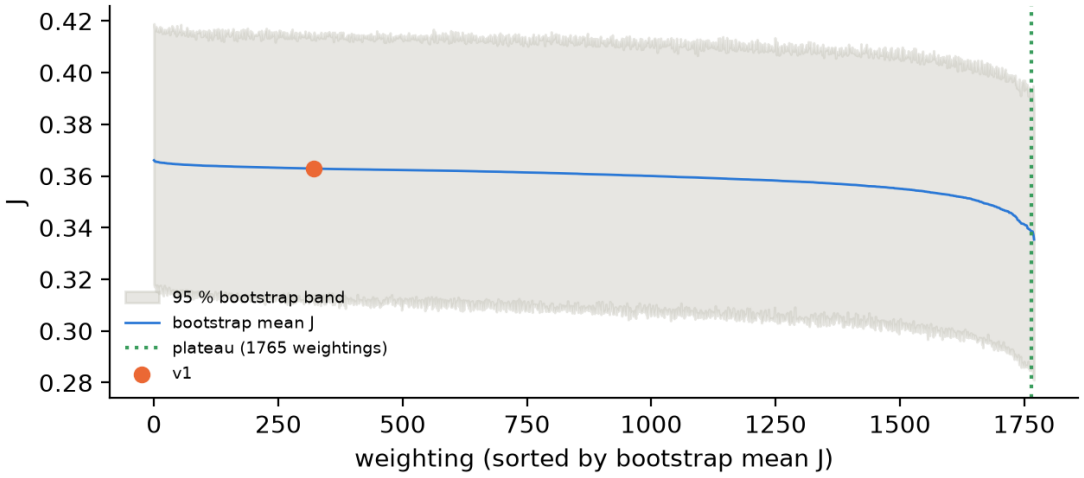


Figure 42: Bootstrap band over sorted weightings and the plateau boundary.

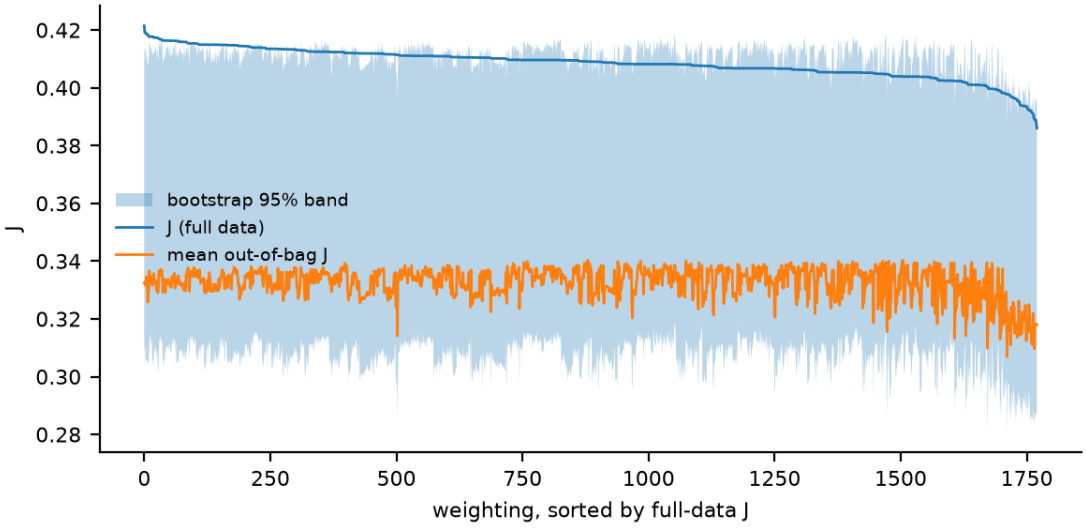


Figure 43: Full-data J, bootstrap band and mean out-of-bag J for every weighting in rank order.

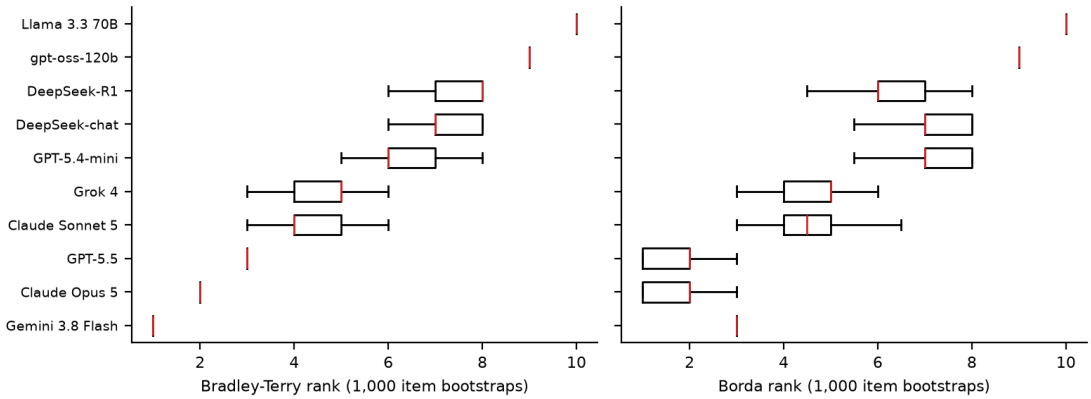


Figure 44: Bootstrap distribution of each model's rank under v1 (1,000 item resamples).

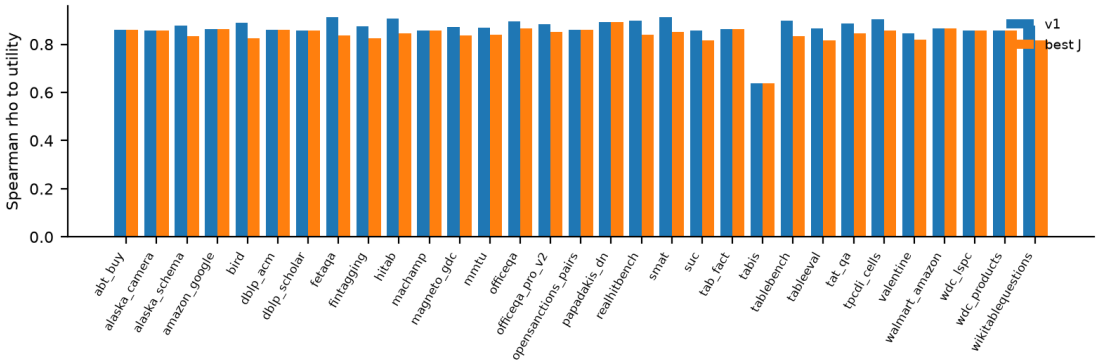


Figure 45: Anchor correctness (Spearman ρ between composite and known utility) per dataset under v1, the best-J and the best-ρ weightings.

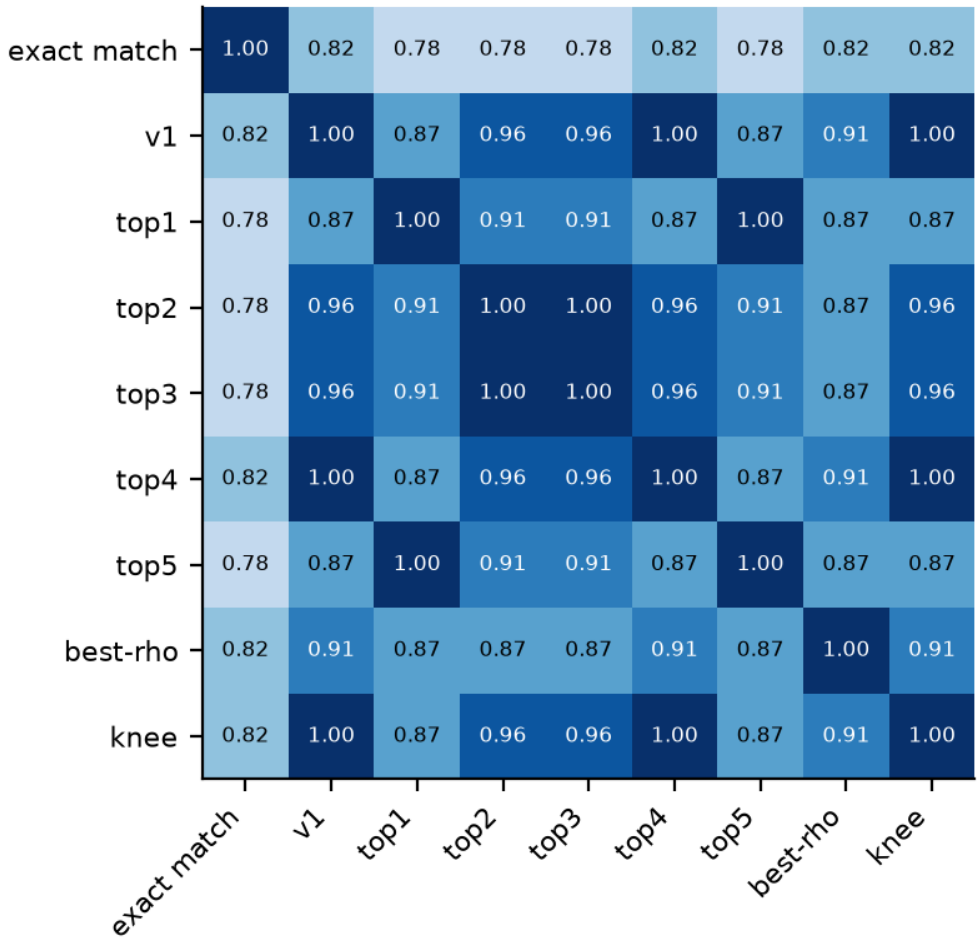


Figure 46: Kendall τ between the Bradley–Terry rankings produced by each pair of rules.

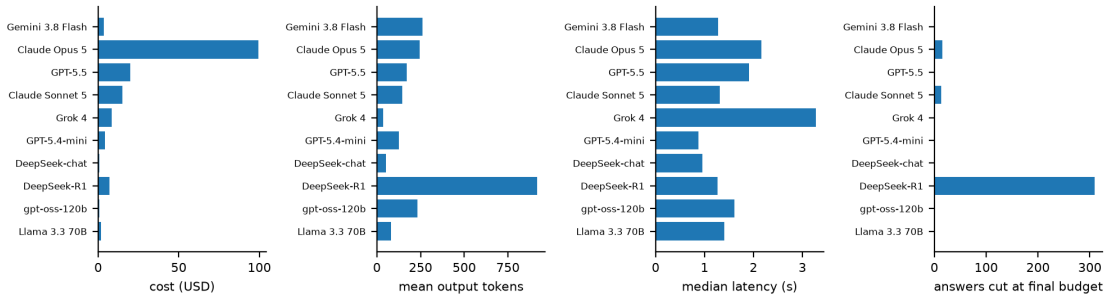


Figure 47: Run facts per model from the prediction cache: cost, output length, latency and answers still cut off at the final budget.

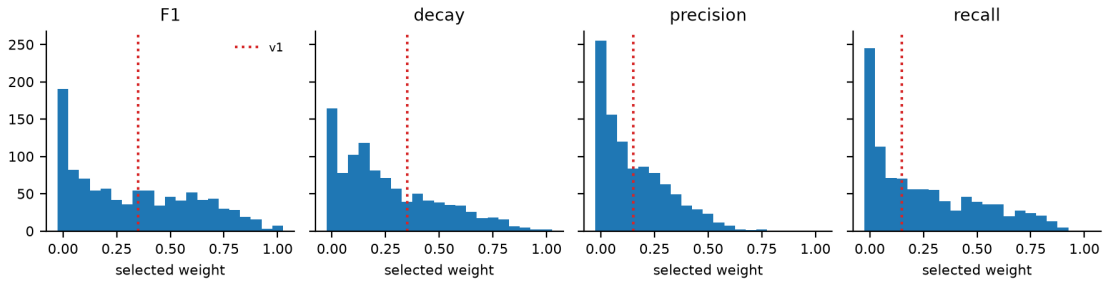


Figure 48: Weights selected as best on the in-bag items across 1,000 bootstrap resamples; dotted line = v1.

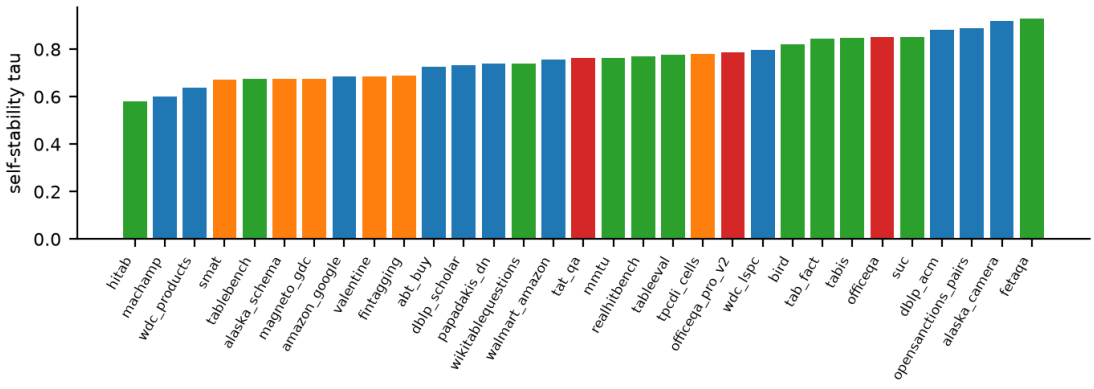


Figure 49: Self-stability of each dataset's model ranking (Kendall τ between bootstrap halves) under v1.

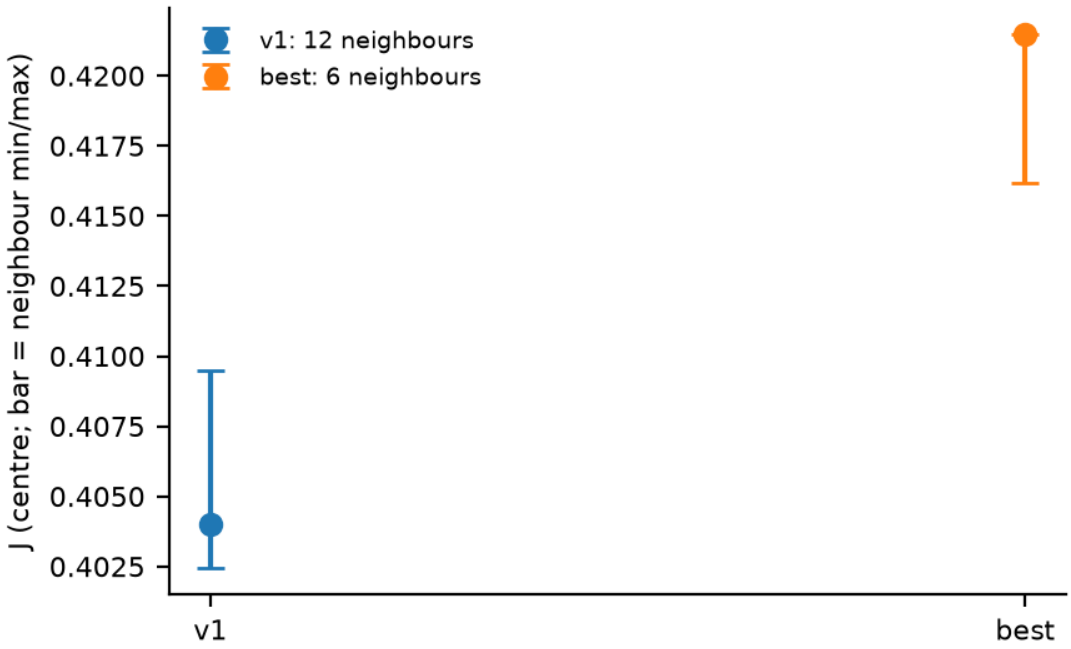


Figure 50: One-step sensitivity: J at v1 and at the best weighting with the range over all lattice neighbours one 0.05 move away.

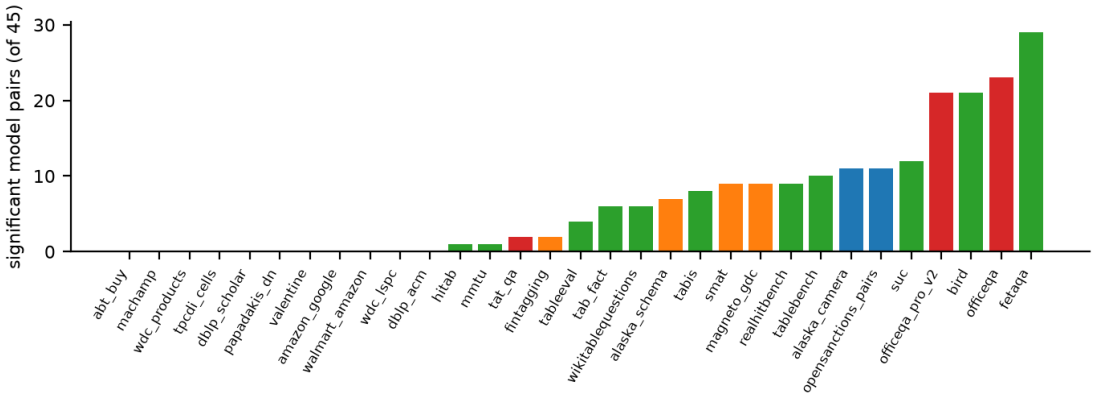


Figure 51: How many of the 45 model pairs each dataset separates at the item level under v1 (colour = family).

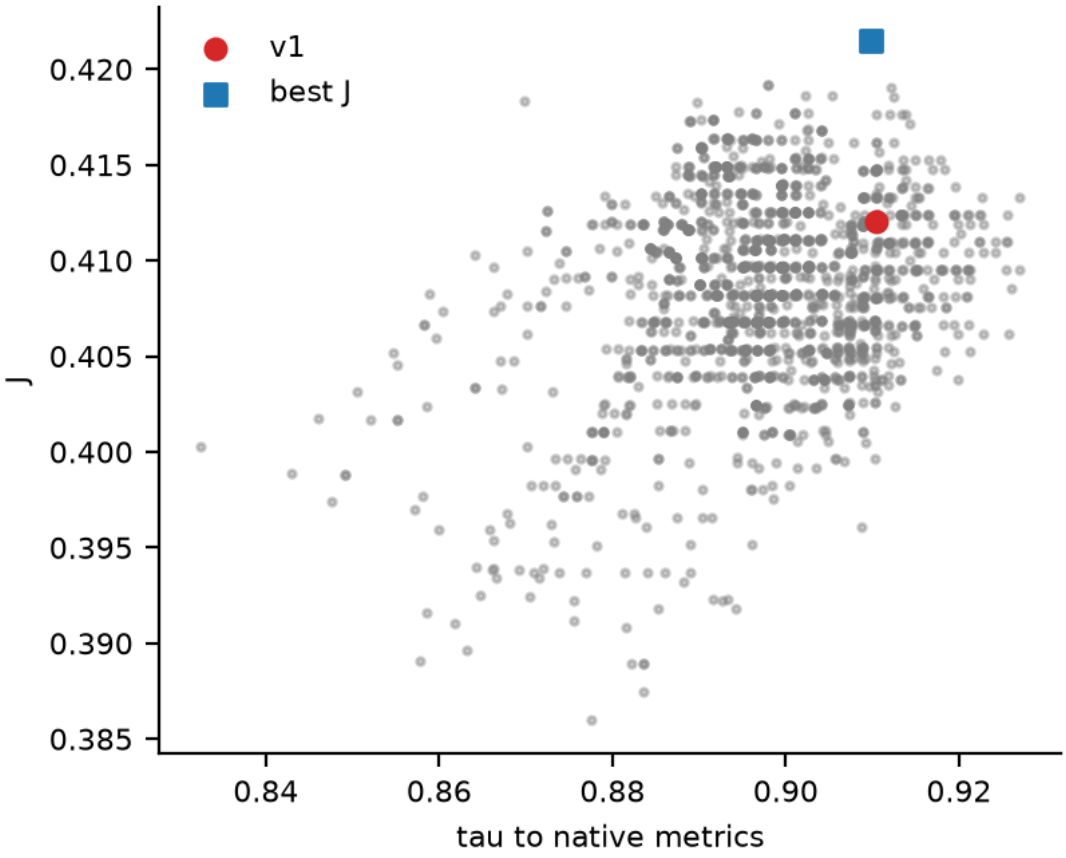


Figure 52: Agreement J against Kendall τ between each weighting's ranking and the native-metric ranking.

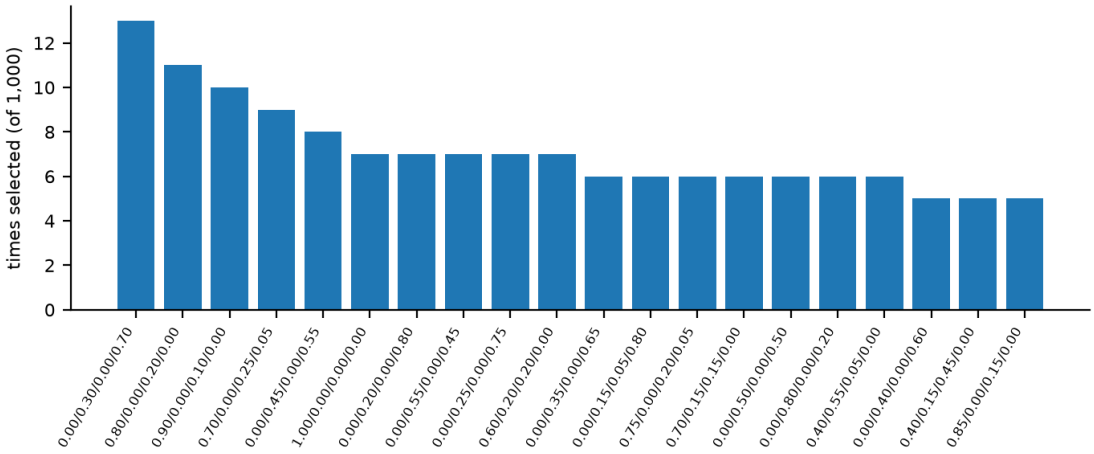


Figure 53: The twenty weightings most often selected as in-bag best (F1 / decay / precision / recall).

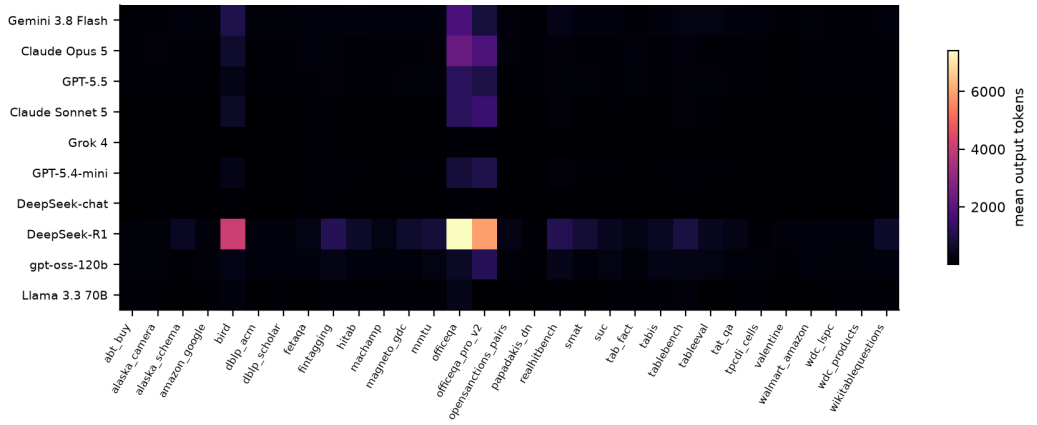


Figure 54: Mean output tokens per model and dataset.

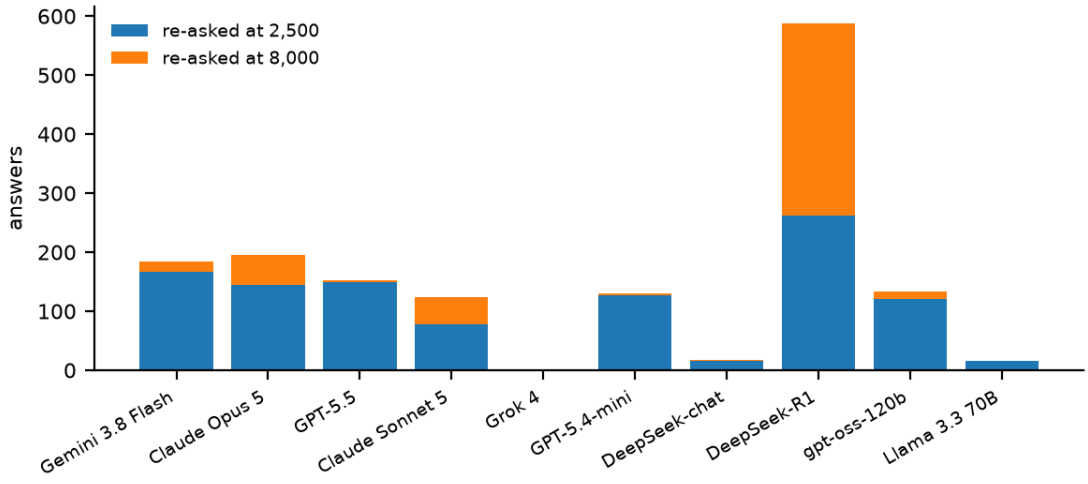


Figure 55: Truncation protocol: answers re-asked at 2,500 and 8,000 output tokens per model.

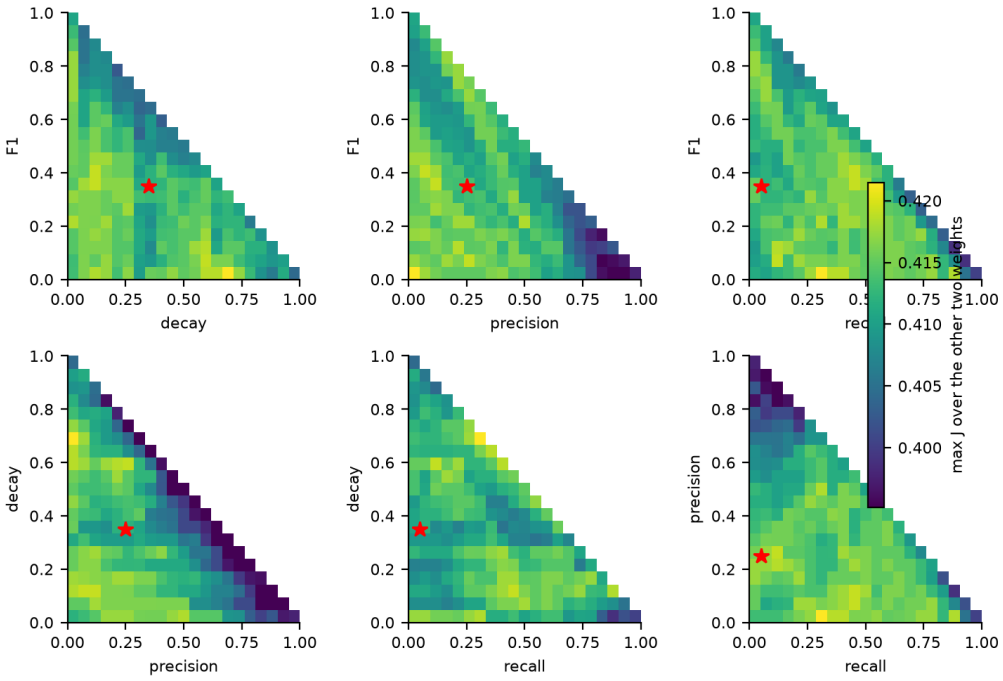


Figure 56: Best J attainable for each pair of weight values (maximised over the remaining two weights); red star = v1.

B All tables

Every saved table, exported from `data/`. Each is a separate file under `tables/`; include any of them in the main text with `\input`.

Table 8: Anchor operators: known utility, mean v1 score, exact-match rate and count.

op	utility	v1	em	n
no_answer	0.000	0.001	0.000	3083
wrong_random	0.000	0.097	0.041	1845
flip	0.000	0.000	0.000	1238
sign_flip	0.050	0.018	0.003	632
magnitude	0.050	0.091	0.013	632
near_miss_large	0.150	0.150	0.000	632
digit_typo	0.200	0.199	0.000	632
list_dump	0.250	0.373	0.013	1845
hedge	0.300	0.493	0.192	2974
shuffled	0.400	0.652	0.009	109
near_miss_medium	0.450	0.293	0.008	632
partial	0.500	0.345	0.061	1213
superset	0.600	0.467	0.045	1213
typo	0.600	0.334	0.101	1104
near_miss_small	0.750	0.405	0.097	632
paraphrase	0.950	0.965	0.930	3083
verbose	1.000	1.000	0.999	3083
identity	1.000	1.000	0.999	3083
tagged	1.000	1.000	0.999	3083
unit_variant	1.000	0.898	0.837	632

Table 9: Anchor correctness for the labelled rules.

label	rank	w_f1	w_decay	w_precision	w_recall	J	tau_native	rho	rho_ds	auc	wrong_mean	hedge_gap
only_f1	627	1.000	0.000	0.000	0.000	0.411	0.893	0.856	0.846	0.989	0.030	-0.496
uniform	768	0.250	0.250	0.250	0.250	0.410	0.903	0.875	0.866	0.994	0.033	-0.480
only_decay	1480	0.000	1.000	0.000	0.000	0.404	0.901	0.772	0.804	0.991	0.040	-0.595
v1_original	1483	0.350	0.350	0.150	0.150	0.404	0.912	0.878	0.868	0.994	0.034	-0.506
only_recall	1673	0.000	0.000	0.000	1.000	0.400	0.832	0.736	0.741	0.989	0.030	-0.244
only_precision	1712	0.000	0.000	1.000	0.000	0.398	0.858	0.848	0.839	0.989	0.031	-0.587

Table 10: Coverage rule outcome per dataset.

dataset	answered items	items kept	models used	models dropped
abt_buy	100	100	10	–
alaska_camera	100	100	10	–
alaska_schema	93	93	10	–
amazon_google	100	100	10	–
bird	100	100	10	–
dblp_acm	100	100	10	–
dblp_scholar	100	100	10	–
fetaqa	100	100	10	–
fintagging	100	100	10	–
hitab	100	100	10	–
machamp	100	100	10	–
magneto_gdc	100	100	10	–
mmtu	100	100	10	–
officeqa	100	100	10	–
officeqa_pro_v2	90	90	10	–
opensanctions_pairs	100	100	10	–
papadakis_dn	100	100	10	–
realhitbench	100	100	10	–
smat	100	100	10	–
suc	100	100	10	–
tab_fact	100	100	10	–
tabis	100	100	10	–
tablebench	100	100	10	–
tableeval	100	100	10	–
tat_qa	100	100	10	–
tpcdi_cells	100	100	10	–
valentine	100	100	10	–
walmart_amazon	100	100	10	–
wdc_lspc	100	100	10	–
wdc_products	100	100	10	–
wikitablequestions	100	100	10	–

Table 11: Datasets: family, items answered, dominant answer type and native metric.

dataset	items	answer_type	native_metric	family
abt_buy	100	boolean	em	EM
alaska_camera	100	boolean	em	EM
alaska_schema	93	text	em	SM
amazon_google	100	boolean	em	EM
bird	100	numeric	em	TQA
dblp_acm	100	boolean	em	EM
dblp_scholar	100	boolean	em	EM
fetaqa	100	freeform	rouge_l	TQA
fintagging	100	text	em	SM
hitab	100	numeric	num_tol	TQA
machamp	100	boolean	em	EM
magneto_gdc	100	text	em	SM
mmtu	100	text	em	TQA
officeqa	100	numeric	num_tol	FIN
officeqa_pro_v2	90	numeric	num_tol	FIN
opensanctions_pairs	100	boolean	em	EM
papadakis_dn	100	boolean	em	EM
realhitbench	100	numeric	num_tol	TQA
smat	100	text	em	SM
suc	100	text	em	TQA
tab_fact	100	boolean	em	TQA
tabis	100	text	em	TQA
tablebench	100	numeric	num_tol	TQA
tableeval	100	numeric	num_tol	TQA
tat_qa	100	numeric	em	FIN
tpcdi_cells	100	numeric	num_tol	SM
valentine	100	text	em	SM
walmart_amazon	100	boolean	em	EM
wdc_lspc	100	boolean	em	EM
wdc_products	100	boolean	em	EM
wikitablequestions	100	numeric	em	TQA

Table 12: Sampled designs: mean and sd over 20 seeds of every coverage and performance measure.

design	best_J_found mean	best_J_found std	regret mean	regret std	centred_L2_discrepancy mean	centred_L2_discrepancy std	min_pairwise_L1 mean	min_pairwise_L1 std	share_with_a_zero_weight mean	share_with_a_zero_weight std	best_rank_in_exhaustive mean	best_rank_in_exhaustive std
dirichlet_lattice	0.4177	0.0008	0.0038	0.0008	0.5382	0.0178	0.1000	0.0000	0.2690	0.0370	18.5500	13.1528
lhs_simplex	0.4180	0.0010	0.0034	0.0010	0.5312	0.0083	0.1000	0.0000	0.2795	0.0268	15.7500	14.6678
uniform_lattice	0.4185	0.0013	0.0029	0.0013	0.5256	0.0177	0.1000	0.0000	0.4500	0.0543	11.4000	11.9226

Table 13: Dataset diagnostics under v1.

dataset	family	mean_score	spread_ across_models	floor	ceiling	share_ constant_ bootstrap_ rankings	ranking_self_ stability_tau	tau_to_ others_mean	n_items	exclude_ from_ objective
abt_buy	EM	0.970	0.060	no	no	0.000	0.726	0.356	100	no
alaska_camera	EM	0.958	0.180	no	no	0.000	0.919	-0.218	100	no
alaska_schema	SM	0.928	0.118	no	no	0.000	0.675	-0.044	93	no
amazon_google	EM	0.936	0.100	no	no	0.000	0.684	-0.012	100	no
bird	TQA	0.187	0.209	no	no	0.000	0.820	0.313	100	no
dblp_acm	EM	0.986	0.090	no	no	0.000	0.884	0.350	100	no
dblp_scholar	EM	0.949	0.060	no	no	0.000	0.733	-0.019	100	no
fetaqa	TQA	0.382	0.192	no	no	0.000	0.930	0.271	100	no
fintagging	SM	0.880	0.069	no	no	0.000	0.689	0.256	100	no
hitab	TQA	0.770	0.068	no	no	0.000	0.581	-0.181	100	no
machamp	EM	0.887	0.070	no	no	0.000	0.600	0.377	100	no
magneto_gdc	SM	0.934	0.218	no	no	0.000	0.677	0.252	100	no
mmtu	TQA	0.867	0.110	no	no	0.000	0.763	0.298	100	no
officeqa	FIN	0.129	0.119	no	no	0.000	0.853	0.276	100	no
officeqa_pro_v2	FIN	0.139	0.215	no	no	0.000	0.785	0.311	90	no
opensanctions_pairs	EM	0.821	0.210	no	no	0.000	0.888	0.030	100	no
papadakis_dn	EM	0.941	0.030	no	no	0.002	0.739	0.067	100	no
realhitbench	TQA	0.673	0.143	no	no	0.000	0.769	0.344	100	no
smat	SM	0.918	0.150	no	no	0.000	0.674	0.200	100	no
suc	TQA	0.914	0.246	no	no	0.000	0.853	0.293	100	no
tab_fact	TQA	0.798	0.200	no	no	0.000	0.845	0.007	100	no
tabis	TQA	0.915	0.210	no	no	0.000	0.849	0.224	100	no
tablebench	TQA	0.818	0.218	no	no	0.000	0.675	0.276	100	no
tableeval	TQA	0.749	0.166	no	no	0.000	0.778	0.195	100	no
tat_qa	FIN	0.803	0.109	no	no	0.000	0.762	0.262	100	no
tpcdi_cells	SM	0.456	0.020	no	no	0.000	0.782	0.298	100	no
valentine	SM	0.984	0.030	no	yes	0.000	0.685	0.229	100	yes
walmart_amazon	EM	0.958	0.060	no	no	0.000	0.755	0.360	100	no
wdc_lspc	EM	0.982	0.040	no	yes	0.004	0.796	0.246	100	yes
wdc_products	EM	0.978	0.020	no	yes	0.006	0.637	0.178	100	yes
wikitablequestions	TQA	0.846	0.132	no	no	0.000	0.739	0.350	100	no

Table 14: Mean v1 composite per model within each family.

model	entity matching	finance	schema matching	table QA
Gemini 3.8 Flash	0.945	0.414	0.854	0.765
Claude Opus 5	0.947	0.439	0.869	0.745
GPT-5.5	0.950	0.388	0.864	0.751
Claude Sonnet 5	0.942	0.391	0.862	0.724
Grok 4	0.948	0.354	0.851	0.738
GPT-5.4-mini	0.947	0.347	0.859	0.716
DeepSeek-chat	0.941	0.342	0.852	0.711
DeepSeek-R1	0.953	0.318	0.853	0.717
gpt-oss-120b	0.915	0.323	0.848	0.698
Llama 3.3 70B	0.936	0.328	0.777	0.633

Table 15: Quantiles of J over the lattice.

count	mean	std	min	1%	5%	10%	25%	50%	75%	90%	95%	99%	max
1771.0000	0.4086	0.0049	0.3860	0.3925	0.3996	0.4025	0.4061	0.4091	0.4119	0.4147	0.4154	0.4176	0.4215

Table 16: v1, the uniform mix and the four single-ingredient corners on the lattice.

label	F1 / decay / prec / rec	rank	J	J Kemeny	J Copeland	J BT	tau native
uniform	0.25 / 0.25 / 0.25 / 0.25	768	0.410	0.414	0.414	0.384	0.903
only_f1	1.00 / 0.00 / 0.00 / 0.00	627	0.411	0.418	0.422	0.375	0.893
only_decay	0.00 / 1.00 / 0.00 / 0.00	1480	0.404	0.408	0.408	0.368	0.901
only_recall	0.00 / 0.00 / 0.00 / 1.00	1673	0.400	0.400	0.403	0.364	0.832
only_precision	0.00 / 0.00 / 1.00 / 0.00	1712	0.398	0.398	0.402	0.335	0.858

Table 17: The 25 best weightings by J with the objective under the other pooling rules.

rank	F1 / decay / prec / rec	J	J Kemeny	J Copeland	J BT	tau native	min agree
1	0.00 / 0.70 / 0.00 / 0.30	0.421	0.421	0.421	0.398	0.910	-0.442
3	0.40 / 0.10 / 0.05 / 0.45	0.419	0.419	0.419	0.385	0.898	-0.396
2	0.35 / 0.10 / 0.10 / 0.45	0.419	0.419	0.419	0.385	0.898	-0.396
4	0.05 / 0.60 / 0.25 / 0.10	0.419	0.419	0.419	0.393	0.912	-0.442
5	0.00 / 0.60 / 0.00 / 0.40	0.419	0.419	0.419	0.402	0.902	-0.442
6	0.00 / 0.65 / 0.00 / 0.35	0.419	0.419	0.419	0.397	0.905	-0.442
7	0.10 / 0.60 / 0.20 / 0.10	0.419	0.419	0.419	0.393	0.912	-0.442
8	0.00 / 0.10 / 0.00 / 0.90	0.418	0.418	0.429	0.400	0.870	-0.303
9	0.20 / 0.00 / 0.50 / 0.30	0.418	0.425	0.430	0.380	0.890	-0.396
10	0.35 / 0.05 / 0.10 / 0.50	0.418	0.418	0.418	0.383	0.895	-0.396
11	0.25 / 0.25 / 0.10 / 0.40	0.418	0.418	0.418	0.383	0.903	-0.396
12	0.45 / 0.15 / 0.00 / 0.40	0.418	0.418	0.418	0.382	0.901	-0.396
13	0.20 / 0.15 / 0.20 / 0.45	0.418	0.418	0.418	0.375	0.901	-0.396
14	0.20 / 0.15 / 0.25 / 0.40	0.418	0.418	0.418	0.375	0.901	-0.396
15	0.40 / 0.10 / 0.10 / 0.40	0.418	0.418	0.422	0.382	0.897	-0.396
16	0.45 / 0.10 / 0.05 / 0.40	0.418	0.418	0.422	0.386	0.897	-0.396
17	0.00 / 0.70 / 0.05 / 0.25	0.418	0.418	0.418	0.394	0.914	-0.442
20	0.15 / 0.60 / 0.05 / 0.20	0.418	0.418	0.418	0.385	0.910	-0.442
18	0.05 / 0.60 / 0.15 / 0.20	0.418	0.418	0.418	0.392	0.913	-0.442
19	0.10 / 0.55 / 0.20 / 0.15	0.418	0.418	0.418	0.394	0.912	-0.442
21	0.70 / 0.00 / 0.25 / 0.05	0.417	0.424	0.429	0.379	0.892	-0.396
22	0.75 / 0.00 / 0.20 / 0.05	0.417	0.424	0.429	0.379	0.892	-0.396
23	0.80 / 0.00 / 0.20 / 0.00	0.417	0.424	0.429	0.379	0.892	-0.396
24	0.85 / 0.00 / 0.15 / 0.00	0.417	0.424	0.429	0.379	0.892	-0.396
25	0.30 / 0.00 / 0.45 / 0.25	0.417	0.424	0.429	0.379	0.890	-0.396

Table 18: Full leaderboard under rule best-rho; weights F1/decay/prec/rec = 0.00 / 0.35 / 0.60 / 0.05.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.104	0.016	1.000	1.000	1.000	0.994	3.000	1.000	3.000	0.035	3.000	3.000	3.000	3.000	2.000	0.814
Claude Opus 5	0.076	0.000	2.000	2.000	2.000	0.006	2.000	1.000	3.000	0.367	1.000	1.000	2.000	2.000	1.000	0.812
GPT-5.5	0.043	0.016	3.000	3.000	5.000	0.000	1.000	1.000	3.000	0.582	2.000	2.000	1.000	1.000	3.000	0.810
Claude Sonnet 5	0.035	0.016	4.000	3.000	6.000	0.000	4.000	4.000	7.000	0.000	4.000	4.000	4.000	4.000	4.000	0.798
GPT-5.4-mini	0.025	0.016	5.000	3.000	6.000	0.000	7.000	4.000	8.000	0.000	7.000	6.000	6.000	7.000	6.000	0.796
Grok 4	0.014	0.016	6.000	4.000	6.000	0.000	5.000	4.000	7.000	0.000	5.000	5.000	5.000	5.000	5.000	0.798
DeepSeek-R1	-0.026	0.016	7.000	7.000	8.000	0.000	6.000	4.000	8.000	0.000	6.000	7.000	7.000	6.000	8.000	0.789
DeepSeek-chat	-0.027	0.016	8.000	7.000	8.000	0.000	8.000	5.000	8.000	0.000	8.000	8.000	8.000	8.000	7.000	0.786
gpt-oss-120b	-0.118	0.016	9.000	9.000	10.000	0.000	9.000	9.000	10.000	0.000	9.000	9.000	9.000	9.000	9.000	0.768
Llama 3.3 70B	-0.125	0.016	10.000	9.000	10.000	0.000	10.000	9.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.744

Table 19: Full leaderboard under rule exact match; weights F1/decay/prec/rec = –.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Claude Opus 5	0.047	0.000	1.000	1.000	3.000	0.764	1.000	1.000	2.000	0.812	1.000	1.000	1.000	1.000	1.000	0.768
GPT-5.5	0.039	0.016	2.000	1.000	3.000	0.152	2.000	1.000	3.000	0.177	2.000	2.000	2.000	2.000	2.000	0.764
Gemini 3.8 Flash	0.037	0.016	3.000	1.000	3.000	0.084	3.000	2.000	5.000	0.004	3.000	3.000	3.000	3.000	3.000	0.763
Grok 4	0.020	0.016	4.000	4.000	6.000	0.000	5.000	3.000	6.012	0.000	5.000	5.000	5.000	5.000	5.000	0.754
Claude Sonnet 5	0.013	0.016	5.000	4.000	7.000	0.000	4.000	3.000	6.000	0.000	4.000	4.000	4.000	4.000	4.000	0.751
GPT-5.4-mini	0.010	0.016	6.000	4.000	7.000	0.000	8.000	6.000	8.000	0.000	7.000	7.000	7.000	8.000	8.000	0.749
DeepSeek-R1	0.008	0.016	7.000	5.000	7.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	6.000	6.000	6.000	0.748
DeepSeek-chat	-0.014	0.016	8.000	8.000	9.000	0.000	7.000	6.000	8.000	0.000	8.000	8.000	8.000	7.000	7.000	0.737
gpt-oss-120b	-0.033	0.016	9.000	8.000	9.000	0.000	9.000	9.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.728
Llama 3.3 70B	-0.128	0.016	10.000	10.000	10.000	0.000	10.000	10.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.680

Table 20: Full leaderboard under rule knee; weights F1/decay/prec/rec = 0.05 / 0.60 / 0.25 / 0.10.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.115	0.016	1.000	1.000	1.000	1.000	3.000	1.000	4.000	0.043	3.000	3.000	3.000	3.000	2.000	0.810
Claude Opus 5	0.083	0.000	2.000	2.000	2.000	0.000	1.000	1.000	3.000	0.447	1.000	1.000	2.000	1.000	1.000	0.810
GPT-5.5	0.049	0.016	3.000	3.000	4.000	0.000	2.000	1.000	3.000	0.504	2.000	2.000	1.000	2.000	3.000	0.806
Claude Sonnet 5	0.032	0.016	4.000	3.000	5.000	0.000	4.000	4.000	7.000	0.000	4.000	4.000	4.000	4.000	4.000	0.794
Grok 4	0.021	0.016	5.000	4.000	6.000	0.000	5.000	4.000	7.000	0.000	5.000	5.000	5.000	5.000	5.000	0.794
GPT-5.4-mini	0.006	0.016	6.000	5.000	7.000	0.000	7.000	5.000	8.000	0.000	7.000	7.000	6.000	7.000	6.000	0.791
DeepSeek-chat	-0.021	0.016	7.000	7.000	8.000	0.000	8.000	6.000	8.000	0.000	8.000	8.000	8.000	8.000	8.000	0.778
DeepSeek-R1	-0.022	0.016	8.000	7.000	8.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	7.000	6.000	7.000	0.783
gpt-oss-120b	-0.115	0.016	9.000	9.000	9.000	0.000	9.000	9.000	10.000	0.000	9.000	9.000	9.000	9.000	9.000	0.763
Llama 3.3 70B	-0.147	0.016	10.000	10.000	10.000	0.000	10.000	9.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.734

Table 21: Full leaderboard under rule top1; weights F1/decay/prec/rec = 0.00 / 0.70 / 0.00 / 0.30.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.117	0.016	1.000	1.000	2.000	0.856	3.000	1.000	4.000	0.029	3.000	3.000	3.000	3.000	2.000	0.809
Claude Opus 5	0.105	0.000	2.000	1.000	2.000	0.144	1.000	1.000	3.000	0.656	1.000	1.000	1.000	1.000	1.000	0.809
GPT-5.5	0.042	0.016	3.000	3.000	4.000	0.000	2.000	1.000	3.000	0.301	2.000	2.000	2.000	2.000	3.000	0.803
Claude Sonnet 5	0.028	0.016	4.000	3.000	5.000	0.000	4.000	4.000	7.000	0.000	4.000	4.000	4.000	4.000	4.000	0.792
Grok 4	0.020	0.016	5.000	4.000	5.000	0.000	5.000	3.500	6.000	0.000	5.000	5.000	5.000	5.000	5.000	0.792
DeepSeek-R1	-0.016	0.016	6.000	6.000	8.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	6.000	6.000	7.000	0.783
DeepSeek-chat	-0.018	0.016	7.000	6.000	8.000	0.000	8.000	6.000	8.000	0.000	8.000	8.000	8.000	8.000	8.000	0.774
GPT-5.4-mini	-0.028	0.016	8.000	6.000	8.000	0.000	7.000	5.000	8.000	0.000	7.000	7.000	7.000	7.000	6.000	0.787
gpt-oss-120b	-0.066	0.016	9.000	9.000	9.000	0.000	9.000	9.000	10.000	0.000	9.000	9.000	9.000	9.000	9.000	0.763
Llama 3.3 70B	-0.183	0.016	10.000	10.000	10.000	0.000	10.000	9.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.731

Table 22: Full leaderboard under rule top2; weights F1/decay/prec/rec = 0.35 / 0.10 / 0.10 / 0.45.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.123	0.016	1.000	1.000	1.000	0.987	3.000	1.000	3.000	0.036	3.000	3.000	3.000	3.000	3.000	0.814
Claude Opus 5	0.100	0.000	2.000	2.000	2.000	0.013	1.000	1.000	3.000	0.537	1.000	1.000	1.000	1.000	1.000	0.813
GPT-5.5	0.050	0.016	3.000	3.000	4.000	0.000	2.000	1.000	3.000	0.415	2.000	2.000	2.000	2.000	2.000	0.811
Claude Sonnet 5	0.027	0.016	4.000	3.000	5.000	0.000	4.000	4.000	7.000	0.000	4.000	4.000	4.000	4.000	4.000	0.797
Grok 4	0.026	0.016	5.000	3.975	5.000	0.000	5.000	4.000	7.000	0.000	5.000	5.000	5.000	5.000	5.000	0.801
DeepSeek-chat	-0.018	0.016	6.000	6.000	8.000	0.000	7.000	5.000	8.000	0.000	7.000	7.000	7.000	7.000	7.000	0.791
GPT-5.4-mini	-0.019	0.016	7.000	6.000	8.000	0.000	8.000	6.000	8.000	0.000	8.000	8.000	8.000	8.000	8.000	0.789
DeepSeek-R1	-0.021	0.016	8.000	6.000	8.000	0.000	6.000	4.000	7.000	0.000	6.000	6.000	6.000	6.000	6.000	0.796
gpt-oss-120b	-0.090	0.016	9.000	9.000	9.000	0.000	9.000	9.000	10.000	0.000	9.000	9.000	9.000	9.000	9.000	0.773
Llama 3.3 70B	-0.177	0.016	10.000	10.000	10.000	0.000	10.000	9.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.746

Table 23: Full leaderboard under rule top3; weights F1/decay/prec/rec = 0.40 / 0.10 / 0.05 / 0.45.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.124	0.016	1.000	1.000	1.000	0.989	3.000	1.000	3.500	0.034	3.000	3.000	3.000	3.000	3.000	0.813
Claude Opus 5	0.099	0.000	2.000	2.000	2.000	0.011	1.000	1.000	3.000	0.547	1.000	1.000	1.000	1.000	1.000	0.813
GPT-5.5	0.050	0.016	3.000	3.000	4.000	0.000	2.000	1.000	3.000	0.410	2.000	2.000	2.000	2.000	2.000	0.811
Claude Sonnet 5	0.027	0.016	4.000	3.000	5.000	0.000	4.000	4.000	7.000	0.000	4.000	4.000	4.000	4.000	4.000	0.797
Grok 4	0.026	0.016	5.000	3.000	5.000	0.000	5.000	4.000	7.000	0.000	5.000	5.000	5.000	5.000	5.000	0.801
DeepSeek-chat	-0.017	0.016	6.000	6.000	8.000	0.000	7.000	5.000	8.000	0.000	7.000	7.000	7.000	7.000	7.000	0.790
GPT-5.4-mini	-0.020	0.016	7.000	6.000	8.000	0.000	8.000	6.000	8.000	0.000	8.000	8.000	8.000	8.000	8.000	0.788
DeepSeek-R1	-0.021	0.016	8.000	6.000	8.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	6.000	6.000	6.000	0.796
gpt-oss-120b	-0.089	0.016	9.000	9.000	9.000	0.000	9.000	9.000	10.000	0.000	9.000	9.000	9.000	9.000	9.000	0.773
Llama 3.3 70B	-0.179	0.016	10.000	10.000	10.000	0.000	10.000	9.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.746

Table 24: Full leaderboard under rule top4; weights F1/decay/prec/rec = 0.05 / 0.60 / 0.25 / 0.10.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.115	0.016	1.000	1.000	1.000	1.000	3.000	1.000	4.000	0.043	3.000	3.000	3.000	3.000	2.000	0.810
Claude Opus 5	0.083	0.000	2.000	2.000	2.000	0.000	1.000	1.000	3.000	0.447	1.000	1.000	2.000	1.000	1.000	0.810
GPT-5.5	0.049	0.016	3.000	3.000	4.000	0.000	2.000	1.000	3.000	0.504	2.000	2.000	1.000	2.000	3.000	0.806
Claude Sonnet 5	0.032	0.016	4.000	3.000	5.000	0.000	4.000	4.000	7.000	0.000	4.000	4.000	4.000	4.000	4.000	0.794
Grok 4	0.021	0.016	5.000	4.000	6.000	0.000	5.000	4.000	7.000	0.000	5.000	5.000	5.000	5.000	5.000	0.794
GPT-5.4-mini	0.006	0.016	6.000	5.000	7.000	0.000	7.000	5.000	8.000	0.000	7.000	7.000	6.000	7.000	6.000	0.791
DeepSeek-chat	-0.021	0.016	7.000	7.000	8.000	0.000	8.000	6.000	8.000	0.000	8.000	8.000	8.000	8.000	8.000	0.778
DeepSeek-R1	-0.022	0.016	8.000	7.000	8.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	7.000	6.000	7.000	0.783
gpt-oss-120b	-0.115	0.016	9.000	9.000	9.000	0.000	9.000	9.000	10.000	0.000	9.000	9.000	9.000	9.000	9.000	0.763
Llama 3.3 70B	-0.147	0.016	10.000	10.000	10.000	0.000	10.000	9.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.734

Table 25: Full leaderboard under rule top5; weights F1/decay/prec/rec = 0.00 / 0.60 / 0.00 / 0.40.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.117	0.016	1.000	1.000	2.000	0.856	3.000	1.000	4.000	0.037	3.000	3.000	3.000	3.000	2.000	0.810
Claude Opus 5	0.105	0.000	2.000	1.000	2.000	0.144	1.000	1.000	3.000	0.636	1.000	1.000	1.000	1.000	1.000	0.810
GPT-5.5	0.042	0.016	3.000	3.000	4.000	0.000	2.000	1.000	3.000	0.316	2.000	2.000	2.000	2.000	3.000	0.805
Claude Sonnet 5	0.028	0.016	4.000	3.000	5.000	0.000	4.000	4.000	6.000	0.000	4.000	4.000	4.000	4.000	4.000	0.793
Grok 4	0.020	0.016	5.000	4.000	5.000	0.000	5.000	4.000	6.000	0.000	5.000	5.000	5.000	5.000	5.000	0.794
DeepSeek-R1	-0.016	0.016	6.000	6.000	8.000	0.000	6.000	4.000	7.500	0.000	6.000	6.000	6.000	6.000	7.000	0.786
DeepSeek-chat	-0.018	0.016	7.000	6.000	8.000	0.000	8.000	6.000	8.000	0.000	8.000	8.000	8.000	8.000	8.000	0.777
GPT-5.4-mini	-0.028	0.016	8.000	6.000	8.000	0.000	7.000	5.488	8.000	0.000	7.000	7.000	7.000	7.000	6.000	0.787
gpt-oss-120b	-0.066	0.016	9.000	9.000	9.000	0.000	9.000	9.000	10.000	0.000	9.000	9.000	9.000	9.000	9.000	0.765
Llama 3.3 70B	-0.183	0.016	10.000	10.000	10.000	0.000	10.000	9.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.734

Table 26: Full leaderboard under rule v1; weights F1/decay/prec/rec = 0.35 / 0.35 / 0.15 / 0.15.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.121	0.016	1.000	1.000	1.000	0.992	3.000	1.000	3.000	0.036	3.000	3.000	3.000	3.000	3.000	0.812
Claude Opus 5	0.095	0.000	2.000	2.000	2.000	0.008	2.000	1.000	3.000	0.476	1.000	1.000	2.000	1.000	1.000	0.811
GPT-5.5	0.052	0.016	3.000	3.000	4.000	0.000	1.000	1.000	3.000	0.479	2.000	2.000	1.000	2.000	2.000	0.808
Claude Sonnet 5	0.029	0.016	4.000	3.000	5.000	0.000	4.000	4.000	6.000	0.000	4.000	4.000	4.000	4.000	4.000	0.795
Grok 4	0.027	0.016	5.000	4.000	5.000	0.000	5.000	4.000	6.000	0.000	5.000	5.000	5.000	5.000	5.000	0.797
GPT-5.4-mini	-0.010	0.016	6.000	6.000	8.000	0.000	8.000	5.988	8.000	0.000	7.000	6.500	7.000	8.000	6.000	0.790
DeepSeek-chat	-0.018	0.016	7.000	6.000	8.000	0.000	7.000	5.000	8.000	0.000	8.000	8.000	8.000	7.000	8.000	0.784
DeepSeek-R1	-0.025	0.016	8.000	6.000	8.000	0.000	6.000	5.000	8.000	0.000	6.000	6.500	6.000	6.000	7.000	0.788
gpt-oss-120b	-0.100	0.016	9.000	9.000	9.000	0.000	9.000	9.000	10.000	0.000	9.000	9.000	9.000	9.000	9.000	0.767
Llama 3.3 70B	-0.170	0.016	10.000	10.000	10.000	0.000	10.000	9.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.738

Table 27: Leave-one-dataset-out: fitted weights and held-out τ for every dataset and weight source.

kind	held_out	weights_from	w_f1	w_decay	w_precision	w_recall	J_rest	tau_heldout_mean	n_heldout
dataset	abt_buy	lodo_weights	0.700	0.000	0.250	0.050	0.413	0.760	1
dataset	abt_buy	full_best	0.000	0.700	0.000	0.300	0.403	0.760	1
dataset	abt_buy	v1	0.350	0.350	0.150	0.150	0.391	0.809	1
dataset	alaska_camera	lodo_weights	0.150	0.250	0.350	0.250	0.450	-0.424	1
dataset	alaska_camera	full_best	0.000	0.700	0.000	0.300	0.444	-0.489	1
dataset	alaska_camera	v1	0.350	0.350	0.150	0.150	0.436	-0.442	1
dataset	alaska_schema	lodo_weights	0.000	0.250	0.400	0.350	0.437	-0.200	1
dataset	alaska_schema	full_best	0.000	0.700	0.000	0.300	0.430	-0.067	1
dataset	alaska_schema	v1	0.350	0.350	0.150	0.150	0.424	-0.200	1
dataset	amazon_google	lodo_weights	0.300	0.200	0.100	0.400	0.435	0.000	1
dataset	amazon_google	full_best	0.000	0.700	0.000	0.300	0.430	-0.072	1
dataset	amazon_google	v1	0.350	0.350	0.150	0.150	0.419	-0.024	1
dataset	bird	lodo_weights	0.350	0.600	0.050	0.000	0.418	0.315	1
dataset	bird	full_best	0.000	0.700	0.000	0.300	0.408	0.422	1
dataset	bird	v1	0.350	0.350	0.150	0.150	0.405	0.539	1
dataset	dblp_acm	lodo_weights	0.000	0.150	0.100	0.750	0.416	0.659	1
dataset	dblp_acm	full_best	0.000	0.700	0.000	0.300	0.414	0.637	1
dataset	dblp_acm	v1	0.350	0.350	0.150	0.150	0.397	0.637	1
dataset	dblp_scholar	lodo_weights	0.050	0.600	0.250	0.100	0.433	0.000	1
dataset	dblp_scholar	full_best	0.000	0.700	0.000	0.300	0.429	-0.047	1
dataset	dblp_scholar	v1	0.350	0.350	0.150	0.150	0.421	0.000	1
dataset	fetaqa	lodo_weights	0.300	0.250	0.100	0.350	0.421	0.494	1
dataset	fetaqa	full_best	0.000	0.700	0.000	0.300	0.420	0.467	1
dataset	fetaqa	v1	0.350	0.350	0.150	0.150	0.407	0.422	1
dataset	fintagging	lodo_weights	0.000	0.150	0.250	0.600	0.419	0.368	1
dataset	fintagging	full_best	0.000	0.700	0.000	0.300	0.405	0.674	1
dataset	fintagging	v1	0.350	0.350	0.150	0.150	0.402	0.467	1
dataset	hitab	lodo_weights	0.000	0.250	0.400	0.350	0.442	-0.315	1
dataset	hitab	full_best	0.000	0.700	0.000	0.300	0.437	-0.289	1
dataset	hitab	v1	0.350	0.350	0.150	0.150	0.433	-0.315	1
dataset	machamp	lodo_weights	0.700	0.000	0.250	0.050	0.416	0.685	1
dataset	machamp	full_best	0.000	0.700	0.000	0.300	0.405	0.685	1
dataset	machamp	v1	0.350	0.350	0.150	0.150	0.393	0.738	1
dataset	magneto_gdc	lodo_weights	0.700	0.000	0.250	0.050	0.421	0.511	1
dataset	magneto_gdc	full_best	0.000	0.700	0.000	0.300	0.412	0.467	1
dataset	magneto_gdc	v1	0.350	0.350	0.150	0.150	0.405	0.539	1
dataset	mmtu	lodo_weights	0.350	0.050	0.100	0.500	0.419	0.494	1
dataset	mmtu	full_best	0.000	0.700	0.000	0.300	0.413	0.689	1
dataset	mmtu	v1	0.350	0.350	0.150	0.150	0.401	0.556	1
dataset	officeqa	lodo_weights	0.200	0.000	0.500	0.300	0.413	0.584	1
dataset	officeqa	full_best	0.000	0.700	0.000	0.300	0.413	0.689	1
dataset	officeqa	v1	0.350	0.350	0.150	0.150	0.399	0.556	1
dataset	officeqa_pro_v2	lodo_weights	0.000	0.600	0.000	0.400	0.416	0.584	1
dataset	officeqa_pro_v2	full_best	0.000	0.700	0.000	0.300	0.411	0.511	1
dataset	officeqa_pro_v2	v1	0.350	0.350	0.150	0.150	0.399	0.556	1
dataset	opensanctions_pairs	lodo_weights	0.700	0.000	0.250	0.050	0.431	0.000	1
dataset	opensanctions_pairs	full_best	0.000	0.700	0.000	0.300	0.427	0.090	1
dataset	opensanctions_pairs	v1	0.350	0.350	0.150	0.150	0.416	0.045	1
dataset	papadakis_dn	lodo_weights	0.000	0.150	0.100	0.750	0.437	0.027	1
dataset	papadakis_dn	full_best	0.000	0.700	0.000	0.300	0.431	0.130	1
dataset	papadakis_dn	v1	0.350	0.350	0.150	0.150	0.417	0.078	1
dataset	realhitbench	lodo_weights	0.000	0.250	0.400	0.350	0.409	0.674	1
dataset	realhitbench	full_best	0.000	0.700	0.000	0.300	0.403	0.733	1
dataset	realhitbench	v1	0.350	0.350	0.150	0.150	0.397	0.600	1

dataset	smat	lodo_weights	0.050	0.100	0.100	0.750	0.429	0.276	1
dataset	smat	full_best	0.000	0.700	0.000	0.300	0.424	0.333	1
dataset	smat	v1	0.350	0.350	0.150	0.150	0.407	0.378	1
dataset	suc	lodo_weights	0.700	0.000	0.250	0.050	0.419	0.584	1
dataset	suc	full_best	0.000	0.700	0.000	0.300	0.413	0.674	1
dataset	suc	v1	0.350	0.350	0.150	0.150	0.398	0.584	1
dataset	tab_fact	lodo_weights	0.000	0.650	0.000	0.350	0.437	-0.068	1
dataset	tab_fact	full_best	0.000	0.700	0.000	0.300	0.437	-0.045	1
dataset	tab_fact	v1	0.350	0.350	0.150	0.150	0.423	-0.045	1
dataset	tabis	lodo_weights	0.200	0.000	0.500	0.300	0.426	0.386	1
dataset	tabis	full_best	0.000	0.700	0.000	0.300	0.420	0.477	1
dataset	tabis	v1	0.350	0.350	0.150	0.150	0.403	0.432	1
dataset	tablebench	lodo_weights	0.000	0.100	0.300	0.600	0.420	0.506	1
dataset	tablebench	full_best	0.000	0.700	0.000	0.300	0.414	0.422	1
dataset	tablebench	v1	0.350	0.350	0.150	0.150	0.403	0.511	1
dataset	tableeval	lodo_weights	0.700	0.000	0.250	0.050	0.429	0.289	1
dataset	tableeval	full_best	0.000	0.700	0.000	0.300	0.424	0.333	1
dataset	tableeval	v1	0.350	0.350	0.150	0.150	0.409	0.333	1
dataset	tat_qa	lodo_weights	0.000	1.000	0.000	0.000	0.413	0.156	1
dataset	tat_qa	full_best	0.000	0.700	0.000	0.300	0.402	0.600	1
dataset	tat_qa	v1	0.350	0.350	0.150	0.150	0.405	0.539	1
dataset	tpcdi_cells	lodo_weights	0.250	0.150	0.200	0.400	0.416	0.565	1
dataset	tpcdi_cells	full_best	0.000	0.700	0.000	0.300	0.412	0.489	1
dataset	tpcdi_cells	v1	0.350	0.350	0.150	0.150	0.403	0.535	1
dataset	valentine	lodo_weights	0.700	0.000	0.250	0.050	0.425	0.386	1
dataset	valentine	full_best	0.000	0.700	0.000	0.300	0.410	0.523	1
dataset	valentine	v1	0.350	0.350	0.150	0.150	0.402	0.477	1
dataset	walmart_amazon	lodo_weights	0.700	0.000	0.250	0.050	0.414	0.722	1
dataset	walmart_amazon	full_best	0.000	0.700	0.000	0.300	0.410	0.768	1
dataset	walmart_amazon	v1	0.350	0.350	0.150	0.150	0.393	0.722	1
dataset	wdc_lspc	lodo_weights	0.200	0.000	0.500	0.300	0.426	0.387	1
dataset	wdc_lspc	full_best	0.000	0.700	0.000	0.300	0.419	0.484	1
dataset	wdc_lspc	v1	0.350	0.350	0.150	0.150	0.405	0.435	1
dataset	wdc_products	lodo_weights	0.700	0.000	0.250	0.050	0.428	0.316	1
dataset	wdc_products	full_best	0.000	0.700	0.000	0.300	0.423	0.369	1
dataset	wdc_products	v1	0.350	0.350	0.150	0.150	0.405	0.369	1
dataset	wikitablequestions	lodo_weights	0.050	0.200	0.000	0.750	0.411	0.736	1
dataset	wikitablequestions	full_best	0.000	0.700	0.000	0.300	0.410	0.778	1
dataset	wikitablequestions	v1	0.350	0.350	0.150	0.150	0.392	0.778	1

Table 28: Leave-one-family-out: fitted weights and held-out τ.

kind	held_out	weights_from	w_f1	w_decay	w_precision	w_recall	J_rest	tau_heldout_mean	n_heldout
family	EM	lodo_weights	0.200	0.000	0.500	0.300	0.481	0.323	11
family	EM	full_best	0.000	0.700	0.000	0.300	0.477	0.287	11
family	EM	v1	0.350	0.350	0.150	0.150	0.456	0.301	11
family	SM	lodo_weights	0.000	0.100	0.000	0.900	0.431	0.351	6
family	SM	full_best	0.000	0.700	0.000	0.300	0.422	0.366	6
family	SM	v1	0.350	0.350	0.150	0.150	0.409	0.252	6
family	TQA	lodo_weights	0.450	0.000	0.400	0.150	0.423	0.379	11
family	TQA	full_best	0.000	0.700	0.000	0.300	0.409	0.395	11
family	TQA	v1	0.350	0.350	0.150	0.150	0.403	0.379	11
family	FIN	lodo_weights	0.000	0.950	0.050	0.000	0.406	0.434	3
family	FIN	full_best	0.000	0.700	0.000	0.300	0.396	0.511	3
family	FIN	v1	0.350	0.350	0.150	0.150	0.381	0.511	3

Table 29: Majority-class baselines for the boolean datasets.

dataset	majority-class accuracy	n
abt_buy	0.510	100
alaska_camera	0.530	100
amazon_google	0.550	100
dblp_acm	0.510	100
dblp_scholar	0.510	100
machamp	0.530	100
opensanctions_pairs	0.510	100
papadakis_dn	0.850	100
tab_fact	0.540	100
walmart_amazon	0.590	100
wdc_lspc	0.540	100
wdc_products	0.540	100

Table 30: Every quantity saved by the nested item bootstrap.

quantity	value
B	1000
n_candidates	1771
in_bag_J_of_selected_mean	0.3802546391161006
out_of_bag_J_of_selected_mean	0.33121789744602803
optimism_gap	0.04903674167007255
out_of_bag_J_v1_mean	0.33883801148330894
out_of_bag_J_full_best_mean	0.3324126063075918
selected_vs_v1_out_of_bag	"mean_diff": -0.007620114037280873, "ci95_lo": -0.0388192860072015, "ci95_hi": 0.01313701503406264, "p_boot_one_sided": 0.721, "share_a_ahead": 0.279
full_best_vs_v1_out_of_bag	"mean_diff": -0.00642540517571714, "ci95_lo": -0.028784469692387554, "ci95_hi": 0.01558572926447651, "p_boot_one_sided": 0.715, "share_a_ahead": 0.285
full_best_vs_v1_in_bag	"mean_diff": -0.004272295355796814, "ci95_lo": -0.02608252614736557, "ci95_hi": 0.01608043685555456, "p_boot_one_sided": 0.664, "share_a_ahead": 0.336
share_resamples_v1_in_top_5pct	0.051
selected_weights_mean	"f1": 0.327, "decay": 0.262, "precision": 0.157, "recall": 0.254
selected_weights_sd	"f1": 0.279, "decay": 0.231, "precision": 0.155, "recall": 0.251
n_distinct_selected	608
plateau_size	1765
plateau_share	0.9966120835686053
v1_in_plateau	True
v1_p_vs_best	0.664
practical_plateau_size_within_0.02	1720
v1_in_practical_plateau	True
practical_plateau_weight_ranges	"f1": [0.0, 1.0], "decay": [0.0, 1.0], "precision": [0.0, 0.85], "recall": [0.0, 0.95]
plateau_weight_ranges	"f1": [0.0, 1.0], "decay": [0.0, 1.0], "precision": [0.0, 1.0], "recall": [0.0, 1.0]

Table 31: Official-split estimates under v1 averaged over the ten models.

dataset	suite_mean	official_estimate	suite_f1_pos	official_precision_pos	official_recall_pos	official_f1_pos	official_positive_share
abt_buy	0.970	0.986	0.968	0.926	0.949	0.936	0.108
alaska_camera	0.958	0.958	0.955	1.000	0.921	0.955	0.500
alaska_schema	0.928	0.928	–	–	–	–	–
amazon_google	0.936	0.932	0.929	0.611	0.942	0.739	0.102
bird	0.187	0.213	–	–	–	–	–
dblp_acm	0.986	0.991	0.985	0.979	0.978	0.976	0.180
dblp_scholar	0.949	0.966	0.948	0.900	0.924	0.910	0.186
fetaqa	0.382	0.382	–	–	–	–	–
fintagging	0.880	0.876	–	–	–	–	–
hitab	0.770	0.773	–	–	–	–	–
machamp	0.887	0.890	0.892	0.644	0.853	0.733	0.175
magneto_gdc	0.934	0.938	–	–	–	–	–
mmtu	0.867	0.829	–	–	–	–	–
officeqa	0.129	0.129	–	–	–	–	–
officeqa_pro_v2	0.139	0.139	–	–	–	–	–
opensanctions_pairs	0.821	0.727	0.773	0.993	0.649	0.777	0.769
papadakis_dn	0.941	0.988	0.833	0.550	0.963	0.683	0.018
realhitbench	0.673	0.673	–	–	–	–	–
smat	0.918	0.916	–	–	–	–	–
suc	0.914	0.912	–	–	–	–	–
tab_fact	0.798	0.789	0.758	0.896	0.674	0.764	0.500
tabis	0.915	0.918	–	–	–	–	–
tablebench	0.818	0.816	–	–	–	–	–
tableeval	0.749	0.754	–	–	–	–	–
tat_qa	0.803	0.758	–	–	–	–	–
tpcdi_cells	0.456	0.461	–	–	–	–	–
valentine	0.984	0.979	–	–	–	–	–
walmart_amazon	0.958	0.975	0.947	0.832	0.927	0.875	0.094
wdc_lspc	0.982	0.983	0.983	0.965	0.979	0.971	0.264
wdc_products	0.978	0.985	0.979	0.980	0.968	0.974	0.280
wikitablequestions	0.846	0.844	–	–	–	–	–

Table 32: Official-split estimates under v1 for every dataset and model.

dataset	model	suite_mean	official_ estimate	official_lo95	official_hi95	effective_n	suite_f1_pos	official_f1_ pos	official_f1_ lo95	official_f1_ hi95	official_ positive_ share
abt_buy	Claude Opus 5	1.000	1.000	1.000	1.000	63.075	1.000	1.000	1.000	1.000	0.108
abt_buy	Claude Sonnet 5	0.970	0.993	0.985	1.000	63.075	0.968	0.968	0.925	1.000	0.108
abt_buy	DeepSeek-chat	0.960	0.976	0.933	0.998	63.075	0.958	0.893	0.735	0.990	0.108
abt_buy	DeepSeek-R1	0.970	0.978	0.931	1.000	63.075	0.969	0.904	0.755	1.000	0.108
abt_buy	Gemini 3.8 Flash	0.980	0.996	0.988	1.000	63.075	0.979	0.979	0.944	1.000	0.108
abt_buy	GPT-5.4-mini	0.960	0.976	0.937	0.998	63.075	0.958	0.893	0.750	0.990	0.108
abt_buy	GPT-5.5	1.000	1.000	1.000	1.000	63.075	1.000	1.000	1.000	1.000	0.108
abt_buy	gpt-oss-120b	0.940	0.987	0.974	0.996	63.075	0.935	0.935	0.879	0.981	0.108
abt_buy	Grok 4	0.960	0.976	0.934	0.998	63.075	0.958	0.893	0.745	0.990	0.108
abt_buy	Llama 3.3 70B	0.960	0.976	0.937	0.998	63.075	0.958	0.893	0.735	0.992	0.108
alaska_camera	Claude Opus 5	0.820	0.820	0.740	0.890	100.000	0.795	0.795	0.687	0.881	0.500
alaska_camera	Claude Sonnet 5	0.980	0.980	0.950	1.000	100.000	0.981	0.981	0.951	1.000	0.500
alaska_camera	DeepSeek-chat	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	DeepSeek-R1	0.990	0.990	0.970	1.000	100.000	0.990	0.990	0.969	1.000	0.500
alaska_camera	Gemini 3.8 Flash	0.890	0.890	0.830	0.950	100.000	0.884	0.884	0.805	0.943	0.500
alaska_camera	GPT-5.4-mini	0.970	0.970	0.930	1.000	100.000	0.971	0.971	0.929	1.000	0.500
alaska_camera	GPT-5.5	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	gpt-oss-120b	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	Grok 4	0.940	0.940	0.890	0.980	100.000	0.940	0.940	0.886	0.982	0.500
alaska_camera	Llama 3.3 70B	0.990	0.990	0.970	1.000	100.000	0.990	0.990	0.967	1.000	0.500
alaska_schema	Claude Opus 5	0.945	0.945	0.900	0.981	93.000	–	–	–	–	–
alaska_schema	Claude Sonnet 5	0.929	0.929	0.878	0.973	93.000	–	–	–	–	–
alaska_schema	DeepSeek-chat	0.951	0.951	0.908	0.986	93.000	–	–	–	–	–
alaska_schema	DeepSeek-R1	0.948	0.948	0.907	0.982	93.000	–	–	–	–	–
alaska_schema	Gemini 3.8 Flash	0.918	0.918	0.866	0.965	93.000	–	–	–	–	–
alaska_schema	GPT-5.4-mini	0.958	0.958	0.916	0.988	93.000	–	–	–	–	–
alaska_schema	GPT-5.5	0.928	0.928	0.882	0.968	93.000	–	–	–	–	–
alaska_schema	gpt-oss-120b	0.934	0.934	0.888	0.976	93.000	–	–	–	–	–
alaska_schema	Grok 4	0.929	0.929	0.876	0.970	93.000	–	–	–	–	–
alaska_schema	Llama 3.3 70B	0.840	0.840	0.787	0.891	93.000	–	–	–	–	–
amazon_google	Claude Opus 5	0.940	0.916	0.841	0.981	67.152	0.936	0.704	0.526	0.907	0.102
amazon_google	Claude Sonnet 5	0.960	0.935	0.865	0.985	67.152	0.957	0.758	0.587	0.933	0.102
amazon_google	DeepSeek-chat	0.960	0.949	0.886	0.998	67.152	0.957	0.796	0.628	0.992	0.102
amazon_google	DeepSeek-R1	0.950	0.932	0.867	0.985	67.152	0.946	0.747	0.583	0.930	0.102
amazon_google	Gemini 3.8 Flash	0.930	0.928	0.863	0.979	67.152	0.923	0.725	0.561	0.926	0.102
amazon_google	GPT-5.4-mini	0.930	0.942	0.876	0.990	67.152	0.921	0.762	0.590	0.962	0.102

amazon_google	GPT-5.5	0.940	0.916	0.840	0.981	67.152	0.936	0.704	0.529	0.917	0.102
amazon_google	gpt-oss-120b	0.860	0.926	0.865	0.975	67.152	0.829	0.676	0.502	0.855	0.102
amazon_google	Grok 4	0.940	0.944	0.886	0.994	67.152	0.933	0.773	0.606	0.968	0.102
amazon_google	Llama 3.3 70B	0.950	0.932	0.867	0.985	67.152	0.946	0.747	0.583	0.932	0.102
bird	Claude Opus 5	0.293	0.323	0.234	0.415	74.082	–	–	–	–	–
bird	Claude Sonnet 5	0.232	0.258	0.176	0.347	74.082	–	–	–	–	–
bird	DeepSeek-chat	0.187	0.203	0.129	0.285	74.082	–	–	–	–	–
bird	DeepSeek-R1	0.178	0.214	0.136	0.297	74.082	–	–	–	–	–
bird	Gemini 3.8 Flash	0.293	0.313	0.224	0.408	74.082	–	–	–	–	–
bird	GPT-5.4-mini	0.178	0.218	0.134	0.310	74.082	–	–	–	–	–
bird	GPT-5.5	0.185	0.204	0.122	0.291	74.082	–	–	–	–	–
bird	gpt-oss-120b	0.084	0.109	0.047	0.186	74.082	–	–	–	–	–
bird	Grok 4	0.122	0.144	0.077	0.223	74.082	–	–	–	–	–
bird	Llama 3.3 70B	0.122	0.144	0.081	0.217	74.082	–	–	–	–	–
dblp_acm	Claude Opus 5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	Claude Sonnet 5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	DeepSeek-chat	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	DeepSeek-R1	0.990	0.996	0.988	1.000	72.166	0.990	0.990	0.966	1.000	0.180
dblp_acm	Gemini 3.8 Flash	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	GPT-5.4-mini	0.990	0.996	0.987	1.000	72.166	0.990	0.990	0.964	1.000	0.180
dblp_acm	GPT-5.5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	gpt-oss-120b	0.910	0.967	0.943	0.987	72.166	0.899	0.899	0.819	0.958	0.180
dblp_acm	Grok 4	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	Llama 3.3 70B	0.970	0.952	0.893	1.000	72.166	0.970	0.882	0.752	1.000	0.180
dblp_scholar	Claude Opus 5	0.950	0.956	0.906	0.993	70.463	0.950	0.888	0.777	0.983	0.186
dblp_scholar	Claude Sonnet 5	0.970	0.976	0.941	1.000	70.463	0.970	0.937	0.836	1.000	0.186
dblp_scholar	DeepSeek-chat	0.960	0.959	0.911	0.996	70.463	0.961	0.898	0.793	0.991	0.186
dblp_scholar	DeepSeek-R1	0.950	0.969	0.930	0.993	70.463	0.949	0.917	0.811	0.990	0.186
dblp_scholar	Gemini 3.8 Flash	0.930	0.961	0.920	0.990	70.463	0.928	0.895	0.795	0.972	0.186
dblp_scholar	GPT-5.4-mini	0.940	0.965	0.922	0.992	70.463	0.939	0.906	0.788	0.980	0.186
dblp_scholar	GPT-5.5	0.960	0.959	0.907	0.996	70.463	0.961	0.898	0.778	0.991	0.186
dblp_scholar	gpt-oss-120b	0.910	0.967	0.943	0.987	70.463	0.903	0.903	0.838	0.962	0.186
dblp_scholar	Grok 4	0.950	0.969	0.930	0.996	70.463	0.949	0.917	0.822	0.990	0.186
dblp_scholar	Llama 3.3 70B	0.970	0.976	0.938	1.000	70.463	0.970	0.937	0.851	1.000	0.186
fetaqa	Claude Opus 5	0.401	0.401	0.373	0.429	100.000	–	–	–	–	–
fetaqa	Claude Sonnet 5	0.373	0.373	0.345	0.400	100.000	–	–	–	–	–
fetaqa	DeepSeek-chat	0.434	0.434	0.407	0.462	100.000	–	–	–	–	–
fetaqa	DeepSeek-R1	0.413	0.413	0.383	0.443	100.000	–	–	–	–	–
fetaqa	Gemini 3.8 Flash	0.445	0.445	0.421	0.469	100.000	–	–	–	–	–
fetaqa	GPT-5.4-mini	0.312	0.312	0.289	0.336	100.000	–	–	–	–	–
fetaqa	GPT-5.5	0.438	0.438	0.412	0.464	100.000	–	–	–	–	–
fetaqa	gpt-oss-120b	0.322	0.322	0.304	0.342	100.000	–	–	–	–	–

fetaqa	Grok 4	0.428	0.428	0.403	0.453	100.000	–	–	–	–	–
fetaqa	Llama 3.3 70B	0.253	0.253	0.225	0.279	100.000	–	–	–	–	–
fintagging	Claude Opus 5	0.909	0.932	0.879	0.976	51.769	–	–	–	–	–
fintagging	Claude Sonnet 5	0.898	0.897	0.838	0.952	51.769	–	–	–	–	–
fintagging	DeepSeek-chat	0.858	0.847	0.779	0.917	51.769	–	–	–	–	–
fintagging	DeepSeek-R1	0.840	0.855	0.768	0.926	51.769	–	–	–	–	–
fintagging	Gemini 3.8 Flash	0.897	0.908	0.848	0.958	51.769	–	–	–	–	–
fintagging	GPT-5.4-mini	0.892	0.863	0.792	0.928	51.769	–	–	–	–	–
fintagging	GPT-5.5	0.893	0.899	0.830	0.957	51.769	–	–	–	–	–
fintagging	gpt-oss-120b	0.863	0.847	0.780	0.913	51.769	–	–	–	–	–
fintagging	Grok 4	0.887	0.876	0.805	0.939	51.769	–	–	–	–	–
fintagging	Llama 3.3 70B	0.858	0.836	0.770	0.902	51.769	–	–	–	–	–
hitab	Claude Opus 5	0.781	0.790	0.719	0.863	94.537	–	–	–	–	–
hitab	Claude Sonnet 5	0.763	0.758	0.678	0.839	94.537	–	–	–	–	–
hitab	DeepSeek-chat	0.814	0.822	0.754	0.888	94.537	–	–	–	–	–
hitab	DeepSeek-R1	0.771	0.771	0.698	0.849	94.537	–	–	–	–	–
hitab	Gemini 3.8 Flash	0.748	0.757	0.678	0.831	94.537	–	–	–	–	–
hitab	GPT-5.4-mini	0.784	0.783	0.707	0.862	94.537	–	–	–	–	–
hitab	GPT-5.5	0.757	0.764	0.684	0.838	94.537	–	–	–	–	–
hitab	gpt-oss-120b	0.750	0.752	0.674	0.834	94.537	–	–	–	–	–
hitab	Grok 4	0.746	0.749	0.662	0.828	94.537	–	–	–	–	–
hitab	Llama 3.3 70B	0.784	0.783	0.706	0.853	94.537	–	–	–	–	–
machamp	Claude Opus 5	0.900	0.912	0.833	0.970	32.491	0.904	0.774	0.623	0.920	0.175
machamp	Claude Sonnet 5	0.900	0.923	0.844	0.971	32.491	0.902	0.793	0.645	0.922	0.175
machamp	DeepSeek-chat	0.890	0.892	0.801	0.959	32.491	0.897	0.742	0.572	0.897	0.175
machamp	DeepSeek-R1	0.900	0.897	0.807	0.959	32.491	0.906	0.751	0.593	0.897	0.175
machamp	Gemini 3.8 Flash	0.900	0.912	0.827	0.967	32.491	0.904	0.774	0.614	0.914	0.175
machamp	GPT-5.4-mini	0.890	0.893	0.805	0.956	32.491	0.895	0.739	0.574	0.897	0.175
machamp	GPT-5.5	0.900	0.897	0.805	0.963	32.491	0.906	0.751	0.593	0.904	0.175
machamp	gpt-oss-120b	0.830	0.794	0.642	0.902	32.491	0.835	0.558	0.372	0.774	0.175
machamp	Grok 4	0.890	0.896	0.812	0.960	32.491	0.895	0.747	0.594	0.887	0.175
machamp	Llama 3.3 70B	0.870	0.880	0.783	0.951	32.491	0.874	0.698	0.541	0.852	0.175
magneto_gdc	Claude Opus 5	0.971	0.974	0.946	0.995	93.027	–	–	–	–	–
magneto_gdc	Claude Sonnet 5	0.955	0.955	0.916	0.986	93.027	–	–	–	–	–
magneto_gdc	DeepSeek-chat	0.953	0.956	0.920	0.986	93.027	–	–	–	–	–
magneto_gdc	DeepSeek-R1	0.970	0.972	0.941	0.994	93.027	–	–	–	–	–
magneto_gdc	Gemini 3.8 Flash	0.963	0.967	0.938	0.992	93.027	–	–	–	–	–
magneto_gdc	GPT-5.4-mini	0.912	0.914	0.862	0.961	93.027	–	–	–	–	–
magneto_gdc	GPT-5.5	0.967	0.972	0.940	0.994	93.027	–	–	–	–	–
magneto_gdc	gpt-oss-120b	0.961	0.965	0.928	0.993	93.027	–	–	–	–	–
magneto_gdc	Grok 4	0.939	0.944	0.905	0.978	93.027	–	–	–	–	–
magneto_gdc	Llama 3.3 70B	0.753	0.759	0.691	0.815	93.027	–	–	–	–	–

mmtu	Claude Opus 5	0.882	0.846	0.759	0.927	86.538	–	–	–	–	–
mmtu	Claude Sonnet 5	0.872	0.838	0.756	0.916	86.538	–	–	–	–	–
mmtu	DeepSeek-chat	0.865	0.831	0.748	0.911	86.538	–	–	–	–	–
mmtu	DeepSeek-R1	0.822	0.770	0.669	0.865	86.538	–	–	–	–	–
mmtu	Gemini 3.8 Flash	0.894	0.856	0.763	0.929	86.538	–	–	–	–	–
mmtu	GPT-5.4-mini	0.882	0.851	0.768	0.923	86.538	–	–	–	–	–
mmtu	GPT-5.5	0.917	0.882	0.800	0.957	86.538	–	–	–	–	–
mmtu	gpt-oss-120b	0.830	0.774	0.680	0.862	86.538	–	–	–	–	–
mmtu	Grok 4	0.901	0.869	0.786	0.941	86.538	–	–	–	–	–
mmtu	Llama 3.3 70B	0.807	0.773	0.684	0.854	86.538	–	–	–	–	–
officeqa	Claude Opus 5	0.187	0.187	0.146	0.235	100.000	–	–	–	–	–
officeqa	Claude Sonnet 5	0.170	0.170	0.128	0.217	100.000	–	–	–	–	–
officeqa	DeepSeek-chat	0.075	0.075	0.040	0.118	100.000	–	–	–	–	–
officeqa	DeepSeek-R1	0.083	0.083	0.053	0.119	100.000	–	–	–	–	–
officeqa	Gemini 3.8 Flash	0.194	0.194	0.160	0.235	100.000	–	–	–	–	–
officeqa	GPT-5.4-mini	0.107	0.107	0.082	0.137	100.000	–	–	–	–	–
officeqa	GPT-5.5	0.168	0.168	0.134	0.206	100.000	–	–	–	–	–
officeqa	gpt-oss-120b	0.079	0.079	0.052	0.114	100.000	–	–	–	–	–
officeqa	Grok 4	0.129	0.129	0.095	0.173	100.000	–	–	–	–	–
officeqa	Llama 3.3 70B	0.097	0.097	0.073	0.127	100.000	–	–	–	–	–
officeqa_pro_v2	Claude Opus 5	0.250	0.250	0.188	0.314	90.000	–	–	–	–	–
officeqa_pro_v2	Claude Sonnet 5	0.133	0.133	0.089	0.182	90.000	–	–	–	–	–
officeqa_pro_v2	DeepSeek-chat	0.121	0.121	0.090	0.161	90.000	–	–	–	–	–
officeqa_pro_v2	DeepSeek-R1	0.036	0.036	0.021	0.054	90.000	–	–	–	–	–
officeqa_pro_v2	Gemini 3.8 Flash	0.198	0.198	0.152	0.246	90.000	–	–	–	–	–
officeqa_pro_v2	GPT-5.4-mini	0.123	0.123	0.094	0.155	90.000	–	–	–	–	–
officeqa_pro_v2	GPT-5.5	0.188	0.188	0.127	0.254	90.000	–	–	–	–	–
officeqa_pro_v2	gpt-oss-120b	0.104	0.104	0.068	0.148	90.000	–	–	–	–	–
officeqa_pro_v2	Grok 4	0.119	0.119	0.083	0.164	90.000	–	–	–	–	–
officeqa_pro_v2	Llama 3.3 70B	0.115	0.115	0.082	0.152	90.000	–	–	–	–	–
opensanctions_pairs	Claude Opus 5	0.940	0.906	0.827	0.969	76.249	0.935	0.935	0.874	0.979	0.769
opensanctions_pairs	Claude Sonnet 5	0.730	0.587	0.478	0.704	76.249	0.630	0.636	0.496	0.761	0.769
opensanctions_pairs	DeepSeek-chat	0.760	0.635	0.521	0.753	76.249	0.684	0.691	0.556	0.792	0.769
opensanctions_pairs	DeepSeek-R1	0.850	0.765	0.655	0.860	76.249	0.819	0.819	0.716	0.905	0.769
opensanctions_pairs	Gemini 3.8 Flash	0.860	0.780	0.669	0.881	76.249	0.833	0.833	0.734	0.909	0.769
opensanctions_pairs	GPT-5.4-mini	0.900	0.865	0.782	0.942	76.249	0.891	0.905	0.837	0.963	0.769
opensanctions_pairs	GPT-5.5	0.750	0.608	0.497	0.724	76.249	0.658	0.658	0.520	0.776	0.769
opensanctions_pairs	gpt-oss-120b	0.780	0.666	0.561	0.770	76.249	0.718	0.725	0.594	0.831	0.769
opensanctions_pairs	Grok 4	0.850	0.776	0.677	0.868	76.249	0.824	0.830	0.733	0.914	0.769
opensanctions_pairs	Llama 3.3 70B	0.790	0.682	0.571	0.789	76.249	0.734	0.741	0.618	0.834	0.769
papadakis_dn	Claude Opus 5	0.920	0.955	0.892	0.996	32.601	0.789	0.322	0.130	0.840	0.018

papadakis_dn	Claude Sonnet 5	0.950	0.994	0.987	0.999	32.601	0.857	0.786	0.569	0.947	0.018
papadakis_dn	DeepSeek-chat	0.940	0.993	0.985	0.998	32.601	0.833	0.757	0.520	0.929	0.018
papadakis_dn	DeepSeek-R1	0.940	0.993	0.985	0.998	32.601	0.833	0.757	0.533	0.927	0.018
papadakis_dn	Gemini 3.8 Flash	0.940	0.992	0.984	0.998	32.601	0.824	0.702	0.459	0.889	0.018
papadakis_dn	GPT-5.4-mini	0.950	0.994	0.987	0.999	32.601	0.857	0.786	0.535	0.936	0.018
papadakis_dn	GPT-5.5	0.950	0.994	0.986	0.999	32.601	0.857	0.786	0.573	0.942	0.018
papadakis_dn	gpt-oss-120b	0.930	0.992	0.983	0.997	32.601	0.788	0.682	0.425	0.881	0.018
papadakis_dn	Grok 4	0.950	0.994	0.987	0.999	32.601	0.857	0.786	0.547	0.939	0.018
papadakis_dn	Llama 3.3 70B	0.940	0.975	0.925	0.998	32.601	0.833	0.463	0.180	0.916	0.018
realhitbench	Claude Opus 5	0.743	0.745	0.668	0.818	98.080	–	–	–	–	–
realhitbench	Claude Sonnet 5	0.681	0.684	0.606	0.766	98.080	–	–	–	–	–
realhitbench	DeepSeek-chat	0.662	0.656	0.569	0.735	98.080	–	–	–	–	–
realhitbench	DeepSeek-R1	0.659	0.662	0.581	0.745	98.080	–	–	–	–	–
realhitbench	Gemini 3.8 Flash	0.746	0.747	0.672	0.818	98.080	–	–	–	–	–
realhitbench	GPT-5.4-mini	0.661	0.658	0.570	0.746	98.080	–	–	–	–	–
realhitbench	GPT-5.5	0.724	0.721	0.642	0.797	98.080	–	–	–	–	–
realhitbench	gpt-oss-120b	0.603	0.602	0.514	0.686	98.080	–	–	–	–	–
realhitbench	Grok 4	0.640	0.638	0.552	0.725	98.080	–	–	–	–	–
realhitbench	Llama 3.3 70B	0.611	0.616	0.538	0.694	98.080	–	–	–	–	–
smat	Claude Opus 5	0.933	0.931	0.883	0.968	98.100	–	–	–	–	–
smat	Claude Sonnet 5	0.938	0.936	0.893	0.973	98.100	–	–	–	–	–
smat	DeepSeek-chat	0.918	0.916	0.872	0.956	98.100	–	–	–	–	–
smat	DeepSeek-R1	0.913	0.909	0.860	0.952	98.100	–	–	–	–	–
smat	Gemini 3.8 Flash	0.921	0.919	0.872	0.961	98.100	–	–	–	–	–
smat	GPT-5.4-mini	0.956	0.958	0.923	0.987	98.100	–	–	–	–	–
smat	GPT-5.5	0.940	0.939	0.894	0.973	98.100	–	–	–	–	–
smat	gpt-oss-120b	0.916	0.913	0.862	0.957	98.100	–	–	–	–	–
smat	Grok 4	0.936	0.933	0.889	0.971	98.100	–	–	–	–	–
smat	Llama 3.3 70B	0.806	0.803	0.743	0.860	98.100	–	–	–	–	–
suc	Claude Opus 5	0.986	0.988	0.963	1.000	98.920	–	–	–	–	–
suc	Claude Sonnet 5	0.911	0.908	0.851	0.955	98.920	–	–	–	–	–
suc	DeepSeek-chat	0.858	0.858	0.796	0.914	98.920	–	–	–	–	–
suc	DeepSeek-R1	0.935	0.932	0.887	0.970	98.920	–	–	–	–	–
suc	Gemini 3.8 Flash	0.976	0.978	0.945	1.000	98.920	–	–	–	–	–
suc	GPT-5.4-mini	0.914	0.910	0.856	0.956	98.920	–	–	–	–	–
suc	GPT-5.5	0.943	0.938	0.895	0.976	98.920	–	–	–	–	–
suc	gpt-oss-120b	0.932	0.930	0.881	0.970	98.920	–	–	–	–	–
suc	Grok 4	0.943	0.938	0.894	0.977	98.920	–	–	–	–	–
suc	Llama 3.3 70B	0.740	0.738	0.662	0.808	98.920	–	–	–	–	–
tab_fact	Claude Opus 5	0.660	0.667	0.571	0.760	99.360	0.673	0.696	0.589	0.785	0.500
tab_fact	Claude Sonnet 5	0.740	0.733	0.646	0.818	99.360	0.698	0.710	0.585	0.809	0.500

tab_fact	DeepSeek-chat	0.840	0.826	0.749	0.902	99.360	0.789	0.789	0.676	0.874	0.500
tab_fact	DeepSeek-R1	0.760	0.754	0.669	0.842	99.360	0.721	0.732	0.625	0.825	0.500
tab_fact	Gemini 3.8 Flash	0.860	0.848	0.781	0.924	99.360	0.821	0.821	0.709	0.909	0.500
tab_fact	GPT-5.4-mini	0.820	0.808	0.723	0.883	99.360	0.769	0.772	0.652	0.867	0.500
tab_fact	GPT-5.5	0.840	0.829	0.745	0.902	99.360	0.800	0.803	0.700	0.892	0.500
tab_fact	gpt-oss-120b	0.830	0.815	0.731	0.891	99.360	0.773	0.773	0.656	0.880	0.500
tab_fact	Grok 4	0.850	0.839	0.756	0.912	99.360	0.810	0.812	0.700	0.897	0.500
tab_fact	Llama 3.3 70B	0.780	0.769	0.690	0.849	99.360	0.725	0.732	0.605	0.835	0.500
tabis	Claude Opus 5	0.960	0.964	0.915	0.994	77.069	–	–	–	–	–
tabis	Claude Sonnet 5	0.920	0.917	0.853	0.970	77.069	–	–	–	–	–
tabis	DeepSeek-chat	0.770	0.710	0.609	0.817	77.069	–	–	–	–	–
tabis	DeepSeek-R1	0.970	0.982	0.958	1.000	77.069	–	–	–	–	–
tabis	Gemini 3.8 Flash	0.950	0.970	0.938	0.994	77.069	–	–	–	–	–
tabis	GPT-5.4-mini	0.900	0.905	0.831	0.960	77.069	–	–	–	–	–
tabis	GPT-5.5	0.960	0.976	0.947	0.994	77.069	–	–	–	–	–
tabis	gpt-oss-120b	0.920	0.951	0.914	0.982	77.069	–	–	–	–	–
tabis	Grok 4	0.980	0.988	0.970	1.000	77.069	–	–	–	–	–
tabis	Llama 3.3 70B	0.820	0.821	0.735	0.908	77.069	–	–	–	–	–
tablebench	Claude Opus 5	0.846	0.846	0.776	0.911	89.187	–	–	–	–	–
tablebench	Claude Sonnet 5	0.877	0.875	0.811	0.928	89.187	–	–	–	–	–
tablebench	DeepSeek-chat	0.833	0.844	0.776	0.905	89.187	–	–	–	–	–
tablebench	DeepSeek-R1	0.783	0.793	0.720	0.863	89.187	–	–	–	–	–
tablebench	Gemini 3.8 Flash	0.835	0.833	0.767	0.902	89.187	–	–	–	–	–
tablebench	GPT-5.4-mini	0.841	0.835	0.765	0.902	89.187	–	–	–	–	–
tablebench	GPT-5.5	0.852	0.850	0.778	0.912	89.187	–	–	–	–	–
tablebench	gpt-oss-120b	0.813	0.808	0.736	0.876	89.187	–	–	–	–	–
tablebench	Grok 4	0.839	0.837	0.764	0.908	89.187	–	–	–	–	–
tablebench	Llama 3.3 70B	0.660	0.641	0.558	0.720	89.187	–	–	–	–	–
tableeval	Claude Opus 5	0.765	0.784	0.684	0.873	69.689	–	–	–	–	–
tableeval	Claude Sonnet 5	0.733	0.764	0.668	0.851	69.689	–	–	–	–	–
tableeval	DeepSeek-chat	0.711	0.708	0.608	0.802	69.689	–	–	–	–	–
tableeval	DeepSeek-R1	0.756	0.757	0.662	0.852	69.689	–	–	–	–	–
tableeval	Gemini 3.8 Flash	0.791	0.805	0.714	0.889	69.689	–	–	–	–	–
tableeval	GPT-5.4-mini	0.742	0.751	0.648	0.838	69.689	–	–	–	–	–
tableeval	GPT-5.5	0.784	0.794	0.690	0.882	69.689	–	–	–	–	–
tableeval	gpt-oss-120b	0.773	0.774	0.683	0.864	69.689	–	–	–	–	–
tableeval	Grok 4	0.802	0.785	0.677	0.881	69.689	–	–	–	–	–
tableeval	Llama 3.3 70B	0.636	0.618	0.514	0.718	69.689	–	–	–	–	–
tat_qa	Claude Opus 5	0.861	0.846	0.773	0.915	60.725	–	–	–	–	–
tat_qa	Claude Sonnet 5	0.845	0.845	0.774	0.909	60.725	–	–	–	–	–
tat_qa	DeepSeek-chat	0.806	0.774	0.684	0.859	60.725	–	–	–	–	–

tat_qa	DeepSeek-R1	0.806	0.763	0.673	0.852	60.725	–	–	–	–	–
tat_qa	Gemini 3.8 Flash	0.829	0.815	0.735	0.889	60.725	–	–	–	–	–
tat_qa	GPT-5.4-mini	0.789	0.719	0.627	0.811	60.725	–	–	–	–	–
tat_qa	GPT-5.5	0.789	0.748	0.669	0.827	60.725	–	–	–	–	–
tat_qa	gpt-oss-120b	0.765	0.684	0.588	0.775	60.725	–	–	–	–	–
tat_qa	Grok 4	0.791	0.710	0.612	0.804	60.725	–	–	–	–	–
tat_qa	Llama 3.3 70B	0.752	0.679	0.573	0.780	60.725	–	–	–	–	–
tpcdi_cells	Claude Opus 5	0.462	0.465	0.377	0.557	95.091	–	–	–	–	–
tpcdi_cells	Claude Sonnet 5	0.462	0.465	0.378	0.563	95.091	–	–	–	–	–
tpcdi_cells	DeepSeek-chat	0.462	0.465	0.377	0.558	95.091	–	–	–	–	–
tpcdi_cells	DeepSeek-R1	0.462	0.466	0.377	0.557	95.091	–	–	–	–	–
tpcdi_cells	Gemini 3.8 Flash	0.462	0.465	0.376	0.555	95.091	–	–	–	–	–
tpcdi_cells	GPT-5.4-mini	0.455	0.458	0.362	0.537	95.091	–	–	–	–	–
tpcdi_cells	GPT-5.5	0.462	0.466	0.370	0.552	95.091	–	–	–	–	–
tpcdi_cells	gpt-oss-120b	0.442	0.453	0.360	0.550	95.091	–	–	–	–	–
tpcdi_cells	Grok 4	0.447	0.456	0.370	0.545	95.091	–	–	–	–	–
tpcdi_cells	Llama 3.3 70B	0.442	0.453	0.368	0.542	95.091	–	–	–	–	–
valentine	Claude Opus 5	1.000	1.000	1.000	1.000	81.736	–	–	–	–	–
valentine	Claude Sonnet 5	0.994	0.989	0.960	1.000	81.736	–	–	–	–	–
valentine	DeepSeek-chat	0.979	0.967	0.928	1.000	81.736	–	–	–	–	–
valentine	DeepSeek-R1	0.990	0.986	0.956	1.000	81.736	–	–	–	–	–
valentine	Gemini 3.8 Flash	0.970	0.988	0.970	1.000	81.736	–	–	–	–	–
valentine	GPT-5.4-mini	0.985	0.978	0.943	1.000	81.736	–	–	–	–	–
valentine	GPT-5.5	1.000	1.000	1.000	1.000	81.736	–	–	–	–	–
valentine	gpt-oss-120b	0.979	0.971	0.928	1.000	81.736	–	–	–	–	–
valentine	Grok 4	0.973	0.957	0.912	0.992	81.736	–	–	–	–	–
valentine	Llama 3.3 70B	0.970	0.958	0.913	0.991	81.736	–	–	–	–	–
walmart_amazon	Claude Opus 5	0.980	0.982	0.948	1.000	70.807	0.976	0.912	0.767	1.000	0.094
walmart_amazon	Claude Sonnet 5	0.950	0.962	0.914	0.997	70.807	0.938	0.823	0.645	0.980	0.094
walmart_amazon	DeepSeek-chat	0.950	0.962	0.912	0.998	70.807	0.938	0.823	0.657	0.977	0.094
walmart_amazon	DeepSeek-R1	0.960	0.978	0.939	0.998	70.807	0.950	0.887	0.737	0.989	0.094
walmart_amazon	Gemini 3.8 Flash	0.990	0.998	0.993	1.000	70.807	0.988	0.988	0.959	1.000	0.094
walmart_amazon	GPT-5.4-mini	0.950	0.975	0.938	0.998	70.807	0.937	0.874	0.722	0.987	0.094
walmart_amazon	GPT-5.5	0.970	0.967	0.917	1.000	70.807	0.964	0.848	0.671	1.000	0.094
walmart_amazon	gpt-oss-120b	0.940	0.973	0.933	0.995	70.807	0.923	0.860	0.698	0.974	0.094
walmart_amazon	Grok 4	0.960	0.978	0.942	0.998	70.807	0.950	0.887	0.720	0.989	0.094
walmart_amazon	Llama 3.3 70B	0.930	0.971	0.932	0.994	70.807	0.909	0.847	0.681	0.965	0.094
wdc_lspc	Claude Opus 5	0.990	0.996	0.987	1.000	75.082	0.991	0.992	0.973	1.000	0.264
wdc_lspc	Claude Sonnet 5	0.980	0.992	0.978	1.000	75.082	0.981	0.984	0.958	1.000	0.264
wdc_lspc	DeepSeek-chat	0.960	0.940	0.868	0.994	75.082	0.964	0.895	0.777	0.990	0.264
wdc_lspc	DeepSeek-R1	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	Gemini 3.8 Flash	0.980	0.992	0.979	1.000	75.082	0.981	0.984	0.956	1.000	0.264

wdc_lspc	GPT-5.4-mini	0.970	0.986	0.967	1.000	75.082	0.971	0.973	0.937	1.000	0.264
wdc_lspc	GPT-5.5	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	gpt-oss-120b	0.980	0.990	0.974	1.000	75.082	0.981	0.981	0.951	1.000	0.264
wdc_lspc	Grok 4	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	Llama 3.3 70B	0.960	0.940	0.872	0.994	75.082	0.964	0.895	0.781	0.990	0.264
wdc_products	Claude Opus 5	0.980	0.990	0.974	1.000	77.046	0.981	0.982	0.953	1.000	0.280
wdc_products	Claude Sonnet 5	0.970	0.986	0.967	1.000	77.046	0.971	0.974	0.936	1.000	0.280
wdc_products	DeepSeek-chat	0.970	0.977	0.944	1.000	77.046	0.972	0.958	0.895	1.000	0.280
wdc_products	DeepSeek-R1	0.980	0.990	0.974	1.000	77.046	0.981	0.982	0.953	1.000	0.280
wdc_products	Gemini 3.8 Flash	0.990	0.995	0.983	1.000	77.046	0.991	0.991	0.968	1.000	0.280
wdc_products	GPT-5.4-mini	0.970	0.986	0.966	1.000	77.046	0.971	0.974	0.940	1.000	0.280
wdc_products	GPT-5.5	0.980	0.979	0.940	1.000	77.046	0.981	0.964	0.902	1.000	0.280
wdc_products	gpt-oss-120b	0.980	0.979	0.945	1.000	77.046	0.981	0.964	0.898	1.000	0.280
wdc_products	Grok 4	0.990	0.995	0.983	1.000	77.046	0.991	0.991	0.969	1.000	0.280
wdc_products	Llama 3.3 70B	0.970	0.977	0.940	1.000	77.046	0.972	0.958	0.901	1.000	0.280
wikitablequestions	Claude Opus 5	0.879	0.878	0.815	0.933	97.686	–	–	–	–	–
wikitablequestions	Claude Sonnet 5	0.863	0.863	0.795	0.922	97.686	–	–	–	–	–
wikitablequestions	DeepSeek-chat	0.843	0.842	0.767	0.906	97.686	–	–	–	–	–
wikitablequestions	DeepSeek-R1	0.846	0.844	0.778	0.908	97.686	–	–	–	–	–
wikitablequestions	Gemini 3.8 Flash	0.881	0.881	0.820	0.934	97.686	–	–	–	–	–
wikitablequestions	GPT-5.4-mini	0.842	0.839	0.770	0.901	97.686	–	–	–	–	–
wikitablequestions	GPT-5.5	0.865	0.862	0.799	0.922	97.686	–	–	–	–	–
wikitablequestions	gpt-oss-120b	0.824	0.821	0.753	0.887	97.686	–	–	–	–	–
wikitablequestions	Grok 4	0.867	0.864	0.803	0.923	97.686	–	–	–	–	–
wikitablequestions	Llama 3.3 70B	0.749	0.747	0.672	0.823	97.686	–	–	–	–	–

Table 33: Cross-dataset pairwise tests under rule best-rho: Friedman $\chi^2 = 73.6$ ($p = 3.0\text{e-}12$), Iman–Davenport $F = 10.7$, Nemenyi CD = 2.43, Kendall's $W = 0.256$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.1613	21	8	2	0.0794	1.0000	no	no
Claude Opus 5	DeepSeek-chat	-2.2581	21	8	2	0.0187	0.4666	no	no
Claude Opus 5	DeepSeek-R1	-1.8065	19	9	3	0.0229	0.5275	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.4677	18	10	3	0.2513	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.0000	23	8	0	0.0187	0.4666	no	no
Claude Opus 5	GPT-5.5	0.2097	13	11	7	0.5300	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.0000	26	4	1	0.0004	0.0144	yes	yes
Claude Opus 5	Grok 4	-1.5806	18	10	3	0.1017	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-4.3548	24	7	0	0.0006	0.0219	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.0968	20	7	4	0.0386	0.7163	no	no
Claude Sonnet 5	DeepSeek-R1	-0.6452	15	14	2	0.4100	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.6935	11	17	3	0.0997	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-0.8387	19	9	3	0.1262	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.3710	9	19	3	0.0303	0.6502	no	no
Claude Sonnet 5	gpt-oss-120b	-2.8387	21	8	2	0.0044	0.1305	yes	no
Claude Sonnet 5	Grok 4	-0.4194	19	10	2	0.4048	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.1935	24	5	2	0.0002	0.0089	yes	yes
DeepSeek-chat	DeepSeek-R1	0.4516	15	15	1	0.8520	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.7903	7	21	3	0.0091	0.2555	no	no
DeepSeek-chat	GPT-5.4-mini	0.2581	12	15	4	0.3777	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.4677	5	22	4	0.0016	0.0527	yes	no
DeepSeek-chat	gpt-oss-120b	-1.7419	20	9	2	0.0334	0.6689	no	no
DeepSeek-chat	Grok 4	0.6774	13	15	3	0.3993	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.0968	23	4	4	0.0004	0.0145	no	yes
DeepSeek-R1	Gemini 3.8 Flash	1.3387	10	19	2	0.0377	0.7163	no	no
DeepSeek-R1	GPT-5.4-mini	-0.1935	16	14	1	0.6804	1.0000	no	no
DeepSeek-R1	GPT-5.5	2.0161	6	21	4	0.0080	0.2324	no	no
DeepSeek-R1	gpt-oss-120b	-2.1935	22	8	1	0.0115	0.3097	no	no
DeepSeek-R1	Grok 4	0.2258	13	14	4	0.4921	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-2.5484	22	6	3	0.0007	0.0254	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.5323	20	10	1	0.0296	0.6502	no	no
Gemini 3.8 Flash	GPT-5.5	0.6774	13	16	2	0.8600	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.5323	27	3	1	0.0000	0.0021	yes	yes
Gemini 3.8 Flash	Grok 4	-1.1129	17	12	2	0.0697	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-3.8871	26	4	1	0.0001	0.0043	yes	yes
GPT-5.4-mini	GPT-5.5	2.2097	6	24	1	0.0023	0.0714	no	no
GPT-5.4-mini	gpt-oss-120b	-2.0000	23	8	0	0.0021	0.0668	no	no
GPT-5.4-mini	Grok 4	0.4194	14	14	3	0.8446	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.3548	25	4	2	0.0001	0.0032	no	yes
GPT-5.5	gpt-oss-120b	-4.2097	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.7903	19	7	5	0.0131	0.3403	no	no
GPT-5.5	Llama 3.3 70B	-4.5645	27	4	0	0.0000	0.0013	yes	yes
gpt-oss-120b	Grok 4	2.4194	5	26	0	0.0001	0.0038	no	yes
gpt-oss-120b	Llama 3.3 70B	-0.3548	18	13	0	0.3081	1.0000	no	no
Grok 4	Llama 3.3 70B	-2.7742	25	5	1	0.0003	0.0109	yes	yes

Table 34: Cross-dataset pairwise tests under rule exact match: Friedman $\chi^2 = 82.5$ ($p = 5.3\text{e-}14$), Iman–Davenport $F = 12.6$, Nemenyi CD = 2.43, Kendall's $W = 0.279$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.3387	21	7	3	0.0266	0.5590	no	no
Claude Opus 5	DeepSeek-chat	-2.6290	22	6	3	0.0049	0.1564	yes	no
Claude Opus 5	DeepSeek-R1	-1.9032	18	6	7	0.0116	0.3133	no	no
Claude Opus 5	Gemini 3.8 Flash	-1.0161	19	8	4	0.0504	0.9126	no	no
Claude Opus 5	GPT-5.4-mini	-2.8065	25	5	1	0.0057	0.1703	yes	no
Claude Opus 5	GPT-5.5	-0.3065	13	9	9	0.3064	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.1935	26	3	2	0.0004	0.0127	yes	yes
Claude Opus 5	Grok 4	-1.7903	18	9	4	0.0559	0.9126	no	no
Claude Opus 5	Llama 3.3 70B	-4.9839	26	5	0	0.0001	0.0049	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.2903	19	5	7	0.0161	0.4034	no	no
Claude Sonnet 5	DeepSeek-R1	-0.5645	14	13	4	0.6305	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.3226	11	14	6	0.3217	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.4677	19	7	5	0.1052	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.0323	7	15	9	0.0397	0.7936	no	no
Claude Sonnet 5	gpt-oss-120b	-2.8548	22	7	2	0.0094	0.2627	yes	no
Claude Sonnet 5	Grok 4	-0.4516	18	9	4	0.4924	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.6452	25	4	2	0.0001	0.0038	yes	yes
DeepSeek-chat	DeepSeek-R1	0.7258	11	13	7	0.6016	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.6129	6	18	7	0.0166	0.4034	no	no
DeepSeek-chat	GPT-5.4-mini	-0.1774	12	14	5	0.4859	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.3226	6	19	6	0.0052	0.1604	no	no
DeepSeek-chat	gpt-oss-120b	-1.5645	19	9	3	0.0827	1.0000	no	no
DeepSeek-chat	Grok 4	0.8387	12	15	4	0.2936	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.3548	22	3	6	0.0001	0.0053	no	yes
DeepSeek-R1	Gemini 3.8 Flash	0.8871	10	17	4	0.0897	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-0.9032	17	12	2	0.9373	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.5968	7	19	5	0.0203	0.4672	no	no
DeepSeek-R1	gpt-oss-120b	-2.2903	21	5	5	0.0038	0.1258	no	no
DeepSeek-R1	Grok 4	0.1129	13	14	4	0.7011	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-3.0806	25	3	3	0.0000	0.0021	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.7903	21	9	1	0.0498	0.9126	no	no
Gemini 3.8 Flash	GPT-5.5	0.7097	10	15	6	0.5755	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.1774	24	2	5	0.0001	0.0033	yes	yes
Gemini 3.8 Flash	Grok 4	-0.7742	15	10	6	0.2506	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-3.9677	26	4	1	0.0001	0.0038	yes	yes
GPT-5.4-mini	GPT-5.5	2.5000	5	24	2	0.0064	0.1852	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.3871	20	10	1	0.0134	0.3494	no	no
GPT-5.4-mini	Grok 4	1.0161	9	16	6	0.3613	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.1774	23	4	4	0.0001	0.0044	no	yes
GPT-5.5	gpt-oss-120b	-3.8871	26	1	4	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.4839	16	6	9	0.0480	0.9126	no	no
GPT-5.5	Llama 3.3 70B	-4.6774	27	4	0	0.0000	0.0009	yes	yes
gpt-oss-120b	Grok 4	2.4032	3	24	4	0.0002	0.0054	no	yes
gpt-oss-120b	Llama 3.3 70B	-0.7903	20	8	3	0.0217	0.4779	no	no
Grok 4	Llama 3.3 70B	-3.1935	24	4	3	0.0001	0.0038	yes	yes

Table 35: Cross-dataset pairwise tests under rule knee: Friedman $\chi^2 = 77.5$ ($p = 5.1\text{e-}13$), Iman–Davenport $F = 11.5$, Nemenyi CD = 2.43, Kendall's $W = 0.269$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.2903	22	7	2	0.0319	0.6211	no	no
Claude Opus 5	DeepSeek-chat	-2.5806	21	8	2	0.0108	0.2817	yes	no
Claude Opus 5	DeepSeek-R1	-1.8387	19	9	3	0.0267	0.5694	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.5000	18	10	3	0.2355	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.2581	25	6	0	0.0068	0.1979	no	no
Claude Opus 5	GPT-5.5	-0.0161	13	11	7	0.5429	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.0968	26	4	1	0.0004	0.0140	yes	yes
Claude Opus 5	Grok 4	-1.5484	18	10	3	0.0640	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-4.5806	25	6	0	0.0002	0.0075	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.2903	20	7	4	0.0248	0.5694	no	no
Claude Sonnet 5	DeepSeek-R1	-0.5484	15	14	2	0.4213	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.7903	11	17	3	0.1313	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-0.9677	20	8	3	0.0653	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.2742	8	20	3	0.0248	0.5694	no	no
Claude Sonnet 5	gpt-oss-120b	-2.8065	22	7	2	0.0030	0.0926	yes	no
Claude Sonnet 5	Grok 4	-0.2581	18	11	2	0.4048	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.2903	25	4	2	0.0001	0.0032	yes	yes
DeepSeek-chat	DeepSeek-R1	0.7419	14	16	1	0.5892	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.0806	6	22	3	0.0068	0.1979	no	no
DeepSeek-chat	GPT-5.4-mini	0.3226	10	17	4	0.2026	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.5645	6	21	4	0.0019	0.0624	yes	no
DeepSeek-chat	gpt-oss-120b	-1.5161	20	9	2	0.0489	0.8305	no	no
DeepSeek-chat	Grok 4	1.0323	11	17	3	0.1825	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.0000	22	5	4	0.0006	0.0203	no	yes
DeepSeek-R1	Gemini 3.8 Flash	1.3387	10	19	2	0.0435	0.7823	no	no
DeepSeek-R1	GPT-5.4-mini	-0.4194	18	12	1	0.8908	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.8226	8	19	4	0.0175	0.4369	no	no
DeepSeek-R1	gpt-oss-120b	-2.2581	20	10	1	0.0103	0.2768	no	no
DeepSeek-R1	Grok 4	0.2903	13	14	4	0.4677	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-2.7419	22	6	3	0.0007	0.0237	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.7581	21	9	1	0.0311	0.6211	no	no
Gemini 3.8 Flash	GPT-5.5	0.4839	14	15	2	0.8293	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.5968	27	3	1	0.0001	0.0022	yes	yes
Gemini 3.8 Flash	Grok 4	-1.0484	17	12	2	0.1669	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.0806	26	4	1	0.0001	0.0027	yes	yes
GPT-5.4-mini	GPT-5.5	2.2419	6	24	1	0.0034	0.1016	no	no
GPT-5.4-mini	gpt-oss-120b	-1.8387	23	8	0	0.0014	0.0461	no	yes
GPT-5.4-mini	Grok 4	0.7097	13	15	3	0.6380	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.3226	26	3	2	0.0000	0.0015	no	yes
GPT-5.5	gpt-oss-120b	-4.0806	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.5323	19	7	5	0.0223	0.5352	no	no
GPT-5.5	Llama 3.3 70B	-4.5645	27	4	0	0.0000	0.0007	yes	yes
gpt-oss-120b	Grok 4	2.5484	4	27	0	0.0001	0.0031	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.4839	19	12	0	0.2724	1.0000	no	no
Grok 4	Llama 3.3 70B	-3.0323	25	5	1	0.0001	0.0055	yes	yes

Table 36: Cross-dataset pairwise tests under rule top1: Friedman $\chi^2 = 75.9$ ($p = 1.1\text{e-}12$), Iman–Davenport $F = 11.2$, Nemenyi CD = 2.43, Kendall's $W = 0.264$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.3387	22	7	2	0.0248	0.5885	no	no
Claude Opus 5	DeepSeek-chat	-2.7419	22	7	2	0.0061	0.1818	yes	no
Claude Opus 5	DeepSeek-R1	-1.6774	18	10	3	0.0241	0.5885	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.5161	18	10	3	0.1792	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.3226	24	7	0	0.0064	0.1818	no	no
Claude Opus 5	GPT-5.5	-0.1290	14	10	7	0.3667	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.0000	26	4	1	0.0003	0.0104	yes	yes
Claude Opus 5	Grok 4	-1.4032	18	10	3	0.0560	0.9839	no	no
Claude Opus 5	Llama 3.3 70B	-4.5806	25	6	0	0.0002	0.0075	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.4032	20	7	4	0.0235	0.5885	no	no
Claude Sonnet 5	DeepSeek-R1	-0.3387	15	14	2	0.5562	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.8226	10	17	4	0.1080	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-0.9839	21	7	3	0.0547	0.9839	no	no
Claude Sonnet 5	GPT-5.5	1.2097	8	20	3	0.0274	0.6025	no	no
Claude Sonnet 5	gpt-oss-120b	-2.6613	21	8	2	0.0063	0.1818	yes	no
Claude Sonnet 5	Grok 4	-0.0645	17	12	2	0.5176	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.2419	25	4	2	0.0001	0.0023	yes	yes
DeepSeek-chat	DeepSeek-R1	1.0645	11	19	1	0.2845	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.2258	6	22	3	0.0061	0.1818	no	no
DeepSeek-chat	GPT-5.4-mini	0.4194	10	17	4	0.2096	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.6129	7	20	4	0.0033	0.1079	yes	no
DeepSeek-chat	gpt-oss-120b	-1.2581	19	10	2	0.0900	1.0000	no	no
DeepSeek-chat	Grok 4	1.3387	11	17	3	0.1524	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-1.8387	21	6	4	0.0010	0.0325	no	yes
DeepSeek-R1	Gemini 3.8 Flash	1.1613	10	19	2	0.0598	0.9839	no	no
DeepSeek-R1	GPT-5.4-mini	-0.6452	18	12	1	0.7537	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.5484	8	19	4	0.0356	0.7474	no	no
DeepSeek-R1	gpt-oss-120b	-2.3226	22	8	1	0.0057	0.1774	no	no
DeepSeek-R1	Grok 4	0.2742	11	16	4	0.3987	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-2.9032	23	5	3	0.0004	0.0136	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.8065	21	9	1	0.0377	0.7542	no	no
Gemini 3.8 Flash	GPT-5.5	0.3871	14	15	2	0.8754	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.4839	26	4	1	0.0001	0.0029	yes	yes
Gemini 3.8 Flash	Grok 4	-0.8871	17	12	2	0.2355	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.0645	26	4	1	0.0001	0.0025	yes	yes
GPT-5.4-mini	GPT-5.5	2.1935	6	24	1	0.0049	0.1572	no	no
GPT-5.4-mini	gpt-oss-120b	-1.6774	22	9	0	0.0081	0.2116	no	no
GPT-5.4-mini	Grok 4	0.9194	12	16	3	0.5177	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.2581	26	3	2	0.0000	0.0013	no	yes
GPT-5.5	gpt-oss-120b	-3.8710	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.2742	18	9	4	0.0499	0.9476	no	no
GPT-5.5	Llama 3.3 70B	-4.4516	27	4	0	0.0000	0.0007	yes	yes
gpt-oss-120b	Grok 4	2.5968	4	27	0	0.0001	0.0036	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.5806	20	11	0	0.1313	1.0000	no	no
Grok 4	Llama 3.3 70B	-3.1774	24	6	1	0.0002	0.0069	yes	yes

Table 37: Cross-dataset pairwise tests under rule top2: Friedman $\chi^2 = 77.2$ ($p = 5.9\text{e-}13$), Iman–Davenport $F = 11.5$, Nemenyi CD = 2.43, Kendall's $W = 0.268$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.4194	22	7	2	0.0224	0.4708	no	no
Claude Opus 5	DeepSeek-chat	-2.3548	21	8	2	0.0151	0.3771	no	no
Claude Opus 5	DeepSeek-R1	-1.8387	20	8	3	0.0177	0.4063	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.6935	19	9	3	0.2060	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.7742	24	7	0	0.0042	0.1264	yes	no
Claude Opus 5	GPT-5.5	-0.0161	14	10	7	0.3879	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.0968	26	4	1	0.0004	0.0123	yes	yes
Claude Opus 5	Grok 4	-1.5484	18	10	3	0.1338	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-4.6129	25	6	0	0.0002	0.0083	yes	yes
Claude Sonnet 5	DeepSeek-chat	-0.9355	19	8	4	0.0899	1.0000	no	no
Claude Sonnet 5	DeepSeek-R1	-0.4194	15	14	2	0.7240	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.7258	12	16	3	0.1263	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.3548	20	8	3	0.0326	0.5538	no	no
Claude Sonnet 5	GPT-5.5	1.4032	8	20	3	0.0274	0.5204	no	no
Claude Sonnet 5	gpt-oss-120b	-2.6774	21	8	2	0.0089	0.2578	yes	no
Claude Sonnet 5	Grok 4	-0.1290	17	12	2	0.8216	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.1935	25	4	2	0.0002	0.0064	yes	yes
DeepSeek-chat	DeepSeek-R1	0.5161	14	16	1	0.9765	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.6613	6	22	3	0.0091	0.2578	no	no
DeepSeek-chat	GPT-5.4-mini	-0.4194	15	12	4	0.5305	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.3387	5	22	4	0.0033	0.1014	no	no
DeepSeek-chat	gpt-oss-120b	-1.7419	21	8	2	0.0163	0.3923	no	no
DeepSeek-chat	Grok 4	0.8065	13	15	3	0.4445	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.2581	23	4	4	0.0001	0.0058	no	yes
DeepSeek-R1	Gemini 3.8 Flash	1.1452	10	19	2	0.0230	0.4708	no	no
DeepSeek-R1	GPT-5.4-mini	-0.9355	20	10	1	0.2553	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.8226	7	20	4	0.0119	0.3103	no	no
DeepSeek-R1	gpt-oss-120b	-2.2581	22	8	1	0.0015	0.0480	no	yes
DeepSeek-R1	Grok 4	0.2903	12	15	4	0.5299	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-2.7742	23	5	3	0.0002	0.0082	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.0806	21	9	1	0.0108	0.2921	no	no
Gemini 3.8 Flash	GPT-5.5	0.6774	14	15	2	1.0000	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.4032	26	4	1	0.0001	0.0036	yes	yes
Gemini 3.8 Flash	Grok 4	-0.8548	17	12	2	0.1238	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-3.9194	25	5	1	0.0001	0.0052	yes	yes
GPT-5.4-mini	GPT-5.5	2.7581	5	25	1	0.0006	0.0206	yes	yes
GPT-5.4-mini	gpt-oss-120b	-1.3226	21	10	0	0.0296	0.5323	no	no
GPT-5.4-mini	Grok 4	1.2258	11	17	3	0.2026	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-1.8387	23	6	2	0.0001	0.0053	no	yes
GPT-5.5	gpt-oss-120b	-4.0806	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.5323	19	7	5	0.0212	0.4659	no	no
GPT-5.5	Llama 3.3 70B	-4.5968	27	4	0	0.0000	0.0008	yes	yes
gpt-oss-120b	Grok 4	2.5484	5	26	0	0.0001	0.0031	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.5161	19	12	0	0.1124	1.0000	no	no
Grok 4	Llama 3.3 70B	-3.0645	26	4	1	0.0002	0.0064	yes	yes

Table 38: Cross-dataset pairwise tests under rule top3: Friedman $\chi^2 = 77.2$ ($p = 5.9\text{e-}13$), Iman–Davenport $F = 11.5$, Nemenyi CD = 2.43, Kendall's $W = 0.268$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.4194	22	7	2	0.0224	0.4708	no	no
Claude Opus 5	DeepSeek-chat	-2.3548	21	8	2	0.0151	0.3771	no	no
Claude Opus 5	DeepSeek-R1	-1.8387	20	8	3	0.0167	0.3923	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.6935	19	9	3	0.2060	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.7742	24	7	0	0.0042	0.1264	yes	no
Claude Opus 5	GPT-5.5	-0.0161	14	10	7	0.3879	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.0968	26	4	1	0.0004	0.0132	yes	yes
Claude Opus 5	Grok 4	-1.5484	18	10	3	0.1338	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-4.6129	25	6	0	0.0002	0.0079	yes	yes
Claude Sonnet 5	DeepSeek-chat	-0.9355	19	8	4	0.0899	1.0000	no	no
Claude Sonnet 5	DeepSeek-R1	-0.4194	15	14	2	0.7240	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.7258	12	16	3	0.1313	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.3548	20	8	3	0.0326	0.5749	no	no
Claude Sonnet 5	GPT-5.5	1.4032	8	20	3	0.0303	0.5749	no	no
Claude Sonnet 5	gpt-oss-120b	-2.6774	21	8	2	0.0094	0.2647	yes	no
Claude Sonnet 5	Grok 4	-0.1290	17	12	2	0.8216	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.1935	25	4	2	0.0002	0.0059	yes	yes
DeepSeek-chat	DeepSeek-R1	0.5161	14	16	1	0.9765	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.6613	6	22	3	0.0091	0.2647	no	no
DeepSeek-chat	GPT-5.4-mini	-0.4194	15	12	4	0.5305	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.3387	5	22	4	0.0031	0.0952	no	no
DeepSeek-chat	gpt-oss-120b	-1.7419	21	8	2	0.0163	0.3923	no	no
DeepSeek-chat	Grok 4	0.8065	13	15	3	0.4445	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.2581	23	4	4	0.0001	0.0058	no	yes
DeepSeek-R1	Gemini 3.8 Flash	1.1452	10	19	2	0.0230	0.4708	no	no
DeepSeek-R1	GPT-5.4-mini	-0.9355	20	10	1	0.2553	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.8226	7	20	4	0.0119	0.3103	no	no
DeepSeek-R1	gpt-oss-120b	-2.2581	22	8	1	0.0016	0.0513	no	no
DeepSeek-R1	Grok 4	0.2903	12	15	4	0.5172	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-2.7742	23	5	3	0.0002	0.0080	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.0806	21	9	1	0.0108	0.2921	no	no
Gemini 3.8 Flash	GPT-5.5	0.6774	14	15	2	0.9844	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.4032	26	4	1	0.0001	0.0036	yes	yes
Gemini 3.8 Flash	Grok 4	-0.8548	17	12	2	0.1215	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-3.9194	25	5	1	0.0001	0.0052	yes	yes
GPT-5.4-mini	GPT-5.5	2.7581	5	25	1	0.0006	0.0206	yes	yes
GPT-5.4-mini	gpt-oss-120b	-1.3226	21	10	0	0.0311	0.5749	no	no
GPT-5.4-mini	Grok 4	1.2258	11	17	3	0.2026	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-1.8387	23	6	2	0.0001	0.0053	no	yes
GPT-5.5	gpt-oss-120b	-4.0806	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.5323	19	7	5	0.0212	0.4659	no	no
GPT-5.5	Llama 3.3 70B	-4.5968	27	4	0	0.0000	0.0008	yes	yes
gpt-oss-120b	Grok 4	2.5484	5	26	0	0.0001	0.0031	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.5161	19	12	0	0.1124	1.0000	no	no
Grok 4	Llama 3.3 70B	-3.0645	26	4	1	0.0002	0.0059	yes	yes

Table 39: Cross-dataset pairwise tests under rule top4: Friedman $\chi^2 = 77.5$ ($p = 5.1\text{e-}13$), Iman–Davenport $F = 11.5$, Nemenyi CD = 2.43, Kendall's $W = 0.269$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.2903	22	7	2	0.0319	0.6211	no	no
Claude Opus 5	DeepSeek-chat	-2.5806	21	8	2	0.0108	0.2817	yes	no
Claude Opus 5	DeepSeek-R1	-1.8387	19	9	3	0.0267	0.5694	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.5000	18	10	3	0.2355	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.2581	25	6	0	0.0068	0.1979	no	no
Claude Opus 5	GPT-5.5	-0.0161	13	11	7	0.5429	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.0968	26	4	1	0.0004	0.0140	yes	yes
Claude Opus 5	Grok 4	-1.5484	18	10	3	0.0640	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-4.5806	25	6	0	0.0002	0.0075	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.2903	20	7	4	0.0248	0.5694	no	no
Claude Sonnet 5	DeepSeek-R1	-0.5484	15	14	2	0.4213	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.7903	11	17	3	0.1313	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-0.9677	20	8	3	0.0653	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.2742	8	20	3	0.0248	0.5694	no	no
Claude Sonnet 5	gpt-oss-120b	-2.8065	22	7	2	0.0030	0.0926	yes	no
Claude Sonnet 5	Grok 4	-0.2581	18	11	2	0.4048	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.2903	25	4	2	0.0001	0.0032	yes	yes
DeepSeek-chat	DeepSeek-R1	0.7419	14	16	1	0.5892	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.0806	6	22	3	0.0068	0.1979	no	no
DeepSeek-chat	GPT-5.4-mini	0.3226	10	17	4	0.2026	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.5645	6	21	4	0.0019	0.0624	yes	no
DeepSeek-chat	gpt-oss-120b	-1.5161	20	9	2	0.0489	0.8305	no	no
DeepSeek-chat	Grok 4	1.0323	11	17	3	0.1825	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.0000	22	5	4	0.0006	0.0203	no	yes
DeepSeek-R1	Gemini 3.8 Flash	1.3387	10	19	2	0.0435	0.7823	no	no
DeepSeek-R1	GPT-5.4-mini	-0.4194	18	12	1	0.8908	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.8226	8	19	4	0.0175	0.4369	no	no
DeepSeek-R1	gpt-oss-120b	-2.2581	20	10	1	0.0103	0.2768	no	no
DeepSeek-R1	Grok 4	0.2903	13	14	4	0.4677	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-2.7419	22	6	3	0.0007	0.0237	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.7581	21	9	1	0.0311	0.6211	no	no
Gemini 3.8 Flash	GPT-5.5	0.4839	14	15	2	0.8293	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.5968	27	3	1	0.0001	0.0022	yes	yes
Gemini 3.8 Flash	Grok 4	-1.0484	17	12	2	0.1669	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.0806	26	4	1	0.0001	0.0027	yes	yes
GPT-5.4-mini	GPT-5.5	2.2419	6	24	1	0.0034	0.1016	no	no
GPT-5.4-mini	gpt-oss-120b	-1.8387	23	8	0	0.0014	0.0461	no	yes
GPT-5.4-mini	Grok 4	0.7097	13	15	3	0.6380	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.3226	26	3	2	0.0000	0.0015	no	yes
GPT-5.5	gpt-oss-120b	-4.0806	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.5323	19	7	5	0.0223	0.5352	no	no
GPT-5.5	Llama 3.3 70B	-4.5645	27	4	0	0.0000	0.0007	yes	yes
gpt-oss-120b	Grok 4	2.5484	4	27	0	0.0001	0.0031	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.4839	19	12	0	0.2724	1.0000	no	no
Grok 4	Llama 3.3 70B	-3.0323	25	5	1	0.0001	0.0055	yes	yes

Table 40: Cross-dataset pairwise tests under rule top5: Friedman $\chi^2 = 76.1$ ($p = 9.4\text{e-}13$), Iman–Davenport $F = 11.3$, Nemenyi CD = 2.43, Kendall's $W = 0.264$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.4032	22	7	2	0.0236	0.5899	no	no
Claude Opus 5	DeepSeek-chat	-2.6774	22	7	2	0.0061	0.1820	yes	no
Claude Opus 5	DeepSeek-R1	-1.5806	18	10	3	0.0267	0.6411	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.4839	17	11	3	0.1922	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.3548	24	7	0	0.0061	0.1820	no	no
Claude Opus 5	GPT-5.5	-0.1452	14	10	7	0.3563	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.0000	26	4	1	0.0003	0.0117	yes	yes
Claude Opus 5	Grok 4	-1.4516	19	9	3	0.0535	0.9099	no	no
Claude Opus 5	Llama 3.3 70B	-4.6129	25	6	0	0.0002	0.0082	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.2742	20	7	4	0.0288	0.6411	no	no
Claude Sonnet 5	DeepSeek-R1	-0.1774	15	14	2	0.6239	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.9194	10	17	4	0.1123	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-0.9516	20	8	3	0.0477	0.8832	no	no
Claude Sonnet 5	GPT-5.5	1.2581	8	20	3	0.0274	0.6411	no	no
Claude Sonnet 5	gpt-oss-120b	-2.5968	20	9	2	0.0084	0.2266	yes	no
Claude Sonnet 5	Grok 4	-0.0484	17	12	2	0.6102	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.2097	25	4	2	0.0001	0.0023	yes	yes
DeepSeek-chat	DeepSeek-R1	1.0968	11	19	1	0.3409	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.1935	6	22	3	0.0068	0.1911	no	no
DeepSeek-chat	GPT-5.4-mini	0.3226	11	16	4	0.3174	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.5323	7	20	4	0.0037	0.1150	yes	no
DeepSeek-chat	gpt-oss-120b	-1.3226	19	10	2	0.0794	1.0000	no	no
DeepSeek-chat	Grok 4	1.2258	12	16	3	0.1890	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-1.9355	21	6	4	0.0008	0.0282	no	yes
DeepSeek-R1	Gemini 3.8 Flash	1.0968	10	19	2	0.0625	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-0.7742	19	11	1	0.6519	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.4355	8	19	4	0.0356	0.7118	no	no
DeepSeek-R1	gpt-oss-120b	-2.4194	23	7	1	0.0029	0.0926	no	no
DeepSeek-R1	Grok 4	0.1290	12	15	4	0.4798	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-3.0323	23	5	3	0.0003	0.0112	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.8710	21	9	1	0.0281	0.6411	no	no
Gemini 3.8 Flash	GPT-5.5	0.3387	14	15	2	0.7242	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.5161	26	4	1	0.0001	0.0025	yes	yes
Gemini 3.8 Flash	Grok 4	-0.9677	17	12	2	0.1923	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.1290	26	4	1	0.0001	0.0027	yes	yes
GPT-5.4-mini	GPT-5.5	2.2097	6	24	1	0.0019	0.0624	no	no
GPT-5.4-mini	gpt-oss-120b	-1.6452	22	9	0	0.0108	0.2815	no	no
GPT-5.4-mini	Grok 4	0.9032	12	16	3	0.4216	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.2581	26	3	2	0.0000	0.0014	no	yes
GPT-5.5	gpt-oss-120b	-3.8548	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.3065	17	9	5	0.0465	0.8832	no	no
GPT-5.5	Llama 3.3 70B	-4.4677	27	4	0	0.0000	0.0007	yes	yes
gpt-oss-120b	Grok 4	2.5484	4	27	0	0.0001	0.0031	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.6129	20	11	0	0.1215	1.0000	no	no
Grok 4	Llama 3.3 70B	-3.1613	25	5	1	0.0002	0.0064	yes	yes

Table 41: Cross-dataset pairwise tests under rule v1: Friedman $\chi^2 = 76.0$ ($p = 1.0e\text{-}12$), Iman–Davenport $F = 11.2$, Nemenyi CD = 2.43, Kendall's $W = 0.264$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.2581	22	7	2	0.0303	0.6068	no	no
Claude Opus 5	DeepSeek-chat	-2.3548	21	8	2	0.0151	0.3620	no	no
Claude Opus 5	DeepSeek-R1	-1.8710	19	9	3	0.0207	0.4550	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.4677	17	11	3	0.2595	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.4194	25	6	0	0.0048	0.1382	no	no
Claude Opus 5	GPT-5.5	0.0161	13	11	7	0.5300	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.0000	26	4	1	0.0003	0.0121	yes	yes
Claude Opus 5	Grok 4	-1.4839	18	10	3	0.0900	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-4.5484	25	6	0	0.0002	0.0082	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.0968	20	7	4	0.0559	0.8948	no	no
Claude Sonnet 5	DeepSeek-R1	-0.6129	15	14	2	0.4680	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.7903	11	17	3	0.0957	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.1613	20	8	3	0.0499	0.8484	no	no
Claude Sonnet 5	GPT-5.5	1.2742	9	19	3	0.0334	0.6342	no	no
Claude Sonnet 5	gpt-oss-120b	-2.7419	21	8	2	0.0044	0.1306	yes	no
Claude Sonnet 5	Grok 4	-0.2258	18	11	2	0.6592	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.2903	25	4	2	0.0001	0.0037	yes	yes
DeepSeek-chat	DeepSeek-R1	0.4839	15	15	1	0.9608	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.8871	6	22	3	0.0068	0.1911	no	no
DeepSeek-chat	GPT-5.4-mini	-0.0645	13	14	4	0.6241	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.3710	6	21	4	0.0025	0.0810	no	no
DeepSeek-chat	gpt-oss-120b	-1.6452	20	9	2	0.0351	0.6342	no	no
DeepSeek-chat	Grok 4	0.8710	13	15	3	0.3270	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.1935	22	5	4	0.0004	0.0141	no	yes
DeepSeek-R1	Gemini 3.8 Flash	1.4032	10	19	2	0.0197	0.4520	no	no
DeepSeek-R1	GPT-5.4-mini	-0.5484	17	13	1	0.9063	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.8871	7	20	4	0.0090	0.2339	no	no
DeepSeek-R1	gpt-oss-120b	-2.1290	22	8	1	0.0082	0.2201	no	no
DeepSeek-R1	Grok 4	0.3871	12	15	4	0.4324	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-2.6774	22	6	3	0.0005	0.0153	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.9516	21	9	1	0.0135	0.3377	no	no
Gemini 3.8 Flash	GPT-5.5	0.4839	14	15	2	0.8909	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.5323	26	4	1	0.0001	0.0024	yes	yes
Gemini 3.8 Flash	Grok 4	-1.0161	17	12	2	0.1145	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.0806	26	4	1	0.0001	0.0034	yes	yes
GPT-5.4-mini	GPT-5.5	2.4355	5	25	1	0.0013	0.0417	yes	yes
GPT-5.4-mini	gpt-oss-120b	-1.5806	22	9	0	0.0042	0.1306	no	no
GPT-5.4-mini	Grok 4	0.9355	12	16	3	0.3468	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.1290	25	4	2	0.0001	0.0025	no	yes
GPT-5.5	gpt-oss-120b	-4.0161	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.5000	18	8	5	0.0212	0.4550	no	no
GPT-5.5	Llama 3.3 70B	-4.5645	27	4	0	0.0000	0.0007	yes	yes
gpt-oss-120b	Grok 4	2.5161	5	26	0	0.0001	0.0032	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.5484	19	12	0	0.1169	1.0000	no	no
Grok 4	Llama 3.3 70B	-3.0645	26	4	1	0.0001	0.0043	yes	yes

Table 42: The (J, ρ) Pareto front.

rank	w__f1	w__decay	w__precision	w__recall	J	rho	auc	wrong__mean	hedge__gap
1	0.000	0.700	0.000	0.300	0.421	0.845	0.994	0.037	-0.489
2	0.350	0.100	0.100	0.450	0.419	0.873	0.994	0.031	-0.402
4	0.050	0.600	0.250	0.100	0.419	0.877	0.994	0.036	-0.553
7	0.100	0.600	0.200	0.100	0.419	0.877	0.994	0.036	-0.548
19	0.100	0.550	0.200	0.150	0.418	0.879	0.994	0.036	-0.531
86	0.200	0.500	0.200	0.100	0.416	0.881	0.994	0.035	-0.538
240	0.150	0.450	0.250	0.150	0.414	0.882	0.994	0.035	-0.525
302	0.250	0.450	0.200	0.100	0.413	0.882	0.994	0.035	-0.533
1660	0.150	0.350	0.500	0.000	0.401	0.882	0.994	0.034	-0.576
1709	0.000	0.350	0.600	0.050	0.398	0.883	0.994	0.034	-0.573

Table 43: Mean score per dataset and model under rule best-rho.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	Claude Sonnet 5	Grok 4	GPT-5.4-mini	DeepSeek-chat	DeepSeek-R1	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.960	0.960	0.960	0.970	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.980	0.940	0.970	1.000	0.990	1.000	0.990
alaska_schema	0.916	0.946	0.925	0.928	0.926	0.957	0.953	0.947	0.931	0.820
amazon_google	0.930	0.940	0.940	0.960	0.940	0.930	0.960	0.950	0.860	0.950
bird	0.297	0.297	0.186	0.232	0.125	0.177	0.184	0.169	0.082	0.119
dblp_acm	1.000	1.000	1.000	1.000	1.000	0.990	1.000	0.990	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.970	0.950	0.940	0.960	0.950	0.910	0.970
fetaqa	0.445	0.390	0.450	0.431	0.429	0.455	0.450	0.444	0.283	0.443
fintagging	0.897	0.908	0.893	0.898	0.887	0.892	0.857	0.835	0.863	0.856
hitab	0.748	0.775	0.751	0.757	0.742	0.778	0.809	0.763	0.744	0.782
machamp	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.900	0.830	0.870
magneto_gdc	0.963	0.970	0.968	0.953	0.939	0.916	0.952	0.968	0.963	0.714
mmtu	0.899	0.882	0.915	0.874	0.902	0.885	0.866	0.823	0.830	0.802
officeqa	0.194	0.187	0.167	0.169	0.129	0.107	0.075	0.081	0.079	0.097
officeqa_pro_v2	0.198	0.253	0.188	0.133	0.119	0.123	0.121	0.030	0.103	0.114
opensanctions_pairs	0.860	0.940	0.750	0.730	0.850	0.900	0.760	0.850	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.950	0.950	0.950	0.940	0.940	0.930	0.940
realhitbench	0.752	0.744	0.728	0.676	0.643	0.662	0.663	0.653	0.611	0.621
smat	0.919	0.932	0.940	0.938	0.934	0.956	0.917	0.911	0.912	0.783
suc	0.976	0.986	0.943	0.911	0.943	0.914	0.858	0.937	0.932	0.741
tab_fact	0.860	0.660	0.840	0.740	0.850	0.820	0.840	0.760	0.830	0.780
tabis	0.950	0.960	0.960	0.920	0.980	0.900	0.770	0.970	0.920	0.820
tablebench	0.838	0.849	0.855	0.880	0.841	0.843	0.835	0.784	0.820	0.655
tableeval	0.792	0.767	0.786	0.732	0.802	0.742	0.704	0.752	0.774	0.632
tat_qa	0.821	0.855	0.786	0.835	0.787	0.790	0.802	0.806	0.760	0.758
tpcdi_cells	0.516	0.516	0.517	0.516	0.499	0.510	0.516	0.517	0.497	0.497
valentine	0.970	1.000	1.000	0.994	0.974	0.984	0.980	0.990	0.980	0.969
walmart_amazon	0.990	0.980	0.970	0.950	0.960	0.950	0.950	0.960	0.940	0.930
wdc_lspc	0.980	0.990	1.000	0.980	1.000	0.970	0.960	1.000	0.980	0.960
wdc_products	0.990	0.980	0.980	0.970	0.990	0.970	0.970	0.980	0.980	0.970
wikitablequestions	0.883	0.884	0.870	0.863	0.870	0.846	0.849	0.851	0.829	0.748

Table 44: Mean score per dataset and model under rule exact match.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	Claude Sonnet 5	Grok 4	GPT-5.4-mini	DeepSeek-chat	DeepSeek-R1	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.960	0.960	0.960	0.970	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.980	0.940	0.970	1.000	0.990	1.000	0.990
alaska_schema	0.892	0.925	0.903	0.914	0.903	0.946	0.925	0.925	0.903	0.731
amazon_google	0.930	0.940	0.940	0.960	0.940	0.930	0.960	0.950	0.860	0.950
bird	0.190	0.180	0.130	0.130	0.080	0.130	0.110	0.110	0.050	0.050
dblp_acm	1.000	1.000	1.000	1.000	1.000	0.990	1.000	0.990	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.970	0.950	0.940	0.960	0.950	0.910	0.970
fetaqa	0.010	0.020	0.010	0.010	0.010	0.000	0.010	0.020	0.000	0.000
fintagging	0.820	0.840	0.820	0.820	0.810	0.810	0.750	0.760	0.760	0.750
hitab	0.680	0.730	0.700	0.700	0.680	0.730	0.760	0.720	0.680	0.710
machamp	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.900	0.830	0.870
magneto_gdc	0.950	0.960	0.960	0.940	0.920	0.890	0.930	0.960	0.950	0.620
mmtu	0.870	0.860	0.900	0.840	0.880	0.850	0.830	0.800	0.790	0.750
officeqa	0.030	0.050	0.030	0.040	0.030	0.010	0.040	0.020	0.020	0.010
officeqa_pro_v2	0.056	0.122	0.100	0.044	0.033	0.011	0.022	0.000	0.033	0.022
opensanctions_pairs	0.860	0.940	0.750	0.730	0.850	0.900	0.760	0.850	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.950	0.950	0.950	0.940	0.940	0.930	0.940
realhitbench	0.650	0.660	0.630	0.610	0.560	0.580	0.580	0.580	0.530	0.480
smat	0.880	0.900	0.910	0.910	0.910	0.930	0.880	0.880	0.880	0.680
suc	0.970	0.980	0.920	0.880	0.920	0.880	0.810	0.910	0.910	0.670
tab_fact	0.860	0.660	0.840	0.740	0.850	0.820	0.840	0.760	0.830	0.780
tabis	0.950	0.960	0.960	0.920	0.980	0.900	0.770	0.970	0.920	0.820
tablebench	0.780	0.800	0.810	0.840	0.800	0.790	0.780	0.710	0.760	0.550
tableeval	0.770	0.740	0.750	0.660	0.770	0.690	0.620	0.720	0.740	0.540
tat_qa	0.740	0.810	0.680	0.750	0.710	0.710	0.730	0.720	0.680	0.670
tpcdi_cells	0.350	0.350	0.350	0.350	0.330	0.340	0.350	0.350	0.330	0.330
valentine	0.970	1.000	1.000	0.990	0.960	0.980	0.970	0.990	0.970	0.960
walmart_amazon	0.990	0.980	0.970	0.950	0.960	0.950	0.950	0.960	0.940	0.930
wdc_lspc	0.980	0.990	1.000	0.980	1.000	0.970	0.960	1.000	0.980	0.960
wdc_products	0.990	0.980	0.980	0.970	0.990	0.970	0.970	0.980	0.980	0.970
wikitablequestions	0.840	0.840	0.820	0.830	0.810	0.810	0.800	0.800	0.770	0.670

Table 45: Mean score per dataset and model under rule knee.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	Claude Sonnet 5	Grok 4	GPT-5.4-mini	DeepSeek-chat	DeepSeek-R1	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.960	0.960	0.960	0.970	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.980	0.940	0.970	1.000	0.990	1.000	0.990
alaska_schema	0.908	0.938	0.918	0.923	0.919	0.953	0.942	0.939	0.921	0.795
amazon_google	0.930	0.940	0.940	0.960	0.940	0.930	0.960	0.950	0.860	0.950
bird	0.320	0.320	0.186	0.259	0.129	0.193	0.204	0.195	0.082	0.144
dblp_acm	1.000	1.000	1.000	1.000	1.000	0.990	1.000	0.990	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.970	0.950	0.940	0.960	0.950	0.910	0.970
fetaqa	0.279	0.254	0.278	0.254	0.269	0.239	0.278	0.274	0.191	0.219
fintagging	0.868	0.882	0.865	0.868	0.858	0.861	0.816	0.809	0.823	0.816
hitab	0.747	0.780	0.755	0.770	0.749	0.785	0.812	0.773	0.745	0.784
machamp	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.900	0.830	0.870
magneto_gdc	0.958	0.967	0.965	0.949	0.932	0.905	0.944	0.966	0.957	0.695
mmtu	0.898	0.892	0.921	0.887	0.905	0.889	0.878	0.827	0.834	0.809
officeqa	0.308	0.281	0.260	0.252	0.200	0.175	0.100	0.120	0.121	0.152
officeqa_pro_v2	0.270	0.318	0.231	0.184	0.165	0.189	0.164	0.051	0.138	0.165
opensanctions_pairs	0.860	0.940	0.750	0.730	0.850	0.900	0.760	0.850	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.950	0.950	0.950	0.940	0.940	0.930	0.940
realhitbench	0.756	0.748	0.733	0.686	0.647	0.671	0.662	0.666	0.609	0.605
smat	0.905	0.920	0.929	0.927	0.926	0.946	0.903	0.900	0.901	0.753
suc	0.974	0.984	0.934	0.901	0.934	0.901	0.843	0.926	0.927	0.718
tab_fact	0.860	0.660	0.840	0.740	0.850	0.820	0.840	0.760	0.830	0.780
tabis	0.950	0.960	0.960	0.920	0.980	0.900	0.770	0.970	0.920	0.820
tablebench	0.839	0.846	0.857	0.883	0.844	0.846	0.837	0.780	0.826	0.670
tableeval	0.785	0.758	0.781	0.730	0.796	0.743	0.694	0.753	0.770	0.625
tat_qa	0.801	0.843	0.779	0.816	0.791	0.792	0.789	0.795	0.769	0.753
tpcdi_cells	0.462	0.462	0.464	0.462	0.446	0.459	0.462	0.464	0.444	0.443
valentine	0.970	1.000	1.000	0.992	0.968	0.983	0.976	0.990	0.976	0.966
walmart_amazon	0.990	0.980	0.970	0.950	0.960	0.950	0.950	0.960	0.940	0.930
wdc_lspc	0.980	0.990	1.000	0.980	1.000	0.970	0.960	1.000	0.980	0.960
wdc_products	0.990	0.980	0.980	0.970	0.990	0.970	0.970	0.980	0.980	0.970
wikitablequestions	0.876	0.877	0.865	0.858	0.864	0.843	0.841	0.844	0.820	0.744

Table 46: Mean score per dataset and model under rule top1.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	Claude Sonnet 5	Grok 4	GPT-5.4-mini	DeepSeek-chat	DeepSeek-R1	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.960	0.960	0.960	0.970	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.980	0.940	0.970	1.000	0.990	1.000	0.990
alaska_schema	0.906	0.934	0.918	0.922	0.919	0.953	0.937	0.938	0.920	0.803
amazon_google	0.930	0.940	0.940	0.960	0.940	0.930	0.960	0.950	0.860	0.950
bird	0.328	0.329	0.188	0.270	0.130	0.200	0.216	0.219	0.084	0.157
dblp_acm	1.000	1.000	1.000	1.000	1.000	0.990	1.000	0.990	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.970	0.950	0.940	0.960	0.950	0.910	0.970
fetaqa	0.216	0.215	0.207	0.170	0.210	0.106	0.210	0.204	0.190	0.069
fintagging	0.856	0.872	0.854	0.856	0.846	0.848	0.800	0.808	0.808	0.801
hitab	0.750	0.787	0.761	0.781	0.756	0.792	0.817	0.785	0.750	0.788
machamp	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.900	0.830	0.870
magneto_gdc	0.957	0.966	0.962	0.948	0.930	0.899	0.941	0.967	0.955	0.723
mmtu	0.896	0.897	0.925	0.891	0.906	0.890	0.882	0.830	0.835	0.816
officeqa	0.354	0.318	0.298	0.286	0.228	0.201	0.110	0.139	0.138	0.174
officeqa_pro_v2	0.298	0.343	0.249	0.204	0.184	0.217	0.182	0.071	0.153	0.185
opensanctions_pairs	0.860	0.940	0.750	0.730	0.850	0.900	0.760	0.850	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.950	0.950	0.950	0.940	0.940	0.930	0.940
realhitbench	0.756	0.750	0.734	0.695	0.649	0.675	0.664	0.683	0.606	0.598
smat	0.901	0.916	0.925	0.923	0.924	0.942	0.898	0.897	0.900	0.758
suc	0.973	0.983	0.930	0.897	0.930	0.895	0.838	0.923	0.925	0.709
tab_fact	0.860	0.660	0.840	0.740	0.850	0.820	0.840	0.760	0.830	0.780
tabis	0.950	0.960	0.960	0.920	0.980	0.900	0.770	0.970	0.920	0.820
tablebench	0.839	0.844	0.858	0.883	0.845	0.847	0.837	0.782	0.826	0.680
tableeval	0.783	0.756	0.778	0.732	0.794	0.745	0.696	0.762	0.769	0.626
tat_qa	0.799	0.843	0.781	0.816	0.795	0.794	0.789	0.795	0.779	0.750
tpcdi_cells	0.423	0.423	0.424	0.423	0.410	0.420	0.423	0.424	0.404	0.404
valentine	0.970	1.000	1.000	0.992	0.966	0.983	0.974	0.990	0.974	0.965
walmart_amazon	0.990	0.980	0.970	0.950	0.960	0.950	0.950	0.960	0.940	0.930
wdc_lspc	0.980	0.990	1.000	0.980	1.000	0.970	0.960	1.000	0.980	0.960
wdc_products	0.990	0.980	0.980	0.970	0.990	0.970	0.970	0.980	0.980	0.970
wikitablequestions	0.875	0.873	0.861	0.857	0.863	0.842	0.837	0.841	0.816	0.746

Table 47: Mean score per dataset and model under rule top2.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	Claude Sonnet 5	Grok 4	GPT-5.4-mini	DeepSeek-chat	DeepSeek-R1	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.960	0.960	0.960	0.970	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.980	0.940	0.970	1.000	0.990	1.000	0.990
alaska_schema	0.930	0.953	0.941	0.936	0.943	0.964	0.961	0.960	0.949	0.905
amazon_google	0.930	0.940	0.940	0.960	0.940	0.930	0.960	0.950	0.860	0.950
bird	0.264	0.268	0.186	0.207	0.116	0.163	0.176	0.180	0.087	0.101
dblp_acm	1.000	1.000	1.000	1.000	1.000	0.990	1.000	0.990	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.970	0.950	0.940	0.960	0.950	0.910	0.970
fetaqa	0.618	0.568	0.601	0.500	0.596	0.381	0.600	0.564	0.491	0.286
fintagging	0.927	0.935	0.921	0.928	0.917	0.924	0.899	0.883	0.903	0.901
hitab	0.754	0.786	0.764	0.763	0.748	0.786	0.820	0.779	0.761	0.790
machamp	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.900	0.830	0.870
magneto_gdc	0.969	0.976	0.968	0.962	0.947	0.918	0.962	0.976	0.965	0.850
mmtu	0.890	0.873	0.912	0.856	0.898	0.874	0.851	0.821	0.826	0.811
officeqa	0.081	0.093	0.075	0.087	0.058	0.040	0.050	0.052	0.037	0.042
officeqa_pro_v2	0.126	0.183	0.146	0.082	0.074	0.058	0.079	0.035	0.070	0.065
opensanctions_pairs	0.860	0.940	0.750	0.730	0.850	0.900	0.760	0.850	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.950	0.950	0.950	0.940	0.940	0.930	0.940
realhitbench	0.737	0.741	0.718	0.680	0.635	0.654	0.666	0.668	0.597	0.621
smat	0.938	0.946	0.953	0.949	0.948	0.966	0.934	0.928	0.934	0.877
suc	0.979	0.989	0.952	0.921	0.952	0.927	0.873	0.945	0.938	0.762
tab_fact	0.860	0.660	0.840	0.740	0.850	0.820	0.840	0.760	0.830	0.780
tabis	0.950	0.960	0.960	0.920	0.980	0.900	0.770	0.970	0.920	0.820
tablebench	0.831	0.847	0.846	0.872	0.834	0.837	0.829	0.791	0.800	0.653
tableeval	0.799	0.775	0.788	0.740	0.810	0.745	0.733	0.770	0.776	0.652
tat_qa	0.862	0.885	0.804	0.881	0.794	0.789	0.830	0.825	0.769	0.753
tpcdi_cells	0.458	0.458	0.458	0.458	0.449	0.449	0.458	0.458	0.437	0.438
valentine	0.970	1.000	1.000	0.995	0.978	0.988	0.982	0.990	0.982	0.974
walmart_amazon	0.990	0.980	0.970	0.950	0.960	0.950	0.950	0.960	0.940	0.930
wdc_lspc	0.980	0.990	1.000	0.980	1.000	0.970	0.960	1.000	0.980	0.960
wdc_products	0.990	0.980	0.980	0.970	0.990	0.970	0.970	0.980	0.980	0.970
wikitablequestions	0.889	0.882	0.866	0.872	0.874	0.842	0.845	0.851	0.831	0.760

Table 48: Mean score per dataset and model under rule top3.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	Claude Sonnet 5	Grok 4	GPT-5.4-mini	DeepSeek-chat	DeepSeek-R1	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.960	0.960	0.960	0.970	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.980	0.940	0.970	1.000	0.990	1.000	0.990
alaska_schema	0.930	0.953	0.941	0.936	0.943	0.964	0.961	0.960	0.949	0.906
amazon_google	0.930	0.940	0.940	0.960	0.940	0.930	0.960	0.950	0.860	0.950
bird	0.264	0.267	0.186	0.207	0.116	0.163	0.176	0.180	0.088	0.102
dblp_acm	1.000	1.000	1.000	1.000	1.000	0.990	1.000	0.990	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.970	0.950	0.940	0.960	0.950	0.910	0.970
fetaqa	0.618	0.568	0.600	0.494	0.595	0.366	0.598	0.560	0.493	0.265
fintagging	0.928	0.936	0.922	0.928	0.918	0.924	0.900	0.883	0.903	0.901
hitab	0.753	0.787	0.764	0.764	0.748	0.787	0.821	0.779	0.761	0.790
machamp	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.900	0.830	0.870
magneto_gdc	0.969	0.977	0.968	0.962	0.947	0.918	0.963	0.976	0.965	0.853
mmtu	0.890	0.873	0.913	0.856	0.898	0.874	0.851	0.821	0.826	0.811
officeqa	0.081	0.093	0.075	0.087	0.058	0.040	0.050	0.052	0.037	0.042
officeqa_pro_v2	0.126	0.182	0.146	0.082	0.075	0.058	0.079	0.035	0.070	0.065
opensanctions_pairs	0.860	0.940	0.750	0.730	0.850	0.900	0.760	0.850	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.950	0.950	0.950	0.940	0.940	0.930	0.940
realhitbench	0.736	0.740	0.717	0.680	0.634	0.654	0.666	0.668	0.596	0.620
smat	0.938	0.946	0.953	0.949	0.949	0.966	0.934	0.928	0.934	0.879
suc	0.979	0.989	0.952	0.921	0.952	0.927	0.873	0.945	0.938	0.762
tab_fact	0.860	0.660	0.840	0.740	0.850	0.820	0.840	0.760	0.830	0.780
tabis	0.950	0.960	0.960	0.920	0.980	0.900	0.770	0.970	0.920	0.820
tablebench	0.831	0.846	0.846	0.871	0.834	0.837	0.829	0.791	0.800	0.653
tableeval	0.799	0.775	0.788	0.740	0.810	0.745	0.734	0.770	0.776	0.652
tat_qa	0.862	0.885	0.804	0.882	0.795	0.789	0.830	0.825	0.770	0.752
tpcdi_cells	0.452	0.452	0.452	0.452	0.443	0.443	0.452	0.452	0.432	0.432
valentine	0.970	1.000	1.000	0.995	0.978	0.988	0.982	0.990	0.982	0.974
walmart_amazon	0.990	0.980	0.970	0.950	0.960	0.950	0.950	0.960	0.940	0.930
wdc_lspc	0.980	0.990	1.000	0.980	1.000	0.970	0.960	1.000	0.980	0.960
wdc_products	0.990	0.980	0.980	0.970	0.990	0.970	0.970	0.980	0.980	0.970
wikitablequestions	0.888	0.882	0.865	0.872	0.873	0.842	0.844	0.850	0.830	0.760

Table 49: Mean score per dataset and model under rule top4.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	Claude Sonnet 5	Grok 4	GPT-5.4-mini	DeepSeek-chat	DeepSeek-R1	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.960	0.960	0.960	0.970	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.980	0.940	0.970	1.000	0.990	1.000	0.990
alaska_schema	0.908	0.938	0.918	0.923	0.919	0.953	0.942	0.939	0.921	0.795
amazon_google	0.930	0.940	0.940	0.960	0.940	0.930	0.960	0.950	0.860	0.950
bird	0.320	0.320	0.186	0.259	0.129	0.193	0.204	0.195	0.082	0.144
dblp_acm	1.000	1.000	1.000	1.000	1.000	0.990	1.000	0.990	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.970	0.950	0.940	0.960	0.950	0.910	0.970
fetaqa	0.279	0.254	0.278	0.254	0.269	0.239	0.278	0.274	0.191	0.219
fintagging	0.868	0.882	0.865	0.868	0.858	0.861	0.816	0.809	0.823	0.816
hitab	0.747	0.780	0.755	0.770	0.749	0.785	0.812	0.773	0.745	0.784
machamp	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.900	0.830	0.870
magneto_gdc	0.958	0.967	0.965	0.949	0.932	0.905	0.944	0.966	0.957	0.695
mmtu	0.898	0.892	0.921	0.887	0.905	0.889	0.878	0.827	0.834	0.809
officeqa	0.308	0.281	0.260	0.252	0.200	0.175	0.100	0.120	0.121	0.152
officeqa_pro_v2	0.270	0.318	0.231	0.184	0.165	0.189	0.164	0.051	0.138	0.165
opensanctions_pairs	0.860	0.940	0.750	0.730	0.850	0.900	0.760	0.850	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.950	0.950	0.950	0.940	0.940	0.930	0.940
realhitbench	0.756	0.748	0.733	0.686	0.647	0.671	0.662	0.666	0.609	0.605
smat	0.905	0.920	0.929	0.927	0.926	0.946	0.903	0.900	0.901	0.753
suc	0.974	0.984	0.934	0.901	0.934	0.901	0.843	0.926	0.927	0.718
tab_fact	0.860	0.660	0.840	0.740	0.850	0.820	0.840	0.760	0.830	0.780
tabis	0.950	0.960	0.960	0.920	0.980	0.900	0.770	0.970	0.920	0.820
tablebench	0.839	0.846	0.857	0.883	0.844	0.846	0.837	0.780	0.826	0.670
tableeval	0.785	0.758	0.781	0.730	0.796	0.743	0.694	0.753	0.770	0.625
tat_qa	0.801	0.843	0.779	0.816	0.791	0.792	0.789	0.795	0.769	0.753
tpcdi_cells	0.462	0.462	0.464	0.462	0.446	0.459	0.462	0.464	0.444	0.443
valentine	0.970	1.000	1.000	0.992	0.968	0.983	0.976	0.990	0.976	0.966
walmart_amazon	0.990	0.980	0.970	0.950	0.960	0.950	0.950	0.960	0.940	0.930
wdc_lspc	0.980	0.990	1.000	0.980	1.000	0.970	0.960	1.000	0.980	0.960
wdc_products	0.990	0.980	0.980	0.970	0.990	0.970	0.970	0.980	0.980	0.970
wikitablequestions	0.876	0.877	0.865	0.858	0.864	0.843	0.841	0.844	0.820	0.744

Table 50: Mean score per dataset and model under rule top5.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	Claude Sonnet 5	Grok 4	GPT-5.4-mini	DeepSeek-chat	DeepSeek-R1	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.960	0.960	0.960	0.970	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.980	0.940	0.970	1.000	0.990	1.000	0.990
alaska_schema	0.911	0.938	0.923	0.925	0.924	0.955	0.941	0.943	0.926	0.827
amazon_google	0.930	0.940	0.940	0.960	0.940	0.930	0.960	0.950	0.860	0.950
bird	0.316	0.319	0.188	0.260	0.127	0.194	0.211	0.218	0.085	0.149
dblp_acm	1.000	1.000	1.000	1.000	1.000	0.990	1.000	0.990	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.970	0.950	0.940	0.960	0.950	0.910	0.970
fetaqa	0.285	0.280	0.273	0.223	0.276	0.141	0.276	0.265	0.253	0.091
fintagging	0.868	0.883	0.865	0.868	0.858	0.861	0.817	0.824	0.824	0.818
hitab	0.751	0.788	0.763	0.781	0.757	0.793	0.819	0.788	0.754	0.790
machamp	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.900	0.830	0.870
magneto_gdc	0.959	0.968	0.963	0.951	0.933	0.902	0.945	0.969	0.956	0.758
mmtu	0.895	0.893	0.923	0.885	0.905	0.887	0.876	0.829	0.834	0.817
officeqa	0.308	0.281	0.261	0.253	0.200	0.175	0.100	0.126	0.121	0.152
officeqa_pro_v2	0.270	0.316	0.232	0.184	0.166	0.190	0.165	0.069	0.139	0.165
opensanctions_pairs	0.860	0.940	0.750	0.730	0.850	0.900	0.760	0.850	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.950	0.950	0.950	0.940	0.940	0.930	0.940
realhitbench	0.753	0.749	0.732	0.694	0.647	0.672	0.666	0.685	0.604	0.602
smat	0.907	0.921	0.930	0.928	0.929	0.946	0.905	0.903	0.907	0.784
suc	0.974	0.984	0.934	0.901	0.934	0.901	0.843	0.927	0.927	0.718
tab_fact	0.860	0.660	0.840	0.740	0.850	0.820	0.840	0.760	0.830	0.780
tabis	0.950	0.960	0.960	0.920	0.980	0.900	0.770	0.970	0.920	0.820
tablebench	0.837	0.844	0.855	0.881	0.843	0.845	0.835	0.785	0.821	0.676
tableeval	0.786	0.759	0.780	0.735	0.797	0.746	0.704	0.767	0.771	0.632
tat_qa	0.811	0.852	0.787	0.830	0.796	0.794	0.798	0.802	0.779	0.750
tpcdi_cells	0.424	0.424	0.425	0.424	0.413	0.420	0.424	0.425	0.405	0.405
valentine	0.970	1.000	1.000	0.992	0.968	0.984	0.975	0.990	0.975	0.967
walmart_amazon	0.990	0.980	0.970	0.950	0.960	0.950	0.950	0.960	0.940	0.930
wdc_lspc	0.980	0.990	1.000	0.980	1.000	0.970	0.960	1.000	0.980	0.960
wdc_products	0.990	0.980	0.980	0.970	0.990	0.970	0.970	0.980	0.980	0.970
wikitablequestions	0.878	0.874	0.862	0.860	0.865	0.842	0.838	0.843	0.819	0.750

Table 51: Mean score per dataset and model under rule v1.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	Claude Sonnet 5	Grok 4	GPT-5.4-mini	DeepSeek-chat	DeepSeek-R1	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.960	0.960	0.960	0.970	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.980	0.940	0.970	1.000	0.990	1.000	0.990
alaska_schema	0.918	0.945	0.928	0.929	0.929	0.958	0.951	0.948	0.934	0.840
amazon_google	0.930	0.940	0.940	0.960	0.940	0.930	0.960	0.950	0.860	0.950
bird	0.293	0.293	0.185	0.232	0.122	0.178	0.187	0.178	0.084	0.122
dblp_acm	1.000	1.000	1.000	1.000	1.000	0.990	1.000	0.990	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.970	0.950	0.940	0.960	0.950	0.910	0.970
fetaqa	0.445	0.401	0.438	0.373	0.428	0.312	0.434	0.413	0.322	0.253
fintagging	0.897	0.909	0.893	0.898	0.887	0.892	0.858	0.840	0.863	0.858
hitab	0.748	0.781	0.757	0.763	0.746	0.784	0.814	0.771	0.750	0.784
machamp	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.900	0.830	0.870
magneto_gdc	0.963	0.971	0.967	0.955	0.939	0.912	0.953	0.970	0.961	0.753
mmtu	0.894	0.882	0.917	0.872	0.901	0.882	0.865	0.822	0.830	0.807
officeqa	0.194	0.187	0.168	0.170	0.129	0.107	0.075	0.083	0.079	0.097
officeqa_pro_v2	0.198	0.250	0.188	0.133	0.119	0.123	0.121	0.036	0.104	0.115
opensanctions_pairs	0.860	0.940	0.750	0.730	0.850	0.900	0.760	0.850	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.950	0.950	0.950	0.940	0.940	0.930	0.940
realhitbench	0.746	0.743	0.724	0.681	0.640	0.661	0.662	0.659	0.603	0.611
smat	0.921	0.933	0.940	0.938	0.936	0.956	0.918	0.913	0.916	0.806
suc	0.976	0.986	0.943	0.911	0.943	0.914	0.858	0.935	0.932	0.740
tab_fact	0.860	0.660	0.840	0.740	0.850	0.820	0.840	0.760	0.830	0.780
tabis	0.950	0.960	0.960	0.920	0.980	0.900	0.770	0.970	0.920	0.820
tablebench	0.835	0.846	0.852	0.877	0.839	0.841	0.833	0.783	0.813	0.660
tableeval	0.791	0.765	0.784	0.733	0.802	0.742	0.711	0.756	0.773	0.636
tat_qa	0.829	0.861	0.789	0.845	0.791	0.789	0.806	0.806	0.765	0.752
tpcdi_cells	0.462	0.462	0.462	0.462	0.447	0.455	0.462	0.462	0.442	0.442
valentine	0.970	1.000	1.000	0.994	0.973	0.985	0.979	0.990	0.979	0.970
walmart_amazon	0.990	0.980	0.970	0.950	0.960	0.950	0.950	0.960	0.940	0.930
wdc_lspc	0.980	0.990	1.000	0.980	1.000	0.970	0.960	1.000	0.980	0.960
wdc_products	0.990	0.980	0.980	0.970	0.990	0.970	0.970	0.980	0.980	0.970
wikitablequestions	0.881	0.879	0.865	0.863	0.867	0.842	0.843	0.846	0.824	0.749

Table 52: Standard error of the per-dataset mean under v1.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	Claude Sonnet 5	Grok 4	GPT-5.4-mini	DeepSeek-chat	DeepSeek-R1	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.014	0.000	0.000	0.017	0.020	0.020	0.020	0.017	0.024	0.020
alaska_camera	0.031	0.038	0.000	0.014	0.024	0.017	0.000	0.010	0.000	0.010
alaska_schema	0.026	0.021	0.024	0.025	0.023	0.019	0.018	0.019	0.022	0.028
amazon_google	0.026	0.024	0.024	0.020	0.024	0.026	0.020	0.022	0.035	0.022
bird	0.037	0.036	0.034	0.033	0.027	0.034	0.031	0.031	0.023	0.023
dblp_acm	0.000	0.000	0.000	0.000	0.000	0.010	0.000	0.010	0.029	0.017
dblp_scholar	0.026	0.022	0.020	0.017	0.022	0.024	0.020	0.022	0.029	0.017
fetaqa	0.012	0.014	0.013	0.014	0.013	0.012	0.014	0.016	0.010	0.014
fintagging	0.022	0.021	0.023	0.022	0.024	0.022	0.025	0.029	0.024	0.025
hitab	0.039	0.037	0.039	0.038	0.039	0.037	0.035	0.038	0.038	0.036
machamp	0.030	0.030	0.030	0.030	0.031	0.031	0.031	0.030	0.038	0.034
magneto_gdc	0.017	0.015	0.016	0.019	0.021	0.025	0.018	0.015	0.017	0.032
mmtu	0.028	0.030	0.026	0.030	0.027	0.029	0.031	0.036	0.034	0.035
officeqa	0.019	0.023	0.019	0.022	0.020	0.014	0.020	0.017	0.016	0.014
officeqa_pro_v2	0.025	0.033	0.032	0.023	0.021	0.016	0.019	0.009	0.021	0.018
opensanctions_pairs	0.035	0.024	0.043	0.044	0.036	0.030	0.043	0.036	0.041	0.041
papadakis_dn	0.024	0.027	0.022	0.022	0.022	0.022	0.024	0.024	0.026	0.024
realhitbench	0.037	0.038	0.039	0.041	0.043	0.042	0.042	0.042	0.044	0.041
smat	0.022	0.021	0.019	0.020	0.021	0.016	0.023	0.024	0.023	0.029
suc	0.014	0.011	0.020	0.025	0.020	0.024	0.031	0.022	0.022	0.038
tab_fact	0.035	0.047	0.037	0.044	0.036	0.038	0.037	0.043	0.038	0.041
tabis	0.022	0.020	0.020	0.027	0.014	0.030	0.042	0.017	0.027	0.038
tablebench	0.033	0.033	0.032	0.030	0.034	0.033	0.034	0.037	0.035	0.040
tableeval	0.039	0.040	0.039	0.040	0.038	0.041	0.040	0.040	0.040	0.042
tat_qa	0.031	0.030	0.033	0.029	0.034	0.035	0.034	0.033	0.036	0.037
tpcdi_cells	0.042	0.042	0.042	0.042	0.041	0.041	0.042	0.042	0.041	0.041
valentine	0.017	0.000	0.000	0.006	0.014	0.011	0.013	0.010	0.013	0.016
walmart_amazon	0.010	0.014	0.017	0.022	0.020	0.022	0.022	0.020	0.024	0.026
wdc_lspc	0.014	0.010	0.000	0.014	0.000	0.017	0.020	0.000	0.014	0.020
wdc_products	0.010	0.014	0.014	0.017	0.010	0.017	0.017	0.014	0.014	0.017
wikitablequestions	0.030	0.030	0.032	0.032	0.030	0.034	0.034	0.033	0.034	0.039

Table 53: Per-family optimum and the family's J under its own optimum, the global best and v1.

family	datasets	best F1 / decay / prec / rec	J own best	J global best	J v1	L1 to global best
EM	11	0.00 / 0.70 / 0.00 / 0.30	0.361	0.361	0.361	0.000
SM	6	0.00 / 1.00 / 0.00 / 0.00	0.580	0.524	0.503	0.600
TQA	11	0.00 / 0.00 / 0.20 / 0.80	0.497	0.464	0.480	1.400
FIN	3	0.05 / 0.05 / 0.50 / 0.40	0.822	0.735	0.689	1.300

Table 54: The 30 best weightings with their bootstrap statistics (full plateau table in data/selection/plateau.csv).

rank_full	w_f1	w_decay	w_precision	w_recall	J_full	J_boot_mean	J_boot_sd	J_boot_lo	J_boot_hi	p_vs_best	J_oob_mean	times_selected
1	0.000	0.700	0.000	0.300	0.421	0.359	0.028	0.303	0.412	1.000	0.332	2
3	0.400	0.100	0.050	0.450	0.419	0.361	0.027	0.311	0.414	0.569	0.333	0
2	0.350	0.100	0.100	0.450	0.419	0.362	0.027	0.311	0.413	0.588	0.333	2
4	0.050	0.600	0.250	0.100	0.419	0.356	0.027	0.305	0.408	0.371	0.332	0
5	0.000	0.600	0.000	0.400	0.419	0.360	0.028	0.306	0.413	0.627	0.335	4
6	0.000	0.650	0.000	0.350	0.419	0.360	0.028	0.305	0.413	0.641	0.334	5
7	0.100	0.600	0.200	0.100	0.419	0.357	0.027	0.305	0.410	0.448	0.333	1
8	0.000	0.100	0.000	0.900	0.418	0.356	0.028	0.304	0.408	0.433	0.326	3
9	0.200	0.000	0.500	0.300	0.418	0.358	0.027	0.306	0.408	0.492	0.329	0
10	0.350	0.050	0.100	0.500	0.418	0.358	0.027	0.305	0.410	0.483	0.330	0
11	0.250	0.250	0.100	0.400	0.418	0.363	0.027	0.310	0.415	0.641	0.337	0
12	0.450	0.150	0.000	0.400	0.418	0.363	0.026	0.315	0.416	0.606	0.335	0
13	0.200	0.150	0.200	0.450	0.418	0.363	0.026	0.314	0.414	0.609	0.335	0
14	0.200	0.150	0.250	0.400	0.418	0.363	0.026	0.313	0.413	0.610	0.335	0
15	0.400	0.100	0.100	0.400	0.418	0.362	0.027	0.311	0.414	0.596	0.333	0
16	0.450	0.100	0.050	0.400	0.418	0.362	0.027	0.313	0.413	0.591	0.333	0
17	0.000	0.700	0.050	0.250	0.418	0.358	0.028	0.305	0.410	0.439	0.332	2
20	0.150	0.600	0.050	0.200	0.418	0.359	0.027	0.306	0.413	0.536	0.334	1
18	0.050	0.600	0.150	0.200	0.418	0.358	0.027	0.303	0.411	0.454	0.334	0
19	0.100	0.550	0.200	0.150	0.418	0.360	0.027	0.307	0.412	0.536	0.336	0
21	0.700	0.000	0.250	0.050	0.417	0.362	0.027	0.309	0.413	0.581	0.333	9
22	0.750	0.000	0.200	0.050	0.417	0.362	0.027	0.309	0.412	0.588	0.333	6
23	0.800	0.000	0.200	0.000	0.417	0.363	0.027	0.310	0.414	0.598	0.333	11
24	0.850	0.000	0.150	0.000	0.417	0.363	0.027	0.310	0.414	0.596	0.333	5
25	0.300	0.000	0.450	0.250	0.417	0.359	0.027	0.306	0.410	0.515	0.330	1
26	0.000	0.150	0.050	0.800	0.417	0.360	0.027	0.309	0.413	0.534	0.331	6
27	0.050	0.150	0.000	0.800	0.417	0.360	0.027	0.309	0.414	0.532	0.331	0
28	0.150	0.100	0.000	0.750	0.417	0.358	0.027	0.303	0.411	0.488	0.329	0
29	0.000	0.750	0.000	0.250	0.417	0.357	0.028	0.302	0.409	0.410	0.331	0
30	0.000	0.450	0.000	0.550	0.417	0.362	0.027	0.308	0.414	0.676	0.337	8

Table 55: Anchor correctness per dataset under three weightings.

dataset	v1	best J
abt_buy	0.860	0.860
alaska_camera	0.857	0.857
alaska_schema	0.877	0.834
amazon_google	0.863	0.863
bird	0.889	0.824
dblp_acm	0.860	0.860
dblp_scholar	0.858	0.858
fetaqa	0.913	0.838
fintagging	0.874	0.825
hitab	0.906	0.844
machamp	0.857	0.857
magneto_gdc	0.872	0.836
mmtu	0.868	0.838
officeqa	0.894	0.865
officeqa_pro_v2	0.885	0.850
opensanctions_pairs	0.860	0.860
papadakis_dn	0.891	0.891
realhitbench	0.899	0.839
smat	0.914	0.850
suc	0.857	0.815
tab_fact	0.862	0.862
tabis	0.639	0.638
tablebench	0.898	0.832
tableeval	0.866	0.817
tat_qa	0.885	0.844
tpcdi_cells	0.905	0.856
valentine	0.846	0.820
walmart_amazon	0.867	0.867
wdc_lspc	0.856	0.856
wdc_products	0.856	0.856
wikitablequestions	0.877	0.816

Table 56: Kendall τ between the Bradley–Terry rankings of the rules.

rule	exact match	v1	top1	top2	top3	top4	top5	best-rho	knee
exact match	1.000	0.822	0.778	0.778	0.778	0.822	0.778	0.822	0.822
v1	0.822	1.000	0.867	0.956	0.956	1.000	0.867	0.911	1.000
top1	0.778	0.867	1.000	0.911	0.911	0.867	1.000	0.867	0.867
top2	0.778	0.956	0.911	1.000	1.000	0.956	0.911	0.867	0.956
top3	0.778	0.956	0.911	1.000	1.000	0.956	0.911	0.867	0.956
top4	0.822	1.000	0.867	0.956	0.956	1.000	0.867	0.911	1.000
top5	0.778	0.867	1.000	0.911	0.911	0.867	1.000	0.867	0.867
best-rho	0.822	0.911	0.867	0.867	0.867	0.911	0.867	1.000	0.911
knee	0.822	1.000	0.867	0.956	0.956	1.000	0.867	0.911	1.000

Table 57: Run facts per model from the prediction cache.

model	answers	cost_usd	mean_out_tokens	median_latency_s	reasked_2500	reasked_8000	still_cut
Gemini 3.8 Flash	3083	3.35	261.60	1.28	166	18	0
Claude Opus 5	3282	99.34	245.15	2.16	144	51	15
GPT-5.5	3283	19.89	169.63	1.91	150	3	0
Claude Sonnet 5	3282	14.86	143.56	1.31	78	46	13
Grok 4	3083	8.33	36.95	3.27	0	0	0
GPT-5.4-mini	3282	4.22	125.36	0.87	127	4	0
DeepSeek-chat	3083	0.62	50.73	0.96	16	1	1
DeepSeek-R1	3083	7.15	916.57	1.27	262	325	309
gpt-oss-120b	3083	0.70	233.36	1.61	120	13	1
Llama 3.3 70B	3083	1.77	82.25	1.40	16	0	0

Table 58: One-step sensitivity around v1 and the best weighting.

point	n_neighbours	J_centre	J_min	J_max	J_mean	flatness_range	share_neighbours_better
v1	12.0000	0.4040	0.4025	0.4095	0.4052	0.0070	0.6154
best	6.0000	0.4215	0.4162	0.4215	0.4176	0.0053	0.0000